Dell PowerFlex v3.6.x

CLI Reference Guide

August 2026
Rev. 2.2

DELL Technologies

Notes, cautions, and warnings

NOTE: A NOTE indicates important information that helps you make better use of your product.

CAUTION: A CAUTION indicates either potential damage to hardware or loss of data and tells you how to avoid the problem.

WARNING: A WARNING indicates a potential for property damage, personal injury, or death.

Copyright © Dell Inc. All Rights Reserved. Dell Technologies, Dell, and other trademarks are trademarks of Dell Inc. or its subsidiaries. Other trademarks may be trademarks of their respective owners.

Contents

Chapter 1: SCLI Basics **10**
Accessing SCLI 10
Command format 10
Syntax 11
Object names 12
Using SCLI in non-secure mode 12
Logging in 12
login 13
logout 14
Getting help 14
help 14

Chapter 2: Acceleration Pool Commands **16**
add_acceleration_pool 16
modify_metadata_acceleration 16
query_acceleration_pool 17
remove_acceleration_pool 18
rename_acceleration_pool 18

Chapter 3: Device Commands **20**
abort_remove_sds_device 20
activate_sds_device 20
add_sds_device 21
clear_sds_device_error 23
modify_sds_device_capacity 24
query_all_device_latency_meters 25
query_device_latency_meters 26
query_device_test 27
query_sds_device_info 28
remove_sds_device 30
rename_device 31
set_sds_device_external_acceleration_type 32
set_sds_device_led 33
set_sds_device_media_type 34
start_device_test 34
set_sds_device_path 36

Chapter 4: Fault Set Commands **37**
add_fault_set 37
clear_fault_set 38
query_all_fault_sets 38
query_fault_set 39
remove_fault_set 40
rename_fault_set 40

Chapter 5: General CLI Commands..... 42
acknowledge_scanner_error_counters..... 42
add_trusted_ca..... 43
extract_root_ca..... 43
list_trusted_ca..... 44
query_license..... 44
query_disconnections..... 44
query_remote_syslog..... 45
query_oscillating_failure_counter_parameters..... 45
remove_trusted_ca..... 46
reset_oscillating_failure_counters..... 47
set_cli_login_banner_acceptance_mode..... 48
set_cli_login_banner_preemptive_acceptance..... 48
set_license..... 49
set_login_banner..... 49
set_oscillating_failure_counter_parameters..... 50
set_remote_read_only_limit_state..... 51
set_syslog_facility..... 52
start_remote_syslog..... 52
stop_remote_syslog..... 53

Chapter 6: LDAP Commands..... 54
add_ldap_service..... 54
assign_ldap_groups_to_roles..... 56
set_user_authentication_method..... 56
reload_ldap_certificate..... 57
remove_ldap_group_from_role_assignment..... 58
remove_ldap_service..... 58
rename_ldap_service..... 59

Chapter 7: Protection Domain Commands..... 60
activate_protection_domain..... 60
add_protection_domain..... 60
disable_fgl_metadata_cache..... 61
enable_fgl_metadata_cache..... 61
inactivate_protection_domain..... 62
query_port_flapping_status..... 63
query_port_flapping_unsupported_components..... 64
query_protection_domain..... 64
remove_protection_domain..... 67
rename_protection_domain..... 67
set_default_fgl_metadata_cache_size..... 68

Chapter 8: Replication Commands..... 69
abort_replication_consistency_group_snapshots_create..... 69
activate_replication_consistency_group..... 69
activate_replication_consistency_group_mappings..... 70
add_replication_consistency_group..... 70

add_replication_pair 72
add_replication_peer_system 73
add_sdr 73
add_sdr_ip 74
clear_broken_replication_consistency_group_alerts 75
create_replication_consistency_group_snapshots 76
enter_sdr_maintenance_mode 77
exit_sdr_maintenance_mode 77
failover_replication_consistency_group 78
failover_test_start_replication_consistency_group 78
failover_test_stop_replication_consistency_group 79
freeze_apply_replication_consistency_group 79
inactivate_replication_consistency_group_mappings 80
modify_replication_consistency_group_rpo 80
modify_replication_consistency_group_target_volume_access_mode 81
modify_replication_peer_system_ip 82
modify_replication_peer_system_port 82
modify_sdr_ip_role 83
modify_sdr_port 84
pause_pair_initial_copy 85
pause_replication_consistency_group 85
print_recreate_replication_consistency_group_script 86
query_all_replicated_volumes 86
query_all_replication_consistency_groups 87
query_all_replication_journal_capacity 88
query_all_replication_pairs 89
query_all_replication_peer_system 89
query_all_sdr 90
query_replication_consistency_group 90
query_replication_pair 92
query_replication_peer_system 92
query_replication_peer_system_destination_volume_candidates 93
query_replication_peer_system_protection_domains 94
query_sdr 95
query_sync_now_replication_consistency_group 96
remove_replication_consistency_group 97
remove_replication_pair 97
remove_replication_peer_system 98
remove_sdr 98
remove_sdr_ip 99
rename_replication_consistency_group 99
rename_replication_peer_system 100
rename_sdr 101
resume_pair_initial_copy 101
resume_replication_consistency_group 102
restore_replication_consistency_group 102
reverse_replication_consistency_group 103
set_pair_initial_copy_queue_position 103
set_replication_consistency_group_consistent 104
set_replication_consistency_group_inconsistent 104

set_replication_journal_capacity....105
switchover_replication_consistency_group....106
sync_now_replication_consistency_group....107
terminate_replication_consistency_group....107
unfreeze_apply_replication_consistency_group....108

Chapter 9: RFcache Commands....109
disable_rfcache....109
enable_rfcache....110
set_rfcache_parameters....110

Chapter 10: SDC Commands....112
add_sdc....112
check_sdc_authentication_status....112
generate_sdc_password....113
query_all_approved_sdc....113
query_all_sdc....114
query_restricted_sdc_mode....115
query_sdc....115
query_sdc_volume_limits....116
remove_sdc....117
rename_sdc....117
reset_sdc_sds_connections....118
set_restricted_sdc_mode....118
set_sdc_authentication....119
set_sdc_volume_limits....120
set_approved_sdc_ips....121

Chapter 11: SDS Commands....122
abort_remove_sds....122
add_sds....122
add_sds_ip....125
disable_sds_rmcache....126
enable_sds_rmcache....127
enter_maintenance_mode....128
exit_maintenance_mode....128
enter_protected_maintenance_mode....129
exit_protected_maintenance_mode....130
modify_sds_ip_role....131
modify_sds_port....132
query_all_sds....132
query_sds_connectivity_status....133
query_sds....134
query_sds_connectivity_status....136
query_sds_network_test_results....137
query_sdc_to_sds_disconnections....137
remove_sds....138
remove_sds_ip....138
rename_sds....139

set_drl_properties....140
set_fgl_metadata_cache_size....140
set_sds_network_limits....141
set_sds_rmcache_size....142
show_certificate....143
start_sds_network_test....144
update_sds_original_paths....145

Chapter 12: Snapshot Commands....146
add_snapshot_policy....146
add_source_volume_to_snapshot_policy....147
align_snapshot_policy_to_current_time....147
clear_snapshot_security....148
lock_auto_snapshot....148
modify_snapshot_policy....149
pause_snapshot_policy....150
query_all_snapshot_policies....150
query_snapshot_policy....151
remove_snapshot_policy....152
remove_snapshots_consistency_group....152
remove_source_volume_from_snapshot_policy....153
rename_snapshot_policy....153
resume_snapshot_policy....154
set_snapshot_security....154
snapshot_volume....155
unlock_auto_snapshot....156

Chapter 13: Storage Pool Commands....158
add_storage_pool....158
disable_background_device_scanner....160
disable_persistent_checksum....160
enable_background_device_scanner....161
enable_persistent_checksum....162
modify_compression_method....163
modify_persistent_checksum....164
modify_zero_padding_policy....164
query_inaccessible_user_address_space....165
query_storage_pool....166
remove_storage_pool....168
rename_storage_pool....168
reset_scanner_error_counters....169
set_capacity_alerts_threshold....170
set_rebalance_mode....171
set_rebalance_policy....172
set_rebuild_mode....173
set_rebuild_policy....174
set_rebuild_rebalance_parallelism....175
set_rmcache_usage....175
set_rmcache_write_handling_mode....176

set_storage_pool_external_acceleration_type....177
set_storage_pool_media_type....178
set_checksum_mode....179
set_protected_maintenance_mode_policy....179
test_inaccessible_data....180

Chapter 14: System Commands....182
abort_upgrade....182
add_certificate....182
add_standby_mdm....183
allow_commands_during_upgrade....184
approve_all_mdm_certificates....184
create_mdm_cluster....184
finalize_upgrade....185
generate_certificate....186
generate_mdm_certificate....186
generate_mdm_csr_file....187
modify_cluster_virtual_ips....188
modify_management_ip....189
modify_spare_policy....190
modify_virtual_ip_interfaces....191
query_all....191
Output description....195
query_cluster....199
query_performance_parameters....200
query_properties....204
Object presets....205
query_remote_read_only_limit_state....206
query_system_limits....207
query_upgrade....208
refresh_mdm_cluster_capabilities....208
remove_standby_mdm....209
rename_mdm....209
rename_system....210
replace_cluster_mdm....210
replace_mdm_security_files....212
set_component_authentication_properties....212
set_management_client_communication....213
set_performance_parameters....213
start_upgrade....215
switch_cluster_mode....215
switch_mdm_ownership....217

Chapter 15: User Commands....218
add_user....218
delete_user....218
disable_admin....219
modify_user....219
query_user....220

query_user_authentication_properties....221
query_users....221
Reset the admin user password....222
reset_admin....222
reset_password....223
set_password....223
set_user_authentication_method....224

Chapter 16: Volume Commands....225
add_volume....225
clear_scsi2_reservation....226
clear_scsi3_reservation....226
map_volume_to_sdc....227
modify_volume_capacity....228
overwrite_volume_content....229
query_all_volumes....229
query_volume....230
query_volume_sdc_meters....231
remove_volume....232
rename_volume....233
set_volume_rmcache_usage....233
unmap_volume_from_sdc....234
set_volume_mapping_access_mode....235
set_volume_access_mode_limit....235

Chapter 17: V-Tree Commands....237
migrate_vtree....237
pause_vtree_migration....238
query_all_vtree_migrations....239
query_vtree....240
query_vtree_migration....241
resume_vtree_migration....241
rollback_vtree_migration....242
set_vtree_migration_policy....243
set_vtree_migration_queue_position....244

Index....246

1

SCLI Basics

This section explains the basic concepts of using the PowerFlex CLI (SCLI), including how to access the SCLI and log in.

Accessing SCLI

PowerFlex CLI (SCLI) is installed as part of the MDM component. It can be found in the following path:

Table 1. SCLI command locations by operating system

Operating System	Path
Linux	`scli`
VMware	`scli`
Citrix Hypervisor (XenServer)	`siocli`

Before using most SCLI commands, you must log in using this command:

```
scli --login --username <username> --password <password>
```

where <username> and <password> are the MDM username and password.

Command format

All SCLI commands use the following format:

Table 2. SCLI command format by operating system

Operating system	Command format
Linux	`scli [--mdm_ip <IP>] --<cmd_name>`
VMware	`scli [--mdm_ip <IP>] --<cmd_name>`
Windows	`scli [--mdm_ip <IP>] --<cmd_name>`
Xen	`siocli [--mdm_ip <IP>] --<cmd_name>`

where:

- `--mdm_ip <IP>`

  is one or more IP addresses of the servers running the primary MDM and secondary MDM. In a non-clustered environment, use the MDM IP address.

  If the CLI does not reside on the MDM, the `--mdm_ip` parameter must be added to every CLI command.

- `--<cmd_name>`

  Is the command to be executed.

Example:

```
scli --mdm_ip 10.10.10.3,10.10.10.4 --query_all
```

Additional rules:

- All parameters are preceded by --
- The order of the parameters in the command is insignificant.
- SCLI commands are lowercase and case-sensitive.

--mdm_ip

The `mdm_ip` parameter indicates the MDM designated to receive and execute the command. If the command is run from the primary MDM, this switch may be omitted.

Example:

```
scli --query_all
```

To avoid using the `--mdm_ip` parameter in every command, or avoid having to install the CLI on other servers, use SSH or RDM to log in to the shell running on the management server.

You cannot execute SCLI commands on the secondary MDM. However, you can send a command from the secondary MDM by adding the `--mdm_ip` parameter with the primary MDM IP address to the command.

Syntax

SCLI syntax follows these rules:

Format

The following command syntax is mandatory:

```
scli --cmd_name
```

Parentheses ( ) indicate a group of required parameters, from which you must choose one.

Angle brackets <> indicate an argument.

Braces { } indicate a series of options for parameter values, from which you must choose one.

A “pipe” | is used to separate items in a series of options (parameters or parameter values). Select one of them.

Square brackets [ ] indicate an optional parameter.

Usage

Each command entry uses the above syntax and looks similar to the following example:

```
scli --cmd_example --r1 (--r2 | --r3 <V1>) [o1 <V2> | o2 {--so1 <V3> | --so2 }] [Options]
```

Parameter description

`--r1`

The `r1` parameter description

`--r2`

The `r2` parameter description

`--r3<V1>`

The `r3` parameter description with possible *<V1>* input values

`--o1<V2>`

The `o1` description with possible *<V2>* input values

--o2 {--so1 <V3> | --so2 }

The o2 description

[Options]

Additional optional parameters:

so1 <V3>

The so1 description with possible *V3* input values

so2

The so2 description

Interpretation

- The text `scli --cmd_example r1` is mandatory. In commands that have no parameters, just `scli --cmd_example` is necessary.
- `(--r2 | --r3 <V1>)` indicates that you must choose one of the options separated by "|". Selecting an option is REQUIRED, as indicated by "( )".
- `[o1 <V2>| o2 {--so1 <V3> | --so2}]` indicates that you may choose one of the options separated by "|". Selecting one of these options is OPTIONAL, as indicated by "[ ]".
- `[Options]` indicates that you may choose from the following additional parameters. It is OPTIONAL, as indicated by "[ ]".

Object names

Define object names according to the following rules:

1. Contains fewer than 32 characters
2. Contains only alphanumeric and punctuation characters
3. Is unique within the object type

PowerFlex objects are assigned a unique ID that can be used to identify the object in CLI commands. You can retrieve the ID via a query or through the object's property sheet in the PowerFlex GUI.

It is recommended that you give each object a meaningful name associated with its operational role.

NOTE: Because the system is a single object, its name is automatically unique.

Using SCLI in non-secure mode

If PowerFlex is running in non-secure mode, you must disable secure communications on every MDM server to enable execution of commands.

- To disable secure communications in Windows, on each MDM open the SCLI `conf.txt` file, and add the following line:

```
cli_use_secure_communication=0
```

- To disable secure communications in Linux, run the following on each MDM:

```
echo cli_use_secure_communication=0 >> ~/.scli/conf.txt
```

NOTE: After renewing certificates, changing from secure mode to non-secure mode is not supported.

Logging in

To access the CLI, you must first log in to the management system using a terminal application.

If the CLI and the MDM do not reside on the same server, add the `--mdm_ip` parameter to all CLI commands.

In a non-clustered environment, use the MDM IP address. In a clustered environment, use the IP addresses of the primary and secondary MDMs, separated by a comma. For example:

```
scli --mdm_ip 10.10.10.3,10.10.10.4 --login --username supervisor1 --password password1
```

You will be prompted to enter the password.

When using LDAP, include the LDAP domain in the command. For example:

```
scli --mdm_ip 10.10.10.3,10.10.10.4 --login --username JohnDoe@ldap.acme.com --password
password1 --ldap_authentication
```

The default user created during setup is the SuperUser, with the *admin* username.

login

Log the specified user into the management system. Every user must log in before performing CLI commands.

When a user is authenticated by the system, all commands will be executed with the respective role until a logout is performed, or until the session expires, by reaching one of the following timeouts:

- Maximum session length (default: 8 hours)
- Session idle time (default: 10 minutes)

Syntax

```
scli --login --username <NAME>
  [--password <PASSWORD>]
  [--ldap_authentication | --native_authentication]
  [(--approve_certificate_once | --approve_certificate]
  [--accept_banner_by_scripts_only]
```

NOTE: Actual command syntax is operating-system dependent.

Parameters

--username

Username

--password

User password. If you do not type your password, you will be prompted to do so.

NOTE: In Linux, to prevent the password from being recorded in the history log, leave out the `password` flag and enter the password interactively.

--ldap_authentication

Log in using the LDAP authentication method. LDAP authentication parameters should be configured and LDAP authentication method should be set.

--native_authentication

Log in using the native authentication method (default).

--approve_certificate_once

One-time approval of the MDM's certificate (without adding the certificate to the truststore)

--approve_certificate

Automatic approval of the MDM's certificate for the next commands (adds the certificate to the truststore)

--accept_banner_by_scripts_only

Preemptive approval of login banner

Examples

```
scli --login --username siouser1 --password 1!2@3A
```

NOTE: During installation using the PowerFlex Installer or the VMware plug-in, the password for the admin user is reset, and you should log in with the new password. If you installed PowerFlex manually, after logging in the first time with the default password (`admin`), you must change the password and log in again. Once that is accomplished, the admin user can create additional users.

When logging in, if a login banner has been configured and enabled in your system, you are prompted to press any key, after which the banner is displayed. To continue, enter "q" to quit the login banner, and then enter "y" to approve the banner.

logout

Log the current user out of the system.

Syntax

```
scli --logout
```

Example

```
scli --logout
```

Getting help

The PowerFlex CLI supports auto-completion. To complete a command or parameters, press the Tab key while typing CLI commands.

NOTE:

In Windows, PowerFlex does not support auto-completion.

Alternately, you can run the `help` command.

help

Display help for the CLI commands.

Syntax

```
scli --help
  [--mdm]
  [--sds]
  [--vol]
  [--sdc]
  [--general]
  [--all]
  [--user]
```

NOTE: To view all help options, run the command without any additional options:

```
scli --help
```

Options

--mdm
Displays help for MDM commands

--sds
Displays help for SDS commands

--vol
Displays help for volume commands

--sdc
Displays help for SDC commands

--general
Displays help for general commands

--all
Displays help for all commands

--user
Displays help for commands related to user management

Example

```
scli --help --mdm
```

2

Acceleration Pool Commands

This section contains commands for managing Acceleration Pools.

add_acceleration_pool

Add an Acceleration Pool to a Protection Domain.

Syntax

```
scli --add_acceleration_pool (--protection_domain_id <ID> | --protection_domain_name
<NAME>) (--media_type <TYPE> {SSD | NVDIMM})
  [--acceleration_pool_name <NAME>]
  [--used_for_rfcache]
```

Parameters

--protection_domain_id *<ID>*
Protection Domain ID

--protection_domain_name *<NAME>*
Protection Domain name

--media_type {SSD | NVDIMM}
Type of devices in the Acceleration Pool: SSD or NVDIMM

--acceleration_pool_name *<NAME>*
Name to assign the Acceleration Pool you are creating

--used_for_rfcache
Assigns the Acceleration Pool for exclusive use with RFcache

Example

```
scli --add_acceleration_pool --protection_domain_name pd1 --media_type SSD --
acceleration_pool_name RfcAccPool --used_for_rfcache
```

modify_metadata_acceleration

Modify the Acceleration Pool that will hold the metadata for the specified Storage Pool.

Syntax

```
scli --modify_metadata_acceleration (((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --storage_pool_name <NAME>) | --storage_pool_id
<ID>) (((--protection_domain_id <ID> | --protection_domain_name <NAME>) --
```

```
acceleration_pool_name <NAME>) | --acceleration_pool_id <ID>)
  [--on_device_metadata_on_missing_capacity]
```

Parameters

--protection_domain_id *<ID>*
Protection Domain ID

--protection_domain_name *<NAME>*
Protection Domain name

--storage_pool_name *<NAME>*
Storage Pool name

--storage_pool_id *<ID>*
Storage Pool ID

--acceleration_pool_name *<NAME>*
Acceleration Pool name

--acceleration_pool_id *<ID>*
Acceleration Pool ID

--on_device_metadata_on_missing_capacity
When there is insufficient capacity in the Acceleration Pool, the metadata of some of the devices will reside on the devices themselves.

Example

```
scli --modify_metadata_acceleration --protection_domain_name pd2 --
storage_pool_name sp4 --protection_domain_name pd2 --acceleration_name ap4 --
on_device_metadata_missing_capacity
```

query_acceleration_pool

Retrieve information about a Protection Domain's Acceleration Pool.

Syntax

```
scli --query_acceleration_pool (((--protection_domain_id <ID> | --protection_domain_name
<NAME>) --acceleration_pool_name <NAME>) | --acceleration_pool_id <ID>)
```

Parameters

--protection_domain_id *<ID>*
Protection Domain ID

--protection_domain_name *<NAME>*
Protection Domain name

--acceleration_pool_name *<NAME>*
Acceleration Pool name

--acceleration_pool_id *ID>*
Acceleration Pool ID

Example

```
scli --query_acceleration_pool --protection_domain_name pd2 --acceleration_pool_name
RFcache_pool
```

Sample output

```
Acceleration Pool RFcache_Pool (Id: 6794008e00000000)
        Media Type: SSD
        Used for RFcache: yes
```

remove_acceleration_pool

Remove an Acceleration Pool from a Protection Domain.

Syntax

```
scli --remove_acceleration_pool (((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --acceleration_pool_name <NAME>) | --acceleration_pool_id
<ID>)
```

Parameters

--protection_domain_id *<ID>*
Protection Domain ID

--protection_domain_name *<NAME>*
Protection Domain name

--acceleration_pool_name *<NAME>*
Acceleration Pool name

--acceleration_pool_id *ID>*
Acceleration Pool ID

Example

```
scli --remove_acceleration_pool --protection_domain_name pd1 --acceleration_pool_name
ap_flash1
```

rename_acceleration_pool

Name or rename an Acceleration Pool.

Syntax

```
scli --rename_acceleration_pool (((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --acceleration_pool_name <NAME>) | --acceleration_pool_id
```

```
<ID>) --new_name <NAME>
```

Parameters

--protection_domain_id *<ID>*
Protection Domain ID

--protection_domain_name *<NAME>*
Protection Domain name

--acceleration_pool_name *<NAME>*
Acceleration Pool name

--acceleration_pool_id *ID>*
Acceleration Pool ID

--new_name *<NAME>*
New name to be assigned to the Acceleration Pool

Example

```
scli --rename_accleration_pool --protection_domain_name pd1 --acceleration_pool_name
ap_1 --new_name ap_flash1
```

3

Device Commands

This section contains commands for managing devices.

abort_remove_sds_device

Stop the process of removing an SDS device from an SDS. If it is performed at too late a stage in the removal process, the command may fail and the device will be removed.

NOTE: Removal of acceleration devices cannot be aborted.

Syntax

```
scli --abort_remove_sds_device (--device_id <ID> ((--sds_id <ID> | sds_name <NAME> |
sds_ip [sds_port <PORT>]) (device_name <NAME>  | device_path <PATH>)))
```

Parameters

--device_id *<ID>*

The ID of the storage device that was requested to be removed from the SDS

--sds_id *<ID>*

The ID of the SDS from which the device was to be removed

--sds_name *<NAME>*

The name of the SDS from which the device was to be removed

--sds_ip *<IP>*

The IP address of the SDS from which the device was to be removed

--sds_port *<PORT>*

The port associated with the SDS

--device_name *<NAME>*

The name of the storage device or file that was requested to be removed from the SDS

--device_path *<PATH>*

The path of the device that was requested to be removed

Example

```
scli --abort_remove_sds_device --sds_ip 192.168.1.6 --device_name /dev/sdb
```

activate_sds_device

Activate SDS devices that have completed their tests.

SDS devices that were added using the `add_sds` or `add_sds_device` command with the `test_only` parameter are not used by the system until activated by this command.

This command does not apply to acceleration devices, which are activated automatically.

NOTE: You can activate one or all SDS devices on an SDS with a single command.

Syntax

```
scli --activate_sds_device (--sds_id <ID> |
--sds_name <NAME> | --sds_ip <IP> [--sds_port <PORT>]) (--device_id <ID> | --device_name
<NAME>  | --device_path <PATH> | --activate_all_devices)
```

Parameters

--sds_id *<ID>*

SDS ID

--sds_name *<NAME>*

SDS name

--sds_ip *<IP>*

SDS IP address

--sds_port *<PORT>*

Port associated with the SDS

--device_id *<ID>*

ID of the SDS device to activate

--device_name *<NAME>*

Storage device name or file name

--device_path *<PATH>*

Path of the SDS device to activate

--activate_all_devices

Activate all of the specified SDS's devices. If all the devices cannot be activated, this command will fail.

Example

```
scli --mdm_ip 192.168.1.200 --activate_sds_device
--sds_ip 192.168.1.5 --activate_all_devices
```

add_sds_device

Add a storage or acceleration device to an SDS.

WARNING: During execution of the command, PowerFlex checks that the device is clear before adding it. If the device is not clear, an error message is returned, and the command fails for that device. If you would like to overwrite existing data on the device by forcing the command, use the `force_device_takeover` flag. Note that this flag destroys all existing data on the device!

Syntax

```
scli --add_sds_device (--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP> [--sds_port
<PORT>]) --device_path <PATH> --device_name <NAME>] (--storage_pool_name <NAME> | --
storage_pool_id <ID> | --acceleration_pool_name <NAME> | --acceleration_pool_id <ID>)
  [--force_device_takeover]
  [--media_type {HDD | SSD}]
  [--external_acceleration {none | read | write | read_and_write}]
```

```
[--test_time <TIME>]
[--test_only | --no_test | <BLANK> ]
```

Parameters

--sds_id *<ID>*

ID of the SDS

--sds_name *<NAME>*

Name of the SDS

--sds_ip *<IP>*

SDS IP address

-sds_port *<PORT>*

Port associated with the SDS

--device_path *<PATH>*

Full path of the device to be added. A device can be a disk, an unmounted partition, or a file that represents free space on a mounted device.

--device_name *<NAME>*

Name to assign to the device

--storage_pool_name *<NAME>*

Storage Pool name

--storage_pool_id *<ID>*

Storage Pool ID

--acceleration_pool_name *<NAME>*

Acceleration Pool name

--acceleration_pool_id *<ID>*

Acceleration Pool ID

--force_device_takeover

Add the device, ignoring data from other SDSs on the device.

WARNING: Use this flag with caution, because all data on the device will be destroyed.

--media_type {HDD | SSD}

Media type of the device, where the device will be HDD or SSD.

--external_acceleration *<TYPE>*

External acceleration for the device where *<TYPE>* can be one of the following:

none

The device is not accelerated by non-PowerFlex cache.

read

The device is accelerated by a non-PowerFlex read cache.

write

The device is accelerated by a non-PowerFlex write cache.

read_and_write

The device is accelerated by both non-PowerFlex read and write cache.

--test_time *<TIME>*

The maximum test run-time in seconds. Default: 10. The test will stop when it reaches either this limit, or the time it takes to complete 128 MB of data read/write, whichever is first. When `no_test` is selected, this switch is ignored.

Test options (choose one):

By default, PowerFlex tests the performance of the device being added before its capacity can be used, and saves the results. Two tests are performed: random writes and random reads. When the tests are

complete, the device capacity is added automatically to the Storage Pool used by the MDM. To modify this behavior, specify one of the test options:

--test_only

Devices will be tested, but not used. To start using their capacity, run the `--activate_sds_device` command.

--no_test

The device capacity will be used without any device testing.

<BLANK>

Read and write test will be run on devices before their capacity is used.

Example

```
scli --mdm_ip 192.168.1.200 --add_sds_device
--sds_ip 192.168.1.6 --device_path /dev/sdb --device_name sd02
```

Devices

The following devices are considered to be not "clear," and thus cannot be added to an SDS:

- Linux: A complete device with either a file system or partition, or a partitioned device with a file system.
- Windows: A complete device with a partition, or a partitioned device with a file system.
- ESX: Same as the above, depending on the operating system of the SVM where the SDS is installed.

Limitations:

- SAN devices will not be prevented from being added to an SDS.
- Devices in an LVM group cannot be added to an SDS.
- Within the database devices, only Oracle ASM devices can be detected and blocked.

SDS device names

Assigning each SDS device a meaningful name facilitates future object identification, because the defined name remains constant even if the path changes. When a name has not been defined, the system may display default system-defined names that use the SDS's first IP address.

Each name should conform to the following rules:

1. Contains fewer than 32 characters
2. Contains only alphanumeric and punctuation characters
3. Is unique within the object type

NOTE: PowerFlex objects are assigned a unique ID that can be used to identify the object in CLI commands. You can retrieve the ID via a query or through the object's property sheet in the PowerFlex GUI.

clear_sds_device_error

Clear an SDS device error.

NOTE: The command does not check whether the device error was fixed. If the error persists, the device will return to an error state as soon as it is accessed. You can verify that the device error has been fixed by running an SDS query.

Syntax

```
scli --clear_sds_device_error (--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP> [--
sds_port <PORT>]) (--device_id <ID> | --device_name <NAME> | --device_path <PATH> | --
clear_all)
```

Parameters

--sds_id *<ID>*
SDS ID

--sds_name *<NAME>*
SDS name

--sds_ip *<IP>*
SDS IP address

--sds_port *<PORT>*
Port associated with the SDS

--device_id *<ID>*
ID of the device

--device_name *<NAME>*
Storage device name or file name

--device_path *<PATH>*
Path of the device

--clear_all
Clear errors on all SDS devices.

Example (single device)

```
scli --mdm_ip 192.168.1.200 --clear_sds_device_error
--device_name /dev/sdb --sds_ip 192.168.1.7
```

Example (all devices)

```
scli --mdm_ip 192.168.1.200 --clear_sds_device_error
--clear_all --sds_ip 192.168.1.7
```

modify_sds_device_capacity

Modify the amount of an SDS device's storage capacity that is made available to PowerFlex.

Use this command to reduce (or increase) the device capacity that is available to PowerFlex, in order to assist in balancing storage, without affecting the physical disk size. Ensure that the modified capacity is less than the total physical size of the device. If you reduce capacity, data will be moved to other devices in an asynchronous fashion.

When reducing capacity, the excess device capacity is not accessible to the user.

Syntax

```
scli --modify_sds_device_capacity (--device_id <ID> | ((--sds_id <ID> | --sds_name
<NAME> | --sds_ip <IP> [--sds_port <PORT>]) (--device_name <NAME> | --device_path
<PATH>))) --size_gb <SIZE>
```

Parameters

--device_id *<ID>*
ID of the SDS device

--sds_id *<ID>*

SDS ID

--sds_name *<NAME>*

SDS name

--sds_ip *<IP>*

SDS IP address

--sds_port *<PORT>*

Port associated with the SDS

--device_name *<NAME>*

SDS storage device name or file

--device_path *<PATH>*

Path of the device whose capacity is to be modified

--size_gb *<SIZE>*

Size in GB of the device to use. The size must be within the device size limits and slightly less than the full device size.

Example

```
scli --mdm_ip 192.168.1.200 --modify_sds_device_capacity --sds_ip 192.168.1.6 --
device_name /dev/sdb --size_gb 150
```

query_all_device_latency_meters

Retrieve device latency meters for all the devices, or for devices in the specified Protection Domain.

NOTE: To query the devices in a specific SDS, use the `query_device_latency_meters` command.

Syntax

```
scli --query_all_device_latency_meters
  [(--protection_domain_id <ID> | --protection_domain_name <NAME>)]
```

NOTE: If the background device scanner is enabled, several device read statistics are dramatically affected.

Parameters

--protection_domain_id *<ID>*

Protection Domain ID

--protection_domain_name *<NAME>*

Protection Domain name

Example

```
scli --query_all_device_latency_meters --protection_domain_name pd10
```

Sample output

```
Protection Domain: fcc9af5500000000, Name: pd1 contains 4 SDS
Storage Pool: 3df1cd9600000000, Name: sp1 contains 8 devices
  SDS sds_idd_1:
    Device: /dev/sdb Average latency read: 836 write: 921 (microseconds)
    Device: /dev/sdc Average latency read: 755 write: 843 (microseconds)

  SDS sds_idd_2:
    Device: /dev/sdb Average latency read: 658 write: 815 (microseconds)
    Device: /dev/sdc Average latency read: 689 write: 732 (microseconds)

  SDS sds_idd_4:
    Device: /dev/sdb Average latency read: 842 write: 811 (microseconds)
    Device: /dev/sdc Average latency read: 852 write: 918 (microseconds)

  SDS sds_idd_5:
    Device: /dev/sdb Average latency read: 844 write: 698 (microseconds)
    Device: /dev/sdc Average latency read: 795 write: 801 (microseconds)

  Devices with maximal read latency:
    SDS: sds_idd_4 Device: /dev/sdc  Latency: 852 (microseconds)
    SDS: sds_idd_5 Device: /dev/sdb  Latency: 844 (microseconds)
    SDS: sds_idd_4 Device: /dev/sdb  Latency: 842 (microseconds)
  Devices with maximal write latency:
    SDS: sds_idd_1 Device: /dev/sdb  Latency: 921 (microseconds)
    SDS: sds_idd_4 Device: /dev/sdc  Latency: 918 (microseconds)
    SDS: sds_idd_1 Device: /dev/sdc  Latency: 843 (microseconds)

Protection Domain: fcc9af5600000001, Name: pd2 contains 1 SDS
Storage Pool: 3df1cd9700000001, Name: sp2 contains 1 devices
  SDS sds_idd_3:
    Device: /dev/sdb Average latency read: 0 write: 0 (microseconds)


Protection Domain: fcc9d66500000002, Name: rrr contains 0 SDS
```

query_device_latency_meters

Retrieve the latency meters for all the devices of the specified SDS.

The SDS maintains statistics about reads and writes to its devices. This command returns the average I/O size and latency for both reads and writes.

NOTE: To query all devices in the system, or all devices in a specific Protection Domain, use the `query_all_device_latency_meters` command.

Syntax

```
scli --query_device_latency_meters (--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP>
[--sds_port <PORT>]) [(--storage_pool_name <NAME> | --storage_pool_id <ID>
```

Parameters

--sds_id *<ID>*
SDS ID

--sds_name *<NAME>*
SDS name

--sds_ip *<IP>*
SDS IP address

--sds_port *<PORT>*

Port associated with the SDS

--storage_pool_name *<NAME>*

Storage Pool name

--storage_pool_id *<ID>*

Storage Pool ID

NOTE: If the background device scanner is enabled, several device read statistics are dramatically affected.

Example

```
scli --mdm_ip 192.168.1.200 --query_device_latency_meters --sds_name sds_idd_2 --
storage_pool_name sp1
```

Sample output

```
SDS sds_idd_2 returned information on 2 devices
Device: /dev/sdc
  Average read size: 277.1 KB (283763 Bytes)
  Average read latency: 873 (microseconds)
  Average write size: 355.6 KB (364140 Bytes)
  Average write latency: 1144 (microseconds)
Device: /dev/sdb
  Average read size: 292.8 KB (299816 Bytes)
  Average read latency: 928 (microseconds)
  Average write size: 362.0 KB (370731 Bytes)
  Average write latency: 1157 (microseconds)
```

query_device_test

Retrieve the SDS device test results. Every SDS saves results for the last two tests per device.

Syntax

```
scli --query_device_test ((--device_id <ID> | ((--sds_id <ID> | --sds_name <NAME> |
--sds_ip <IP> [--sds_port <PORT>]) (--device_name <NAME> | --device_path <PATH>)))
| (--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP> [--sds_port <PORT>]) | (((--
protection_domain_id <ID> | --protection_domain_name <NAME>) --storage_pool_name <NAME>)
| --storage_pool_id <ID>) | (((--protection_domain_id <ID> | --protection_domain_name
<NAME>) --acceleration_pool_name <NAME>) | --acceleration_pool_id <ID>))
```

Parameters

--device_id *<ID>*

Device ID

--sds_id *<ID>*

SDS ID

--sds_name *<NAME>*

SDS name

--sds_ip *<IP>*

SDS IP address

--sds_port *<PORT>*

Port associated with the SDS

--device_id *<ID>*

Device ID

--device_name *<NAME>*

SDS storage device name or file name

--device_path *<PATH>*

File path to storage device

--protection_domain_id *<ID>*

Protection Domain ID

--protection_domain_name *<NAME>*

Protection Domain name

--storage_pool_name *<NAME>*

Storage Pool name

--storage_pool_id *<ID>*

Storage Pool ID

--acceleration_pool_name *<NAME>*

Storage Pool name

--acceleration_pool_id *<ID>*

Storage Pool ID

NOTE: If you use the `storage_pool_id` parameter in the command, you do not need to also specify the Protection Domain ID or name, and the Storage Pool name.

Example

```
scli --mdm_ip 192.168.1.200 --query_device_test
--sds_ip 192.168.1.6 --device_name /dev/sdb
```

query_sds_device_info

Retrieve detailed information for the specified SDS device, or for all devices in the specified SDS.

Syntax

```
scli --query_sds_device_info (--device_id <ID> | ((--sds_id <ID> | --sds_name <NAME> |
--sds_ip <IP> [--sds_port <PORT>]) (--device_name <NAME> | --device_path <PATH>)))
  [--all_devices]
```

Parameters

--device_id *<ID>*

ID assigned to the storage or RFcache device

--sds_id *<ID>*

SDS ID

--sds_name *<NAME>*

SDS name

--sds_ip *<IP>*

SDS IP address

--sds_port *<PORT>*

Port associated with the SDS

--device_name *<NAME>*

Name assigned to the storage or RFcache device

--device_path *<PATH>*

Device path or file path of the storage or RFcache device

--all_devices

Return detailed information for all devices in the specified SDS.

Example

```
scli --query_sds_device_info --sds_id 06b108a400000004 --all_devices
```

Sample output

```
Storage Device Information (total 2 devices):
        Device ID: c8bea95800040000 Name: N/A Path: /dev/sdb
                ScaleIO Device Configuration:
                        Original Path: /dev/sdb
                        Storage Pool Name: sp1
                        Capacity Limit: 148.9 GB (152500 MB)
                        Device State: Normal
                        Data Layout: Medium Granularity
                        MG Persistent checksum state: PROTECTED
                Physical Device Information:
                        Device Type: UNKNOWN
                        Media Type: HDD
                        Auto Detected Media Type: SSD
                        External Acceleration Type: NONE
                        Queue Depth: 16
                        Preset Queue Depth: HDD
                        Vendor Name: N/A
                        Model Name: N/A
                        Serial Number: N/A
                        Slot Number: N/A
                        Firmware Version: N/A
                        Cache Look-ahead: not Active
                        Write Cache: not Active
                        ATA Security: not Active
                        Logical Sector Size: 0 B
                        Physical Sector Size: 0 B
                        Capacity: 0 GB
                        LED Setting: OFF
                Background Device Scanner Information:
                        Scanned: 1873793 MB
                        Unresolved Read Errors: 0/0
                        Unresolved Compare Errors: 0/0
                SMART Information:
                        Aggregated State: NEVER_FAILED
                        Temperature State: NEVER_FAILED
                                Current Value: 0 Worst Value: 0 Threshold: 0
                        Media Wearout Indicator State: NEVER_FAILED
                                Current Value: 0 Worst Value: 0 Threshold: 0
                RAID Controller Information:
                        Serial Number: N/A
                        RAID vDisk Status: N/A
                        RAID vDisk Type: N/A
                        RAID vDisk Cache: N/A

        Device ID: c8bea95900040001 Name: N/A Path: /dev/sdc
                ScaleIO Device Configuration:
                        Original Path: /dev/sdc
                        Storage Pool Name: sp1
                        Capacity Limit: 148.9 GB (152500 MB)
```

```
                        Device State: Normal
                        Data Layout: Medium Granularity
                        MG Persistent checksum state: PROTECTED
                Physical Device Information:
                        Device Type: UNKNOWN
                        Media Type: HDD
                        Auto Detected Media Type: SSD
                        External Acceleration Type: NONE
                        Queue Depth: 16
                        Preset Queue Depth: HDD
                        Vendor Name: N/A
                        Model Name: N/A
                        Serial Number: N/A
                        Slot Number: N/A
                        Firmware Version: N/A
                        Cache Look-ahead: not Active
                        Write Cache: not Active
                        ATA Security: not Active
                        Logical Sector Size: 0 B
                        Physical Sector Size: 0 B
                        Capacity: 0 GB
                        LED Setting: OFF
                Background Device Scanner Information:
                        Scanned: 1873886 MB
                        Unresolved Read Errors: 0/0
                        Unresolved Compare Errors: 0/0
                SMART Information:
                        Aggregated State: NEVER_FAILED
                        Temperature State: NEVER_FAILED
                                Current Value: 0 Worst Value: 0 Threshold: 0
                        Media Wearout Indicator State: NEVER_FAILED
                                Current Value: 0 Worst Value: 0 Threshold: 0
                RAID Controller Information:
                        Serial Number: N/A
                        RAID vDisk Status: N/A
                        RAID vDisk Type: N/A
                        RAID vDisk Cache: N/A


Acceleration Device Information (total 0 devices):

RAID Controllers Information (total 0 controllers):
```

remove_sds_device

Initiate removal of a storage device from an SDS. Note that this process is asynchronous, and will run in the background.

An SDS device can be removed at any time, and no downtime is required. The removal process is asynchronous and runs in the background.

During execution of this command, the associated data is replicated to different nodes. Therefore, the process may take a long time.

When removing an SDS device on a VMware server, use the vSphere client.

NOTE: If the capacity of this SDS is still used by volumes, and cannot be replaced, the command will fail.

Syntax

```
scli --remove_sds_device (--device_id <ID> | ((--sds_id <ID> | --sds_name <NAME> | --
sds_ip <IP> [--sds_port <PORT>]) (--device_name <NAME> | --device_path <PATH>)))
```

Parameters

--device_id *<ID>*

ID of the device to be removed

--sds_id *<ID>*

ID of the SDS from which the device should be removed

--sds_name *<NAME>*

Name of the SDS from which the device should be removed

--sds_ip *<IP>*

IP address of the SDS from which the device should be removed

--sds_port *<PORT>*

Port associated with the SDS

--device_name *<NAME>*

Name of the storage device or file to be removed from the SDS

--device_path *<PATH>*

Path of the storage device to be removed from the SDS

Example

```
scli --mdm_ip 192.168.1.200 --remove_sds_device
--sds_ip 192.168.1.6 --device_name /dev/sdb
```

Related operations

To abort the removal, run the command `scli --abort_remove_sds_device`.

To track progress of the removal operation, use either of the following queries:

- `scli --query_sds_device_info`
- `scli --query_sds`

rename_device

Assign a name to, or rename, an SDS device.

Syntax

```
scli --rename_device (--device_id <ID> | ((--sds_id <ID> | --sds_name <NAME> | --sds_ip
<IP> [--sds_port <PORT>]) (--device_name <NAME> | --device_path <PATH>))) --new_name
<NAME>
```

Parameters

None.

--device_id *<ID>*

ID of the SDS device

--sds_id *<ID>*

SDS ID

--sds_name *<NAME>*

SDS name

--sds_ip *<IP>*

SDS IP address

--sds_port *<PORT>*
Port associated with the SDS

--device_name *<NAME>*
Name of the SDS device

--device_path *<PATH>*
Path of the SDS device

--new_name *<NAME>*
New name to be assigned to the SDS device

Example

```
scli --rename_device --sds_ip 8.8.8.100 --device_name flash_18 --new_name flash_10018
```

set_sds_device_external_acceleration_type

Set the external acceleration type of an SDS device.

Syntax

```
scli --set_sds_device_external_acceleration_type (--device_id <ID> | ((--sds_id <ID>
| --sds_name <NAME> | --sds_ip <IP> [--sds_port <PORT>]) (--device_name <NAME> | --
device_path <PATH>))) --external_acceleration <TYPE>
```

Parameters

--device_id *<ID>*
Device ID

--sds_id *<ID>*
SDS ID

--sds_name *<NAME>*
SDS name

--sds_ip *<IP>*
SDS IP address

--sds_port *<PORT>*
Port associated with the SDS

--device_name *<NAME>*
Device name

--device_path *<PATH>*
SDS storage device path or file path

--external_acceleration *<TYPE>*
External acceleration for the device, where TYPE is one of the following:

none
The device is not accelerated by a non-PowerFlex read or write cache.

read
The device is accelerated by a non-PowerFlex read cache.

write
The device is accelerated by a non-PowerFlex write cache.

read_and_write

The device is accelerated by both non-PowerFlex read and write cache.

Example

```
scli --set_sds_device_external_acceleration_type --sds_name sds_23 --device_name
storage_2310 --external_acceleration read
```

set_sds_device_led

Turn the LED on a storage or cache device on or off.

Syntax

```
scli --set_sds_device_led (--device_id <ID> | ((--sds_id <ID> | --sds_name <NAME> |
--sds_ip <IP> [--sds_port <PORT>]) (--device_name <NAME> | --device_path <PATH>))) (--
led_on | --led_off)
```

Parameters

--device_id *<ID>*
: ID assigned to the storage or RFcache device

--sds_id *<ID>*
: SDS ID

--sds_name *<NAME>*
: SDS name

--sds_ip *<IP>*
: SDS IP address

--sds_port *<PORT>*
: Port associated with the SDS

--device_id *<ID>*
: Device ID. Can also be an RFcache device ID

--device_name *<NAME>*
: Name assigned to the storage or RFcache device

--device_path *<PATH>*
: Device path or file path of a storage or RFcache device

--led_on
: Turn the LED on.

--led_off
: Turn the LED off.

Example

```
scli --set_sds_device_led --sds_ip  192.168.1.5 --sds_port 7072 --device_name slot5 --
led_on
```

set_sds_device_media_type

Set the media type of an SDS device.

Syntax

```
scli --set_sds_device_media_type (--device_id <ID> | ((--sds_id <ID> | --sds_name <NAME>
| --sds_ip <IP> [--sds_port <PORT>]) (--device_name <NAME> | --device_path <PATH>))) --
media_type <TYPE>
```

Parameters

--device_id *<ID>*
Device ID

--sds_id *<ID>*
SDS ID

--sds_name *<NAME>*
SDS name

--sds_ip *<IP>*
SDS IP address

--sds_port *<PORT>*
Port associated with the SDS

--device_name *<NAME>*
Device name

--device_path *<PATH>*
SDS storage device path or file path

--media_type *<TYPE>*
Media type of the device where TYPE is one of the following:

HDD
The device is an HDD device.

SSD
The device is an SSD device.

Example

```
scli --set_sds_device_media_type --sds_name sds22 --device_name storage_210 --media_type
HDD
```

start_device_test

Initiate a device test for one or all SDS devices.

Syntax

```
cli --start_device_test ((--device_id <ID> | ((--sds_id <ID> | --sds_name <NAME> |
--sds_ip <IP> [--sds_port <PORT>]) (--device_name <NAME> | --device_path <PATH>)))
| (--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP> [--sds_port <PORT>]) | (((--
protection_domain_id <ID> | --protection_domain_name <NAME>) --storage_pool_name <NAME>)
```

```
|--storage_pool_id <ID>)) | (((--protection_domain_id <ID> | --protection_domain_name
<NAME>) --acceleration_pool_name <NAME>) | --acceleration_pool_id <ID>)) (Options)
(Control)
  [--i_am_sure]
```

Parameters

--device_id *<ID>*

Device ID

--sds_id *<ID>*

SDS ID

--sds_name *<NAME>*

SDS name

--sds_ip *<IP>*

IP address associated with the SDS

--sds_port *<PORT>*

Port associated with the SDS

--device_name *<NAME>*

SDS storage device name or file

--device_path *<PATH>*

SDS storage device path or file path

--protection_domain_id *<ID>*

Protection Domain ID

--protection_domain_name *<NAME>*

Protection Domain name

--storage_pool_name *<NAME>*

Storage Pool name

--storage_pool_id *<ID>*

Storage Pool ID

--acceleration_pool_name *<NAME>*

Acceleration Pool name

--acceleration_pool_id *<ID>*

Acceleration Pool ID

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

Options

Choose several:

--io_type_write

Use write I/O type (default is read). If the storage device is in use, only read testing can be performed.

--io_pattern_squential

Use sequential I/O pattern (default is random).

--io_size_kb *<SIZE>*

The read or write I/O size in KB. Range is 4 - 512 KB. Default: 8 KB.

Control

Choose several:

--test_time *<TIME>*

Maximum test run time in seconds. If not specified, the test will be limited by I/O size only.

--io_total_size_mb *<SIZE>*

The total size of the test reads or writes in MB. Range is 1 - 32768 MB, which specifies the total I/O size in MB. Default: 128 MB

<blank>

Use defaults

Example

```
scli --mdm_ip 192.168.1.200 --start_device_test
--sds_ip 192.168.1.6 --device_path /dev/sdb --io_type_write --io_size_kb 8
```

set_sds_device_path

Configure the path of the given storage device.

Syntax

```
scli --set_sds_device_path (--device_id <ID> | ((--sds_id <ID> | --sds_name <NAME> |
--sds_ip <IP> [--sds_port <PORT>]) (--device_name <NAME> | --device_path <PATH>))) --
new_device_path <PATH>
```

Parameters

--sds_id *<ID>*

SDS ID

--sds_name *<NAME>*

SDS name

--sds_ip *<IP>*

SDS IP address

--sds_port *<PORT>*

Port assigned to the SDS

--device_id *<ID>*

Device ID

--device_name *<NAME>*

Device name

--device_path *<PATH>*

SDS storage device path or file path. The file path is used only when the device is available at this path

--new_device_path *<PATH>*

The new path to configure

4

Fault Set Commands

This section contains commands for managing Fault Sets.

add_fault_set

Add an empty Fault Set to a Protection Domain.

Fault Sets are subsets of a given Protection Domain. They provide additional safeguards for protecting your data against hardware failure. When you define SDSs later on, you can add them to a Fault Set.

Fault Sets are enabled by default.

Syntax

```
scli --add_fault_set (--protection_domain_id <ID> | --protection_domain_name <NAME>)
  [--fault_set_name <NAME>]
```

Parameters

--protection_domain_id *<ID>*

ID of the Protection Domain to which the Fault Set will be added

--protection_domain_name *<NAME>*

Name of the Protection Domain to which the Fault Set will be added

--fault_set_name *<NAME>*

Name of the new Fault Set

Example

```
scli --add_fault_set --protection_domain_name pd1 --fault_set_name fs1-mg
```

Fault Set names

Assign each Fault Set a meaningful name. When a name has not been defined, the system may display default system-defined names that use the Fault Set's ID.

Each Fault Set name must conform to the following rules:

1. Contains fewer than 32 characters
2. Contains only alphanumeric and punctuation characters
3. Is unique within the object type

NOTE: PowerFlex objects are assigned a unique ID that can be used to identify the object in CLI commands. You can retrieve the ID via a query or through the object's property sheet in the PowerFlex GUI.

clear_fault_set

Remove all SDSs from the specified Fault Set.

Syntax

```
scli --clear_fault_set (--fault_set_id <ID> | ((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --fault_set_name <NAME>))
  [--i_am_sure]
```

Parameters

--fault_set_id *<ID>*
Fault Set ID

--protection_domain_id *<ID>*
Protection Domain ID

--protection_domain_name *<NAME>*
Protection Domain name

--fault_set_name *<NAME>*
Fault Set name

--i_am_sure
Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

Example

```
scli --clear_fault_set --protection_domain_name pd18 --fault_set_name sio-fs1
```

query_all_fault_sets

Retrieve information for all the Fault Sets in the specified Protection Domain.

Syntax

```
scli --query_all_fault_sets (--protection_domain_id <ID> | --protection_domain_name
<NAME>)
```

Parameters

--protection_domain_id *<ID>*
Protection Domain ID

--protection_domain_name *<NAME>*
Protection Domain name

Example

```
scli --query_all_fault_sets --protection_domain_name pd1
```

Sample output

```
Protection Domain has 1 Fault Sets:
Protection Domain ID: fcc9af5500000000 Fault Set: ID: 36c9d76800000000 Name: fs1_mg SDS
Count: 2
  SDS ID: 06b108a400000004 Name: sds_idd_1
  SDS ID: 06b108a300000003 Name: sds_idd_2
```

query_fault_set

Retrieve information about the specified Fault Set.

Syntax

```
scli --query_fault_set (--fault_set_id <ID> | ((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --fault_set_name <NAME>))
```

Parameters

--fault_set_id *<ID>*
: Fault Set ID

--protection_domain_id *<ID>*
: Protection Domain ID

--protection_domain_name *<NAME>*
: Protection Domain name

--fault_set_name *<NAME>*
: Name of the new Fault Set

Example

```
scli --query_fault_set --protection_domain_name pd18 --fault_set_name fs1_mg
```

Sample output

```
Protection Domain ID: fcc9af5500000000 Fault Set: ID: 36c9d76800000000 Name: fs1_mg SDS
Count: 2
  SDS ID: 06b108a400000004 Name: sds_idd_1
  SDS ID: 06b108a300000003 Name: sds_idd_2
```

remove_fault_set

Remove the specified Fault Set.

Ensure that the Fault Set is empty before executing this command. You can remove SDSs from the Fault Set using the `clear_fault_set` command.

Syntax

```
scli --remove_fault_set (--fault_set_id <ID> | ((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --fault_set_name <NAME>))
```

Parameters

--fault_set_id *<ID>*
Fault Set ID

--protection_domain_id *<ID>*
Protection Domain ID

--protection_domain_name *<NAME>*
Protection Domain name

--fault_set_name *<NAME>*
Name of the new Fault Set

Example

```
scli --remove_fault_set --protection_domain_name pd18 --fault_set_name sio-fs1
```

rename_fault_set

Rename the specified Fault Set.

Syntax

```
scli --rename_fault_set (--fault_set_id <ID> | ((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --fault_set_name <NAME>)) --new_name <NAME>
```

Parameters

--fault_set_id *<ID>*
Fault Set ID

--protection_domain_id *<ID>*
Protection Domain ID

--protection_domain_name *<NAME>*
Protection Domain name

--fault_set_name *<NAME>*
Fault Set name

--new_name *<NAME>*

New name to assign to the Fault Set

Example

```
scli --rename_fault_set --protection_domain_name pd18 --fault_set_name sio-fs-test --
new_name sio-fs1
```

5

General CLI Commands

This section contains general CLI commands, including commands relating to licenses, certificates, login banner, and syslog.

acknowledge_scanner_error_counters

Acknowledge read and/or compare errors so that alerts will no longer be shown for the current counter value. Acknowledgment can be performed at Storage Pool level or at device level, and for counter type.

Syntax

```
scli --acknowledge_scanner_error_counters (((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --storage_pool_name <NAME>) | --storage_pool_id
<ID>) | (--device_id <ID> | ((--sds_id <ID> | --sds_name <NAME> | --sds_ip
<IP> [--sds_port <PORT>]) (--device_name <NAME> | --device_path <PATH>))) ) [--
acknowledge_compare_errors] [--acknowledge_read_errors]
```

Parameters

--protection_domain_id *<ID>*
Protection Domain ID

--protection_domain_name *<NAME>*
Protection Domain name

--storage_pool_name *<NAME>*
Storage Pool name

--storage_pool_id *<ID>*
Storage Pool ID

--sds_id *<ID>*
SDS ID

--sds_name *<NAME>*
SDS name

--sds_ip *<IP>*
SDS IP address

--sds_port *<PORT>*
Port assigned to the SDS

--device_id *<ID>*
Device ID

--device_name *<NAME>*
Device name

--device_path *<PATH>*
SDS storage device path or file path

--acknowledge_read_errors
Acknowledge read error counter

--acknowledge_compare_errors

Acknowledge compare error counter

Example

```
scli --acknowledge_scanner_error_counters --sds_name sds25 --device_name abc123 --
acknowledge_compare_errors
```

add_trusted_ca

Upload the certificate from the specified file as a trusted certificate.

Syntax

```
scli --add_trusted_ca --certificate_file <FILE> --comment <COMMENT>
```

Parameters

--certificate_file *<FILE>*

The certificate file to upload as trusted

--comment *<COMMENT>*

Optional free text to describe the certificate

Example

```
scli --mdm_ip 192.168.1.200 --add_trusted_ca --certificate_file system2_ca.cer --comment
"Certificate for system 2"
```

extract_root_ca

Extract the MDM root certificate to the specified file. This is the first step in the process of exchanging certificates between replication peer systems.

Syntax

```
scli --extract_root_ca --certificate_file <FILE>
```

Parameters

--certificate_file *<FILE>*

The file to write the certificate to

Example

```
scli --mdm_ip 192.168.1.200 --extract_root_ca --certificate_file system1_ca.cer
```

list_trusted_ca

Display a list of the currently trusted certificates in the system.

Syntax

```
scli --list_trusted_ca
```

Parameters

None.

Example

```
scli --mdm_ip 192.168.1.200 --list_trusted_ca
```

query_license

Retrieve information about the current license and restrictions.

Syntax

```
scli --query_license
```

Parameters

None.

Example

```
scli --mdm_ip 192.168.1.200 --query_license
```

query_disconnections

Query a variety of connection statuses.

Syntax

```
scli --query_disconnections [--disconnection_type <TYPE>] [--print_all] [--print_proxy]
```

Parameters

--disconnection_type <TYPE>

Print disconnections of the given type only. The following options are available:

sdc_sds

SDC-SDS connections

sdc_sdr

SDC-SDR connections

sdr_sds

SDR-SDS connections

sdr_sdr

SDR-SDR connections

--print_all

Use this flag to print all disconnections

--print_proxy

When Proxy Access mode is active, print SDSs that can not serve as a proxy.

Example

```
scli --mdm_ip 192.168.1.200 --query_disconnections --disconnection_type sdr_sdr
```

query_remote_syslog

Retrieve the remote syslog module state and configuration.

Syntax

```
scli --query_remote_syslog
```

Parameters

None.

Example

```
scli --query_remote_syslog
```

query_oscillating_failure_counter_parameters

Retrieve oscillating failure counter parameters.

Syntax

```
scli --query_oscillating_failure_counter_parameters --failure_counter (<COUNTER FROM
GROUP A> (--protection_domain_id <ID> | --protection_domain_name <NAME>) | <COUNTER FROM
```

```
GROUP B> | <COUNTER FROM GROUP C> | --all_counters(((----protection_domain_id <ID> | --
protection_domain_name <NAME>) --storage_pool_name <NAME>) | --storage_pool_id <ID>))
```

Parameters

--failure_counter *<COUNTER>*

The oscillating failure counter to be queried. Counters are grouped as follows:

Counter Group A

MDM and SDS-related counters:

- mdm_sds_network_disconnections
- sds_sds_network_disconnections
- sds_decoupled
- sds_configuration_failures
- sds_receive_buffer_allocation_failures

Counter Group B

SDC-related counters:

- sdc_mdm_network_disconnections
- sdc_sds_network_disconnections
- sdc_long_operations
- sdc_memory_allocation_failures
- sdc_socket_allocation_failures

Counter Group C

SDS device-related counters:

- sds_device_long_successful_ios

--all_counters

Query all oscillating failure counters parameters of the selected group

--protection_domain_id *<ID>*

Protection Domain ID

--protection_domain_name *<NAME>*

Protection Domain name

--storage_pool_name *<NAME>*

Storage Pool name

--storage_pool_id *<ID>*

Storage Pool ID

Example

```
scli --query_oscillating_failure_counter_parameters --failure_counter
sds_device_long_successful_ios --protection_domain_name pd10 --storage_pool_name sp18
```

remove_trusted_ca

Remove the certificate with the given fingerprint.

Syntax

```
scli --remove_trusted_ca --fingerprint <FINGERPRINT>
```

Parameters

--fingerprint <FINGERPRINT>

The fingerprint of the certificate to be removed

Example

```
scli --mdm_ip 192.168.1.200 --remove_trusted_ca --fingerprint
B5:F0:43:1E:9B:7B:55:7F:05:98:B7:D9:7E:50:D4:9F:6D:1B:A7:5F
```

reset_oscillating_failure_counters

Reset the specified oscillating failure counters to zero. This command is useful when you have fixed a problem and want to ensure that an alert is no longer active in the system.

Syntax

```
scli --reset_oscillating_failure_counters (--failure_counter (<COUNTER FROM GROUP A> ((--
protection_domain_id <ID> | --protection_domain_name <NAME>) | --all_sds) | <COUNTER
FROM GROUP B> --all_sdc | <COUNTER FROM GROUP C> (((----protection_domain_id <ID> | --
protection_domain_name <NAME>) --storage_pool_name <NAME>) | --storage_pool_id <ID>) |
--all_devices) | --all_counters)
```

Parameters

--failure_counter *<COUNTER>*

The oscillating failure counter to be reset to zero. Counters are grouped as follows:

Counter Group A

MDM and SDS-related counters:

- mdm_sds_network_disconnections
- sds_sds_network_disconnections
- sds_decoupled
- sds_configuration_failures
- sds_receive_buffer_allocation_failures

Counter Group B

SDC-related counters:

- sdc_mdm_network_disconnections
- sdc_sds_network_disconnections
- sdc_long_operations
- sdc_memory_allocation_failures
- sdc_socket_allocation_failures

Counter Group C

SDS device-related counters:

- sds_device_long_successful_ios

--protection_domain_id *<ID>*

Protection Domain ID

--protection_domain_name *<NAME>*

Protection Domain name

--storage_pool_name *<NAME>*

Storage Pool name

--storage_pool_id *<ID>*
Storage Pool ID

--all_sds
Reset the counter parameters on all SDSs.

--all_sdc
Reset the counter parameters on all SDCs.

--all_devices
Reset the counter parameters on all SDS devices.

--all_counters
Reset all counters to zero.

Example

```
scli --reset_oscillating_failure_counters --failure_counter
mdm_sds_network_disconnections --all_sds
```

set_cli_login_banner_acceptance_mode

Set the login banner acceptance mode.

Syntax

```
scli --set_cli_login_banner_acceptance_mode (--password_only | --accept_banner)
```

Parameters

--password_only
Allow CLI users to accept the login banner by entering a valid password.

--accept_banner
Require CLI users to accept the login banner by pressing 'y'.

Example

```
scli --set_cli_login_banner_acceptance_mode --password_only
```

set_cli_login_banner_preemptive_acceptance

Enable or prevent CLI users from preemptively accepting the login banner.

Syntax

```
scli --set_cli_login_banner_preemptive_acceptance (--enable | --disable)
```

Parameters

--enable

Allow CLI users to preemptively accept the login banner

--disable

Prevent CLI users from preemptively accepting the login banner

Example

```
scli --set_cli_login_banner_preemptive_acceptance --enable
```

set_license

Set the PowerFlex license.

Setting the license is a precondition for all other configuration commands. To set the system license, you must first obtain a license from Dell.

Syntax

```
scli --set_license --license_file <LICENSE_FILE>
```

Parameters

--license_file *<LICENSE_FILE>*

Path to the file containing the license key value supplied by Dell.

Example

```
scli --mdm_ip 192.168.1.200 --set_license
--license_file /tmp/0239SH4SS89023T6.lic
```

set_login_banner

Set the login banner that is displayed at every login to the system, or remove an existing login banner.

The maximum file size for the banner is 16 KB. Only ASCII text files are supported.

Syntax

```
--set_login_banner (--filename <FILE> | --remove_banner)
```

Parameters

--filename *<FILE>*

File containing the login banner to set

--remove_banner

Remove the login banner.

Example

```
scli --set_login_banner --filename login_banner.txt
```

set_oscillating_failure_counter_parameters

Modify oscillating failure counter parameters. You can use this command to configure the time interval associated with each window type and the number of failures allowed before reporting commences for each window type, per counter.

Syntax

```
scli --set_oscillating_failure_counter_parameters --failure_counter (<COUNTER FROM GROUP
A> ((--protection_domain_id <ID> | --protection_domain_name <NAME>) | --all_sds) |
<COUNTER FROM GROUP B> --all_sdc | <COUNTER FROM GROUP C> (((----protection_domain_id
<ID> | --protection_domain_name <NAME>) --storage_pool_name <NAME>) | --storage_pool_id
<ID>) | --all_devices) --window_type {SHORT | MEDIUM | LONG} --window_interval
<INTERVAL> --failures_threshold<ID> <NUM OF FAILURES IN INTERVAL>
```

Parameters

--failure_counter *<COUNTER>*

The oscillating failure counter to be reset to zero. Counters are grouped as follows:

Counter Group A

MDM and SDS-related counters:

- `mdm_sds_network_disconnections`
- `sds_sds_network_disconnections`
- `sds_decoupled`
- `sds_configuration_failures`
- `sds_receive_buffer_allocation_failures`

Counter Group B

SDC-related counters:

- `sdc_mdm_network_disconnections`
- `sdc_sds_network_disconnections`
- `sdc_long_operations`
- `sdc_memory_allocation_failures`
- `sdc_socket_allocation_failures`

Counter Group C

SDS device-related counters:

- `sds_device_long_successful_ios`

--protection_domain_id *<ID>*

Protection Domain ID

--protection_domain_name *<NAME>*

Protection Domain name

--storage_pool_name *<NAME>*

Storage Pool name

--storage_pool_id *<ID>*

Storage Pool ID

--all_sds

Apply the counter parameters to all SDSs.

--all_sdc

Apply the counter parameters to all SDCs.

--all_devices

Apply the counter parameters to all SDS devices.

--window_type {SHORT | MEDIUM | LONG}

Window type to be modified: short, medium, or long

--window_interval *<INTERVAL>*

New window time interval in seconds. The valid interval range is 1-86400. If the interval is set to 0, the window type is disabled.

--failures_threshold*<ID> <NUM OF FAILURES IN INTERVAL>*

Maximum number of failures per window type to be ignored before error reporting begins

Example

```
scli --set_oscillating_failure_counter_parameters --failure_counter
mdm_sds_network_disconnections --window_interval 120 --failures_threshold 10 --
window_type short --all_sds
```

set_remote_read_only_limit_state

Configure the PowerFlex system to restrict access from remote clients to read-only operations on the MDM. When this restriction is enabled, remote client users cannot change any configurations on the MDM. However, users on a local host may still make configuration changes while the restriction is enforced.

NOTE: A local user is one that successfully communicates with the MDM using the IP address 127.0.0.1 (the default IP address used by SCLI). If you try to issue commands from the local machine using other local IP addresses, you will not be able configure the system.

Syntax

```
scli --set_remote_read_only_limit_state --remote_read_only_limit_state {enabled |
disabled}
```

Parameters

--remote_read_only_limit_state {enabled | disabled}

Enable or disable restricted state. Default: disabled.

Example

```
scli --set_remote_read_only_limit_state --remote_read_only_limit_state enabled
```

set_syslog_facility

Set the facility field of the syslog events. Legal values are 0 to 23.

Syntax

```
scli --set_syslog_facility --remote_syslog_server_ip <IP> --syslog_facility <FACILITY>
```

Parameters

--remote_syslog_server_ip *<IP>*

Comma-separated list of IP addresses or hostnames of syslog servers. Omit the space after each comma.

--syslog_facility *<FACILITY>*

Control the facility field of the event. Default is 16.

Example

```
scli --set_syslog_facility --remote_syslog_server_ip 192.168.1.201 --syslog_facility 16
```

start_remote_syslog

Start posting events to a remote syslog server.

Syntax

```
scli --start_remote_syslog --remote_syslog_server_ip <IP>
  [--remote_syslog_server_port <PORT>]
  [--syslog_facility <FACILITY>]
  [--attach_event_code]
```

Parameters

--remote_syslog_server_ip *<IP>*

Comma-separated list of IP addresses or hostnames of syslog servers. Omit the space after each comma.

--remote_syslog_server_port *<PORT>*

Syslog server port. Default is 1468.

--syslog_facility *<FACILITY>*

Control the facility field of the event. Default is 16.

--attach_event_code

Add the posted event code to the event message. This parameter is disabled by default.

Example

```
scli --start_remote_syslog --remote_syslog_server_ip 192.168.1.201 --syslog_facility 16
```

stop_remote_syslog

Stop posting events to a remote syslog server.

Syntax

```
scli --stop_remote_syslog (--remote_syslog_server_ip <IP> | --all_remote_syslog_servers)
```

Parameters

--remote_syslog_server_ip *<IP>*

Comma-separated list of IP addresses or hostnames of syslog servers. Omit the space after each comma.

--all_remote_syslog_servers

Stop posting events to all remote syslog servers.

Example

```
scli --stop_remote_syslog --remote_syslog_server_ip 192.168.1.201
```

6

LDAP Commands

This section contains commands relating to Lightweight Directory Access Protocol (LDAP) and LDAP services.

For detailed information on LDAP and its implementation in PowerFlex, see the *PowerFlex User Roles and LDAP Usage Technical Notes*.

add_ldap_service

Add the service to the MDM to be used for authentication. In return, the user receives the ID of the LDAP service.

NOTE: LDAP should be configured on all the MDMs in the system in order to support switch ownership scenarios.

NOTE: PowerFlex systems support authentication by up to eight LDAP servers. When multiple LDAP servers are used, add each one separately using this command.

Syntax

```
scli --add_ldap_service --ldap_service_uri <URI> --ldap_base_dn <LDAP_DN> [--ldap_fqdn
<FQDN>]
  [--ldap_service_name <LDAP_NAME>]
  [--username_dn_format <FORMAT>]
  [--search_filter_format <FORMAT>]
  [--authorizer_dn <USER_DN>]
  [--authorizer_password <PASSWORD>
```

NOTE: `--ldap_fqdn`, `--username_dn_format`, and `--search_filter_format` parameters are used when the LDAP server is running on Linux.

Parameters

`--ldap_service_uri <URI>`

URI of the LDAP service:

<LDAP_SCHEMA>://<LDAP_HOSTNAME>: [*<PORT_NUMBER>*]

Where:

<LDAP_SCHEMA>

Defines the connection protocol:

- LDAPS: Secure LDAP connection (recommended)
- LDAP: non-secure LDAP connection

<LDAP_HOSTNAME>

LDAP hostname

<PORT_NUMBER>

LDAP service port (optional, default: 389 for LDAP and 636 for LDAPS)

This parameter must start with ldap:// or ldaps:// followed by the host name.

Example: ldaps://my.ldaphost.com:636

When using the global catalog, one should specify the port number of the global catalog. The standard ports for global catalog are 3268 (non-secure) and 3269 (secure).

NOTE: No extra validation is performed at this stage.

--ldap_base_dn *<LDAP_DN>*

Base Distinguished Name (DN) of users in the domain. Must be a valid DN containing the DC substring. For example, if a user corporate login is johnd@ecme.corp.com, the DC string would be DC=ecme, DC=corp, DC=com.

NOTE: Configuring multiple LDAP servers with the same base DN is not supported.

NOTE: On Active Directory Windows servers, use the dsquery tool to find LDAP Base DN information. To see available options, in the command line type `dsquery /?`

On Linux servers, from the command line, use `ldapsearch`. (ldapsearch may need to be installed.)

--ldap_fqdn *<FQDN>*

The FQDN is used to identify the LDAP service. By default it is derived from the base-DN, but there are cases that it must be defined explicitly.

If you want to support multiple FQDNs for one service, this parameter can contain a list of sub-strings in square brackets. For example: `[us.,eu.,as.,]dell.ldap` will include users with the following suffixes: us.dell.ldap, eu.dell.ldap, as.dell.ldap, dell.ldap.

--ldap_service_name *<LDAP_NAME>*

LDAP service name

--username_dn_format *<FORMAT>*

The username format in DN format (must contain [USER] as a place holder for the username)

--search_filter_format *<FORMAT>*

A search filter for the LDAP query. This is required only if it is different from the default. It must contain [USER] and [GROUP] as place holders for username and group-DN. Example: `(&(objectClass=user)(sAMAccountName=[USER])(memberOf:1.2.840.113556.1.4.1941:=[GROUP]))`

--authorizer_dn *<USER_DN>*

The authorizer username for groups search

--authorizer_password *<PASSWORD>*

Password of the authorizer user

Examples

```
scli --add_ldap_service --ldap_service_uri "ldaps://ldaps.ecme.com:636" --ldap_base_dn
"OU=SIO_OU_1,DC=ldaps,DC=local"
```

where:

- `ldaps://ldaps.ecme.com` is the host name of the authentication server.
- 636 is the port number.
- `OU=SIO_OU_1` is a specific organizational unit group defined in the Active Directory.
- `DC=ldaps, DC=local` are the domain component parts of the Base DN.

```
scli --add_ldap_service --ldap_service_uri ldaps://ldaps.ecme.com:3269 --ldap_base_dn
ou=sio_ou_1,dc=ldaps,dc=ecme,dc=com --ldap_fqdn [na.,eu.,as.,]ldaps.ecme.com
```

where:

- The access will be to the global catalog through port 3269.
- `ou=sio_ou_1` is a specific organizational unit group defined in the Active Directory.
- `dc=ldaps,dc=ecme,dc=com` are the domain component parts of the Base DN.
- The following FQDNs servers will be mapped to the same URI ldaps://ldaps.ecme.com(): ldaps://na.ldaps.ecme.com, ldaps://eu.ldaps.ecme.com, ldaps://as.ldaps.ecme.com, and ldaps://ldaps.ecme.com.

assign_ldap_groups_to_roles

Map LDAP groups to PowerFlex system roles.

The LDAP service must be configured before using this command. Once you have mapped the roles, you can assign users in the Active Directory to the relevant LDAP groups.

NOTE: To enable LDAP users to use the PowerFlex GUI or vSphere Plug-in, you must assign all LDAP groups the Monitor role.

Syntax

```
scli --assign_ldap_groups_to_roles (--ldap_service_id <LDAP_SERVICE_ID> | --
ldap_service_name <LDAP_SERVICE_NAME>)
  [--administrator_role_dn <LDAP_GROUP_DN>]
  [--security_role_dn <LDAP_GROUP_DN>]
  [--backend_config_role_dn <LDAP_GROUP_DN>]
  [--frontend_config_role_dn <LDAP_GROUP_DN>]
  [--monitor_role_dn <LDAP_GROUP_DN>]
  [--allow_overwrite ]
```

Parameters

--ldap_service_id *<LDAP_SERVICE_ID>*

ID of the LDAP service

--ldap_service_name *<LDAP_SERVICE_NAME>*

Name of the LDAP service

--administrator_role_dn *<LDAP_GROUP_DN>*

LDAP group that has users with administration privileges

--security_role_dn *<LDAP_GROUP_DN>*

LDAP group that has users with security privileges

--backend_config_role_dn *<LDAP_GROUP_DN>*

LDAP group that has users with backend configuration privileges

--frontend_config_role_dn *<LDAP_GROUP_DN>*

LDAP group that has users with frontend configuration privileges

--monitor_role_dn *<LDAP_GROUP_DN>*

LDAP group that has users with monitoring privileges

--allow_overwrite

Overwrites the role's LDAP group

Example

```
scli --assign_ldap_groups_to_roles --ldap_service_id 0xAABBCCDDEEFF0011
--administrator_role_dn "CN=SIO_GRP_1,OU=SIO_OU_1,DC=ldaps,DC=ecme,DC=com" --
monitor_role_dn "CN=SIO_GRP_2,OU=SIO_OU_1,DC=ldaps,DC=ecme,DC=com"
```

set_user_authentication_method

Set the user authentication method for the system.

CAUTION: Use this command with caution. The operation is complex to roll back.

Syntax

```
scli --set_user_authentication_method (--ldap_authentication | --native_authentication |
--native_and_ldap_authentication | allow_ldap_without_admin)
  [--i_am_sure]
```

Parameters

--ldap_authentication

LDAP-based authentication method where users are managed on an LDAP-compliant server. Configure LDAP service and LDAP user before switching to this authentication method.

--native_authentication

Native authentication method where users are managed locally in the system

--native_and_ldap_authentication

A hybrid authentication method. Both LDAP and local users may log in to the system after it is set.

--allow_ldap_without_admin

Allow setting LDAP athentication method even if there is no LDAP service with administrator role.

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

Example

```
scli --set_user_authentication_method --native_and_ldap_authentication --i_am_sure
```

reload_ldap_certificate

Reload the LDAP certificate

Syntax

```
scli reload_ldap_certificate
```

Parameters

None.

Example

```
scli --reload_ldap_certificate
```

remove_ldap_group_from_role_assignment

Remove LDAP groups from LDAP user role assignments.

Syntax

```
scli --remove_ldap_group_from_role_assignment (--ldap_service_id <LDAP_SERVICE_ID> | --
ldap_service_name  <LDAP_SERVICE_NAME>)
  [--administrator_role]
  [--security_role]
  [--backend_config_role]
  [--frontend_config_role]
  [--monitor_role]
```

Parameters

--ldap_service_id *<LDAP_SERVICE_ID>*

ID of the LDAP service

--ldap_service_name *<LDAP_SERVICE_NAME>*

Name of the LDAP service

Options:

(choose at least one of the following)

--administrator_role

LDAP group containing users with administration privileges

--security_role

LDAP group containing users with security privileges

--backend_config_role

LDAP group that containing with backend configuration privileges

--frontend_config_role

LDAP group that containing with frontend configuration privileges

--monitor_role

LDAP group that containing with monitoring privileges

Example

```
scli --remove_ldap_group_from_role_assignment --ldap_service_id 0xAABBCCDDEEFF0011 --
administrator_role "CN=SIO_GRP_1,OU=SIO_OU_1,DC=ldaps,DC=ecme,DC=com"
```

remove_ldap_service

Remove an LDAP service from the system.

Syntax

```
scli --remove_ldap_service (--ldap_service_id <LDAP_SERVICE_ID> | --ldap_service_name
<LDAP_SERVICE_NAME> | --remove_all)
```

Parameters

--ldap_service_id *<LDAP_SERVICE_ID>*

ID of the LDAP service

--ldap_service_name *<LDAP_SERVICE_NAME>*

Name of the LDAP service

--remove_all

Remove all LDAP services.

Example

```
scli --remove_ldap_service --ldap_service_name ldap1
```

rename_ldap_service

Assign a name to, or rename, an LDAP service.

Syntax

```
scli --rename_ldap_service (--ldap_service_id <LDAP_SERVICE_ID> | --ldap_service_name
<LDAP_SERVICE_NAME>) --new_name <NAME>
```

Parameters

--ldap_service_id *<LDAP_SERVICE_ID>*

ID of LDAP service

--ldap_service_name *<LDAP_SERVICE_NAME>*

Name of LDAP service

--new_name *<NAME>*

New name to be assigned to the LDAP service

Example

```
scli --rename_ldap_service --ldap_service_name prevLSName --new_name newLSName
```

7

Protection Domain Commands

This section contains commands for managing Protection Domains.

activate_protection_domain

Activate a Protection Domain.

Protection Domains are activated by default. Use this command to activate a Protection Domain that was inactivated.

Syntax

```
scli --activate_protection_domain (--protection_domain_id <ID> | --
protection_domain_name <NAME>)
  [--force_activate]
  [--i_am_sure]
```

Parameters

--protection_domain_id *<ID>*

Protection Domain ID

--protection_domain_name *<NAME>*

Protection Domain name

--force_activate

Activates the Protection Domain, ignoring current system errors

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

Example

```
scli --activate_protection_domain --protection_domain_name pd1
```

add_protection_domain

Add a Protection Domain.

Syntax

```
scli --add_protection_domain
  [--protection_domain_name <NAME>]
```

Parameters

--protection_domain_name *<NAME>*

Unique name to assign the Protection Domain

Example

```
Example
scli --mdm_ip 192.168.1.200 --add_protection_domain
--protection_domain_name rack_1.1
```

Protection Domain names

Assign each Protection Domain a meaningful name associated with its operational role. When a name has not been defined, the system may display default system-defined names that use the volume's ID.

Each Protection Domain name should conform to the following rules:

1. Contains fewer than 32 characters
2. Contains only alphanumeric and punctuation characters
3. Is unique within the object type

NOTE: PowerFlex objects are assigned a unique ID that can be used to identify the object in CLI commands. You can retrieve the ID via a query or through the object's property sheet in the PowerFlex GUI.

disable_fgl_metadata_cache

Disable fine granularity metadata caching for the specified Protection Domain.

Syntax

```
scli --disable_fgl_metadata_cache (--protection_domain_id <ID> | --
protection_domain_name <NAME>)
```

Parameters

--protection_domain_id *<ID>*

Protection Domain ID

--protection_domain_name *<NAME>*

Protection Domain name

Example

```
scli --mdm_ip 192.168.1.200 --disable_fgl_metadata_cache --protection_domain_name PD1
```

enable_fgl_metadata_cache

Enable fine granularity metadata caching for the specified Protection Domain. Fine granularity metadata caching helps improve the performance of the system when using the fine granularity layout.

Syntax

```
scli --enable_fgl_metadata_cache (--protection_domain_id <ID> | --protection_domain_name
<NAME>)
```

Parameters

--protection_domain_id *<ID>*
Protection Domain ID

--protection_domain_name *<NAME>*
Protection Domain name

Example

```
scli --mdm_ip 192.168.1.200 --enable_fgl_metadata_cache --protection_domain_name PD1
```

inactivate_protection_domain

Inactivate the specified Protection Domain. Using this command is a much more effective way to shut down nodes and is preferable to shutting them down manually.

NOTE: When you inactivate a Protection Domain, the data remains on the SDSs. It is therefore preferable to remove a Protection Domain if you no longer need it.

When this feature is in effect, the system may perform the following activities in the background:

- Determine whether there are any current rebuild/rebalance activities taking place. If so, the shutdown will be delayed (unless it is forced) until they are finished.
- Block future rebuild/rebalance activities.
- Pause application I/O and disable access to volumes.
- Move the DRL mode of all SDSs to harden in preparation for rebooting the server.
- Reload all SDSs before re-enabling data access.

Syntax

```
scli --inactivate_protection_domain (--protection_domain_id <ID> | --
protection_domain_name <NAME>)
  [--force_inactivate]
  [--i_am_sure]
```

Parameters

--protection_domain_id *<ID>*
Protection Domain ID

--protection_domain_name *<NAME>*
Protection Domain name

--force_inactivate
Inactivate the Protection Domain, ignoring current system errors.

--i_am_sure
Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

Example

```
scli --inactivate_protection_domain --protection_domain_name pd1
```

query_port_flapping_status

Query the system's port flapping status.

Syntax

```
scli --query_port_flapping_status [--weak] (--all | --protection_domain_id <ID> | --
protection_domain_name <NAME> | --sds_id <ID> | --sds_name <NAME> | --sdc_id <ID> | --
sdc_name <NAME> | --apply_to_mdm | --peer_system_name <NAME> | --peer_system_object_id
<ID>)
```

Parameters

--weak
Include reported paths with weak ERT violations

--all
Query all Protection Domains in the system

--protection_domain_id *<ID>*
Protection Domain ID

--protection_domain_name *<NAME>*
Protection Domain name

--sds_id *<ID>*
SDS ID

--sds_name *<NAME>*
SDS name

--sdc_id *<ID>*
SDC ID

--sdc_name *<NAME>*
SDC name

--apply_to_mdm
Display MDM only

--peer_system_name *<NAME>*
Replication peer system name

--peer_system_id *<ID>*
Replication peer system ID

Example

```
scli --mdm_ip 192.168.1.200 --query_port_flapping_status --all
```

Sample output

```
0 Protection Domains with disqualified portals

All components support port flapping feature
```

query_port_flapping_unsupported_components

Query components that don't support the port flapping feature, and/or have it disabled.

Syntax

```
scli --query_port_flapping_unsupported_components [(--full | --protection_domain_id <ID>
| --protection_domain_name <NAME>) [--raw]] [(--only_disabled | --only_unsupported)]
```

Parameters

--full

Display a full unsummarized report

--protection_domain_id *<ID>*

Protection Domain ID

--protection_domain_name *<NAME>*

Protection Domain name

--raw

Outputs results in raw format with one component displayed per line

--only_disabled

Query only the the components that have the port flapping feature disabled

--only_unsupported

Query only the the components that do not support the port flapping feature. Omit components that have the feature disabled.

Example

```
scli --mdm_ip 192.168.1.200 --query_port_flapping_unsupported_components --full
```

query_protection_domain

Retrieve aggregated information about all objects in a single Protection Domain.

Syntax

```
scli --query_protection_domain (--protection_domain_id <ID> | --protection_domain_name
<NAME>
```

Parameters

--protection_domain_id *<ID>*

Protection Domain ID

--protection_domain_name *<NAME>*

Protection Domain name

Example

```
scli --mdm_ip 192.168.1.200 --query_protection_domain --protection_domain_name pd2
```

Sample output

```
Protection Domain pd2 (Id: fcc9af5600000001) has 1 storage pools, 1 acceleration pools,
0 Fault Sets, 1 SDS nodes and 0 volumes
        Free net capacity: 22.3 GB (22875 MB), Volume allocation limit: 0 Bytes
        Operational state is Active
        RFcache enabled, Mode: Write miss, Page Size 64 KB, Max IO size 128 KB,
Acceleration Pool ID: 6794008e00000000
        I/O flow control is enabled
        Fine granularity default number of concurrent writes: 1000
        FGL metadata cache is disabled
        FGL default metadata cache size: 0 MB
        Replication capacity utilization is in Normal level

Storage Pool sp2 (Id: 3df1cd9700000001) has 0 volumes and 22.3 GB (22875 MB) free net
capacity. 0 Bytes volume allocation limit.
        The number of parallel rebuild/rebalance jobs: 2
        Rebuild is enabled and using Limit-Concurrent-IO policy with the following
parameters:
          Number of concurrent IOs per device: 1
        Rebalance is enabled and using Favor-Application-IO policy with the following
parameters:
          Number of concurrent IOs per device: 1, Bandwidth limit per device: 10240 KB
per second
        VTree Migration is enabled and using Favor-Application-IO policy with the
following parameters:
          Number of concurrent IOs per device: 1, Bandwidth limit per device: 10240 KB
per second
        Protected Maintenance Mode is enabled and using Limit-Concurrent-IO policy with
the following parameters:
          Number of concurrent IOs per device: 1
        Fragmentation: Enabled
        Background device scanner: Enabled, Read Error Action: report and fix, Compare
Error Action: report and fix, Bandwidth Limit 3072 KBps per device
        Zero padding: Enabled
        Spare policy: 70% out of total
        Inflight Checksum: Disabled
        Doesn't use RAM Read Cache
        Uses Read Flash Cache
        Capacity alert thresholds: High: 80, Critical: 90
        Media Type: SSD, External Acceleration Type: NONE, Inflight Requests Factor:
115, Inflight Bandwidth Factor: 115
        Data layout: Medium granularity
        Replication Capacity Max Allowed Ratio: 10%
        Replication Capacity Max Allowed: 3.0 GB (3072 MB)
        Replication Allocated Capacity: 0 Bytes
        Replication Used Capacity: 0 Bytes
        Number of devices with inaccessible capacity: 0
        Medium granularity persistent checksum:
                Persistent checksum is enabled
                State is PROTECTED
                Checksum builder limit is: 3072 KBps
                Validation on read is disabled
```

```
Acceleration Pool RFcache_Pool (Id: 6794008e00000000)
        Media Type: SSD
        Used for RFcache: yes

SDR-SDS connectivity status: All connected

SDS Summary:
        Total 1 SDS Nodes
        1 SDS nodes have membership state 'Joined'
        1 SDS nodes have connection state 'Connected'

Physical data:
        148.9 GB (152500 MB) total capacity
                0 Bytes in-use capacity
                        0 Bytes user data
                        0 Bytes metadata
                        0 Bytes fine granularity spares
                44.7 GB (45750 MB) unused capacity
                104.2 GB (106750 MB) spare capacity
                0 Bytes decreased capacity
        Capacity Health:
                0 Bytes protected capacity
                0 Bytes failed capacity
                0 Bytes degraded-failed capacity
                0 Bytes degraded-healthy capacity
                0 Bytes unreachable-unused capacity
                0 Bytes semi-protected capacity
                0 Bytes in-maintenance capacity
                0 Bytes temporary capacity

User data:
        0 Bytes total user data
                0 Bytes medium granularity user data
                0 Bytes fine granularity user data
                        0 Bytes compressed data (Compression ratio: Not applicable)
                        0 Bytes uncompressed data
        0 Bytes trimmed capacity
        0 Bytes user data without trimmed capacity
        Total compression ratio: Not applicable
        Thin and snapshot saving: Not applicable
        Overall saving: Not applicable
        Data Movement:
                0 Bytes rebalance capacity
                0 Bytes forward-rebuild capacity
                0 Bytes backward-rebuild capacity
                0 Bytes enter protected maintenance mode capacity
                0 Bytes exit protected maintenance mode capacity
                0 Bytes at-rest capacity
                0 Bytes total moving capacity
                        0 Bytes active moving capacity
                                0 Bytes active rebalance capacity
                                0 Bytes active forward-rebuild capacity
                                0 Bytes active backward-rebuild capacity
                                0 Bytes active enter protected maintenance mode capacity
                                0 Bytes active exit protected maintenance mode capacity
                        0 Bytes pending moving capacity
                                0 Bytes pending rebalance capacity
                                0 Bytes pending forward-rebuild capacity
                                0 Bytes pending backward-rebuild capacity
                                0 Bytes pending enter protected maintenance mode capacity
                                0 Bytes pending exit protected maintenance mode capacity

Bandwidth:
        Primary-reads                           0 IOPS 0 Bytes per second
        Primary-writes                          0 IOPS 0 Bytes per second
        Secondary-reads                         0 IOPS 0 Bytes per second
        Secondary-writes                        0 IOPS 0 Bytes per second
        Backward-rebuild-reads                  0 IOPS 0 Bytes per second
        Backward-rebuild-writes                 0 IOPS 0 Bytes per second
        Forward-rebuild-reads                   0 IOPS 0 Bytes per second
        Forward-rebuild-writes                  0 IOPS 0 Bytes per second
        Rebalance-reads                         0 IOPS 0 Bytes per second
        Rebalance-writes                        0 IOPS 0 Bytes per second
```

```
        Volume-migration-reads                    0 IOPS 0 Bytes per second
        Volume-migration-writes                   0 IOPS 0 Bytes per second
        Enter-protected-maintenance-mode-reads    0 IOPS 0 Bytes per second
        Enter-protected-maintenance-mode-writes   0 IOPS 0 Bytes per second
        Exit-protected-maintenance-mode-reads     0 IOPS 0 Bytes per second
        Exit-protected-maintenance-mode-writes    0 IOPS 0 Bytes per second

Volumes summary:
        0 Bytes all volumes size
        0 Bytes base volumes size
        0 Volumes.
```

remove_protection_domain

Remove a Protection Domain from the PowerFlex system. You can only remove a Protection Domain if no SDS and no Storage Pool is associated with it.

Syntax

```
scli --remove_protection_domain (--protection_domain_id <ID> | --protection_domain_name
<NAME>)
```

Parameters

--protection_domain_id *<ID>*
Protection Domain ID

--protection_domain_name *<NAME>*
Protection Domain name

Example

```
scli --mdm_ip 192.168.1.200 --remove_protection_domain
--protection_domain_name my_protection_domain
```

rename_protection_domain

Assign a name to, or rename, a Protection Domain.

Syntax

```
scli --rename_protection_domain (--protection_domain_id <ID> | --protection_domain_name
<NAME>) --new_name <NAME>
```

Parameters

--protection_domain_id *<ID>*
Protection Domain ID

--protection_domain_name *<NAME>*
Protection Domain name

--new_name *<NAME>*

Example

```
scli --mdm_ip 192.168.1.200 --rename_protection_domain --protection_domain_name pd1 --
new_name protection_domain_1.
```

Protection Domain names

Assign the Protection Domain a meaningful name associated with its operational role. The Protection Domain name should conform to the following rules:

1. Contains fewer than 32 characters
2. Contains only alphanumeric and punctuation characters
3. Is unique within the object type

set_default_fgl_metadata_cache_size

Configure the default fine granularity metadata cache size for a Protection Domain. New SDSs in the Protection Domain will inherit this value when created.

Syntax

```
scli --set_default_fgl_metadata_cache_size (--protection_domain_id <ID> | --
protection_domain_name <NAME>) --metadata_cache_size_mb <SIZE>
```

Parameters

--protection_domain_id *<ID>*
Protection Domain ID

--protection_domain_name *<NAME>*
Protection Domain name

--metadata_cache_size_mb *<SIZE>*
Define the default size of metadata cache for the protection domain in MB

Example

```
scli --mdm_ip 192.168.1.200 --set_default_fgl_metadata_cache_size --
protection_domain_name PD1 --metadata_cache_size_mb 10248
```

8

Replication Commands

This section contains commands for managing replication.

abort_replication_consistency_group_snapshots_create

Abort a previous request to create a snapshot of a replication consistency group (RCG).

NOTE: This command cannot be performed when the specified RCG is inactive.

Syntax

```
scli --abort_replication_consistency_group_snapshots_create (--
replication_consistency_group_id <ID> | --replication_consistency_group_name <NAME>)
```

Parameters

--replication_consistency_group_id *<ID>*

Replication Consistency Group ID

--replication_consistency_group_name *<NAME>*

Replication Consistency Group name

Example

```
scli --mdm_ip 192.168.1.200 --abort_replication_consistency_group_snapshots_create --
replication_consistency_group_name rcg27
```

activate_replication_consistency_group

Activate the specified replication consistency group (RCG).

Notes

When activating the RCG, replication is enabled. The initial copy process is started as soon as there are sufficient system resources.

Syntax

```
scli --activate_replication_consistency_group (--replication_consistency_group_id <ID> |
--replication_consistency_group_name <NAME>)
```

Parameters

--replication_consistency_group_id *<ID>*

Replication Consistency Group ID

--replication_consistency_group_name *<NAME>*

Replication Consistency Group name

Example

```
scli --mdm_ip 192.168.1.200 --activate_replication_consistency_group --
replication_consistency_group_name rcg27
```

activate_replication_consistency_group_mappings

Activate Replication Consistency Group mappings on the peer system addressed by the command.

Notes

In general, the disaster recovery processes of failover, switchover, recover, and restore perform all the required operations, including activating mappings. This command is useful in situations when you choose to perform each of the individual operations directly. You can only activate mappings when the replication consistency group (RCG) role is neutral or source. You cannot activate mappings if the RCG role is target.

Syntax

```
scli --activate_replication_consistency_group_mappings (--
replication_consistency_group_id <ID> | --replication_consistency_group_name <NAME>)
```

Parameters

--replication_consistency_group_id *<ID>*

Replication Consistency Group ID

--replication_consistency_group_name *<NAME>*

Replication Consistency Group name

Example

```
scli --mdm_ip 192.168.1.200 --activate_replication_consistency_group_mappings --
replication_consistency_group_name rcg27
```

add_replication_consistency_group

Add a Replication Consistency Group (RCG).

Notes

You may only add an RCG after the replication peer system has been added using `add_replication_peer_system`.

Syntax

```
scli --add_replication_consistency_group (--destination_system_id <ID> |
--destination_system_name <NAME> | --destination_system_ip <IP> [--
destination_system_port <PORT>] | --destination_system_object_id <ID>) --rpo
<PERIOD> [--replication_consistency_group_name <NAME>] (--protection_domain_id
<ID> | --protection_domain_name <NAME>) (--remote_protection_domain_id
<ID> | --remote_protection_domain_name <NAME>) [--target_volume_access_mode
<TARGET_VOLUME_ACCESS_MODE>] [--replication_consistency_group_activity_mode
<ACTIVITY_MODE>]
```

Parameters

--destination_system_id *<ID>*

Destination Replication System ID

--destination_system_name *<NAME>*

Destination Replication System Name

--destination_system_ip *<IP>*

Destination Replication System IP address

--destination_system_port *<PORT>*

Port assigned to the Destination Replication System

--destination_system_object_id *<ID>*

Destination Replication System internal object ID

--rpo *<PERIOD>*

Desired RPO in units of time. To specify the time period, use s|m|h respectively indicating seconds, minutes, or hours. Examples: 30s (30 seconds), 15m (15 minutes), or 1h (one hour).

--replication_consistency_group_name *<NAME>*

A unique name to be assigned to the Replication Consistency Group

--protection_domain_id *<ID>*

Protection Domain ID

--protection_domain_name *<NAME>*

Protection Domain name

--remote_protection_domain_id *<ID>*

Remote Protection Domain ID

--remote_protection_domain_name *<NAME>*

Remote Protection Domain name

--target_volume_access_mode *<TARGET_VOLUME_ACCESS_MODE>*

Target volume access mode. Can be one of the following: `no_access` (default) or `read_only`

--replication_consistency_group_activity_mode *<ACTIVITY_MODE>*

Replication Consistency Group activity mode. Can be one of the following:

active

(Default). When the RCG is active, replication is enabled and initial copy begins as soon as there are sufficient system resources.

inactive

The RCG is created in inactive state and data is not copied to the target. The RCG can be activated later using the `activate_replication_consistency_group` command.

Example

```
scli --mdm_ip 192.168.1.200 --add_replication_consistency_group --destination_system_id
459865f0774b81e6 --rpo 15m --protection_domain_id 1caf743100000000 --
remote_protection_domain_id dc65bd9900000000 --target_volume_access_mode read_only
```

add_replication_pair

Add a replication pair consisting of a volume from the source system and a volume from the target system.

Notes

A volume can be a member of at most one replication pair. When the replication pair is created, both the source volume and the target volume must be the same size. It is recommended, but not mandatory, that the volumes in the volume pair have the same attributes (including zero padding and granularity). Not doing so can impact performance and capacity.

Syntax

```
scli --add_replication_pair (--replication_consistency_group_id <ID> | --
replication_consistency_group_name <NAME>) (--source_volume_id <ID> | --
source_volume_name <NAME>) (--destination_volume_id <ID> | --destination_volume_name
<NAME>) [--replication_pair_name <NAME>]
```

Parameters

--source_volume_id *<ID>*
: Source volume ID

--source_volume_name *<NAME>*
: Source volume name

--destination_volume_id *<ID>*
: Target volume ID

--destination_volume_name *<NAME>*
: Target volume name

--replication_consistency_group_id *<ID>*
: Replication Consistency Group ID

--replication_consistency_group_name *<NAME>*
: Replication Consistency Group name

--replication_pair_name *<NAME>*
: A unique name to be assigned to the replication pair

Example

```
scli --mdm_ip 192.168.1.200 --add_replication_pair --replication_consistency_group_name
RCG1 --source_volume_name s_vol1 --destination_volume_name t_vol1 --copy_type full --
replication_pair_name RP1
```

add_replication_peer_system

The peer system must be added to each of the systems involved in replication. There must be network connectivity between the two systems.

Syntax

```
scli --add_replication_peer_system --peer_system_ip <IP> [--peer_system_port <PORT>] --
peer_system_id <ID> [--peer_system_name <NAME>]
```

Parameters

--peer_system_ip *<IP>*

A comma-separated list of IP addresses assigned to the peer system. These are necessary for the replication to survive failures.

--peer_system_port *<PORT>*

Port assigned to the peer system. If this is omitted, the port used is the default peer listening port.

--peer_system_id *<ID>*

Peer system ID

--peer_system_name *<NAME>*

A name to be assigned to the peer system. The name should be unique.

Example

```
scli --mdm_ip 192.168.1.200 --add_replication_peer_system --peer_system_ip 10.55.105.21
--peer_system_id 7d897dd0269f5b0f --peer_system_name Durham1
```

add_sdr

Add an SDR to the specified Protection Domain. The SDR is one of the main components that must be installed on each of the peer Protection Domains for replication. It transfers data from the source volume to the target volume.

Syntax

```
scli --add_sdr --sdr_ip <IP> [--sdr_ip_role <ROLE>] [--sdr_name <NAME>] (--
protection_domain_id <ID> | --protection_domain_name <NAME>) [--sdr_port <PORT>] [--
profile <PROFILE>] [--force_clean [--i_am_sure]]
```

Parameters

--sdr_ip *<IP>*

A comma-separated list of IP addresses assigned to the SDR

--sdr_ip_role *<ROLE>*

The role assigned to each SDR IP address where *<ROLE>* can be one of the following:

application_only

Application only, for communicating with SDC

storage_only

Storage only, for communicating with SDS

external_only

External only, for communicating with remote SDR

application_and_storage

Application and storage, for communicating with SDC and SDS

application_and_external

Application and external, for communicating with SDC and remote SDR

storage_and_external

Storage and external, for communicating with SDS and remote SDR

all

Application, storage, and external, for communicating with SDC, SDS, and remote SDR

--sdr_name *<NAME>*

A name to be assigned to the SDR

--protection_domain_id *<ID>*

Protection Domain ID

--protection_domain_name *<NAME>*

Protection Domain name

--sdr_port *<PORT>*

Port assigned to the SDR

--profile {compact | high_performance}

Set the performance profile as compact or high performance. The high performance profile configures a predefined set of parameters for very high performance use cases.

--force_clean

Clean a previous SDR configuration. Use this if the SDR was previously part of a PowerFlex system.

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

Example

```
scli --mdm_ip 192.168.1.200 --add_sdr --sdr_ip 10.55.105.41,10.76.30.14 --sdr_name SDR2
--protection_domain_id 2c3071a200000001
```

SDR names

Assign each SDR a meaningful name. When a name has not been defined, the system may display default system-defined names that use the SDR’s IDs.

Each name should conform to the following rules:

1. Contains fewer than 32 characters
2. Contains only alphanumeric and punctuation characters
3. Is unique within the object type

NOTE: PowerFlex objects are assigned a unique ID that can be used to identify the object in CLI commands. You can retrieve the ID via a query or through the object’s property sheet in the PowerFlex GUI.

add_sdr_ip

Add an IP address to the specified SDR.

Syntax

```
scli --add_sdr_ip (--sdr_id <ID> | --sdr_name <NAME> | --sdr_ip <IP> [--sdr_port
<PORT>]) --new_sdr_ip <IP> [--sdr_ip_role <ROLE>]
```

Parameters

--sdr_id *<ID>*
SDR ID

--sdr_name *<NAME>*
SDR name

--sdr_ip *<IP>*
IP address assigned to the SDR

--sdr_port *<PORT>*
Port assigned to the SDR

--new_sdr_ip *<IP>*
New IP address to be assigned to the SDR

--sdr_ip_role *<ROLE>*
The role assigned to each SDR IP address where *<ROLE>* can be one of the following:

application_only
Application only, for communicating with SDC

storage_only
Storage only, for communicating with SDS

external_only
External only, for communicating with remote SDR

application_and_storage
Application and storage, for communicating with SDC and SDS

application_and_external
Application and external, for communicating with SDC and remote SDR

storage_and_external
Storage and external, for communicating with SDS and remote SDR

all
Application, storage, and external, for communicating with SDC, SDS, and remote SDR

Example

```
scli --mdm_ip 192.168.1.200 --add_sdr_ip --sdr_name SDR27 --new_sdr_ip 10.76.30.14 --
sdr_ip_role application_only
```

clear_broken_replication_consistency_group_alerts

Clear broken replication consistency group (RCG) indications for all volumes associated with the specified Protection Domain. This is useful in situations when the SDRs run out of journal capacity and RCG auto-termination cannot be applied (such as when the remote MDM doesn't support it). At that point, an alert is produced for every replicated volume in the affected Protection Domain. These alerts can be cleared using this command.

Syntax

```
scli --clear_broken_replication_consistency_group_alerts (--protection_domain_id <ID> |
--protection_domain_name <NAME>)
```

Parameters

--protection_domain_id *<ID>*

Protection Domain ID

--protection_domain_name *<NAME>*

Protection Domain name

Example

```
scli --mdm_ip 192.168.1.200 --clear_broken_replication_consistency_group_alerts --
protection_domain_name pd1
```

create_replication_consistency_group_snapshots

Initiate creation of Replication Consistency Group snapshots on the target.

Notes

The resulting snapshots are guaranteed to contain the latest consistent image. If a consistent image cannot be created, the command will fail. The `--force` option can be used to create a snapshot even if a consistent image is not available. This option should be used very carefully as the resulting snapshot may be crash inconsistent and may not be usable by applications.

Syntax

```
scli --create_replication_consistency_group_snapshots (--
replication_consistency_group_id <ID> | --replication_consistency_group_name <NAME>) [--
force]
```

Parameters

--replication_consistency_group_id *<ID>*

Replication Consistency Group ID

--replication_consistency_group_name *<NAME>*

Replication Consistency Group name

--force

Force creation of a Replication Consistency Group snapshot regardless of its consistency state.

Example

```
scli --mdm_ip 192.168.1.200 --create_replication_consistency_group_snapshots --
replication_consistency_group_name rcg27
```

enter_sdr_maintenance_mode

Enter the SDR into maintenance mode to allow for maintenance of the SDR. Note that there must be at least two other active SDRs joined to the system for replication to continue. When maintenance mode is activated, the MDM changes the mapping and distributes the I/Os to the remaining SDRs.

Syntax

```
scli --enter_sdr_maintenance_mode (--sdr_id <ID> | --sdr_name <NAME> | --sdr_ip <IP> [--
sdr_port <PORT>])
```

Parameters

--sdr_id *<ID>*
SDR ID

--sdr_name *<NAME>*
SDR name

--sdr_ip *<IP>*
IP address assigned to the SDR

--sdr_port *<PORT>*
Port assigned to the SDR

Example

```
scli --mdm_ip 192.168.1.200 --enter_sdr_maintenance_mode --sdr_name SDR27
```

exit_sdr_maintenance_mode

Move the SDR out of maintenance mode.

Notes

An SDR exiting maintenance mode requires free journal capacity in order to become active in the replication.

Syntax

```
scli --exit_sdr_maintenance_mode (--sdr_id <ID> | --sdr_name <NAME> | --sdr_ip <IP> [--
sdr_port <PORT>])
```

Parameters

--sdr_id *<ID>*
SDR ID

--sdr_name *<NAME>*
SDR name

--sdr_ip *<IP>*

IP address assigned to the SDR

--sdr_port *<PORT>*

Port assigned to the SDR

Example

```
scli --mdm_ip 192.168.1.200 --exit_sdr_maintenance_mode --sdr_name SDR27
```

failover_replication_consistency_group

Fail over the specified Replication Consistency Group. During failover, the application I/Os are stopped at the source and the access mode of the source volumes is changed to inaccessible. The target volumes are brought to the newest consistent image available on the destination and their access mode is changed to read-write. Recovery from failover state can be achieved by either restoring the replication or reversing it.

Syntax

```
scli --failover_replication_consistency_group (--replication_consistency_group_id <ID> |
--replication_consistency_group_name <NAME>)[--i_am_sure]
```

Parameters

--replication_consistency_group_id *<ID>*

Replication Consistency Group ID

--replication_consistency_group_name *<NAME>*

Replication Consistency Group name

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

Example

```
scli --mdm_ip 192.168.1.200 --failover_replication_consistency_group --
replication_consistency_group_name rcg27  --i_am_sure
```

failover_test_start_replication_consistency_group

Start a test failover of the specified replication consistency group (RCG). This allows you to demonstrate failover mode without interrupting data synchronization.

Syntax

```
scli --failover_test_start_replication_consistency_group (--
replication_consistency_group_id <ID> | --replication_consistency_group_name <NAME>)
```

Parameters

--replication_consistency_group_id *<ID>*

Replication Consistency Group ID

--replication_consistency_group_name *<NAME>*

Replication Consistency Group name

Example

```
scli --mdm_ip 192.168.1.200 --failover_test_start_replication_consistency_group --
replication_consistency_group_name rcg27
```

failover_test_stop_replication_consistency_group

Stop a failover test for the specified replication consistency group (RCG).

Syntax

```
scli --failover_test_stop_replication_consistency_group (--
replication_consistency_group_id <ID> | --replication_consistency_group_name <NAME>)
```

Parameters

--replication_consistency_group_id *<ID>*

Replication Consistency Group ID

--replication_consistency_group_name *<NAME>*

Replication Consistency Group name

Example

```
scli --mdm_ip 192.168.1.200 --failover_test_stop_replication_consistency_group --
replication_consistency_group_name rcg27
```

freeze_apply_replication_consistency_group

Apply the latest consistent image available at the destination to the target volumes and freeze the Replication Consistency Group (RCG). While the RCG is frozen, replication remains active and I/Os are replicated to the target journal. The I/Os are stored in the target journal and are not applied to the target volumes. Remaining in a frozen state for a long time runs the risk of running out of space in the target journal.

Syntax

```
scli --freeze_apply_replication_consistency_group (--replication_consistency_group_id
<ID> | --replication_consistency_group_name <NAME>)
```

Parameters

--replication_consistency_group_id *<ID>*

Replication Consistency Group ID

--replication_consistency_group_name *<NAME>*

Replication Consistency Group name

Example

```
scli --mdm_ip 192.168.1.200 --freeze_apply_replication_consistency_group --
replication_consistency_group_name rcg27
```

inactivate_replication_consistency_group_mappings

Inactivate replication consistency group mappings on the peer to which the command is applied.

Notes

In general, the disaster recovery processes of failover, switchover, recover, and restore perform all the required operations, including inactivating the replication consistency group mappings. This command is useful in situations when you choose to perform each of the individual operations directly.

Syntax

```
scli --inactivate_replication_consistency_group_mappings (--
replication_consistency_group_id <ID> | --replication_consistency_group_name <NAME>)
```

Parameters

--replication_consistency_group_id *<ID>*

Replication Consistency Group ID

--replication_consistency_group_name *<NAME>*

Replication Consistency Group name

Example

```
scli --mdm_ip 192.168.1.200 --inactivate_replication_consistency_group_mappings --
replication_consistency_group_name rcg27w
```

modify_replication_consistency_group_rpo

Modify the recovery point objective (RPO) of the Replication Consistency Group.

The RPO is the maximum data loss (in units of time) that you are willing to lose. When you set the RPO, you are setting the goal of replication.

Syntax

```
scli --modify_replication_consistency_group_rpo (--replication_consistency_group_id <ID>
| --replication_consistency_group_name <NAME>) --rpo <PERIOD>
```

Parameters

--replication_consistency_group_id *<ID>*

Replication Consistency Group ID

--replication_consistency_group_name *<NAME>*

Replication Consistency Group name

--rpo *<PERIOD>*

Desired RPO in units of time, between 30 seconds and 1 hour. To specify the time period, use s|m|h respectively indicating seconds, minutes, or hours. Examples: 30s (30 seconds), 15m (15 minutes), or 1h (one hour).

Example

```
scli --mdm_ip 192.168.1.200 --modify_replication_consistency_group_rpo --
replication_consistency_group_name rcg27 --rpo 20m
```

modify_replication_consistency_group_target_volume_access_mode

Modify the target volume access mode of a Replication Consistency Group.

Syntax

```
scli --modify_replication_consistency_group_target_volume_access_mode (--
replication_consistency_group_id <ID> | --replication_consistency_group_name <NAME>) --
target_volume_access_mode <TARGET_VOLUME_ACCESS_MODE>
```

Parameters

--replication_consistency_group_id *<ID>*

Replication Consistency Group ID

--replication_consistency_group_name *<NAME>*

Replication Consistency Group name

--target_volume_access_mode *<TARGET_VOLUME_ACCESS_MODE>*

Target volume access mode. Can be one of the following: `no_access` (default), `read_only`

Example

```
scli --mdm_ip 192.168.1.200
--modify_replication_consistency_group_target_volume_access_mode --
replication_consistency_group_name rcg27 --target_volume_access_mode read_only
```

modify_replication_peer_system_ip

Modify an IP address of a replication peer system.

Syntax

```
scli --modify_replication_peer_system_ip (--peer_system_id <ID> | --peer_system_name
<NAME> | --peer_system_ip <IP> [--peer_system_port <PORT>] | --peer_system_object_id
<ID>) --new_peer_system_ip <IP>
```

Parameters

--peer_system_id *<ID>*
: Replication peer system ID

--peer_system_name *<NAME>*
: Replication peer system name

--peer_system_ip *<IP>*
: Replication peer system IP address

--peer_system_port *<PORT>*
: Port assigned to the replication peer system

--peer_system_object_id *<ID>*
: Replication peer system internal object ID

--new_peer_system_ip *<IP>*
: A comma-separated list of new IP addresses assigned to the peer system

Example

```
scli --mdm_ip 192.168.1.200 --modify_replication_peer_system_ip --peer_system_name RPS1
--new_peer_system_ip 10.55.105.21
```

modify_replication_peer_system_port

Modify the port of the specified replication peer system.

Notes

Before running this SCLI command to modify port assigned to the replication peer system, you must first edit the `/opt/emc/scaleio/mdm/cfg/conf.txt` configuration file and set the `mdm_external_port` field to the new port. After you've saved the configuration file, restart the MDM service. Then run this command to modify the port assigned to the replication peer system.

Syntax

```
scli --modify_replication_peer_system_port (--peer_system_id <ID> | --peer_system_name
<NAME> | --peer_system_ip <IP> [--peer_system_port <PORT>] | --peer_system_object_id
<ID>) --new_peer_system_port <PORT>
```

Parameters

--peer_system_id *<ID>*
Replication peer system ID

--peer_system_name *<NAME>*
Replication peer system name

--peer_system_ip *<IP>*
IP address of the replication peer system

--peer_system_port *<PORT>*
Port assigned to the replication peer system

--peer_system_object_id *<ID>*
Replication peer system internal object ID

--new_peer_system_port *<PORT>*
New port assigned to the peer system. When this parameter is omitted, the default listening port is used.

Example

```
scli --mdm_ip 192.168.1.200 --modify_replication_peer_system_port --peer_system_name RP1
--new_peer_system_port 7611
```

modify_sdr_ip_role

Modify the role of the specified SDR's IP address.

Syntax

```
scli --modify_sdr_ip_role (--sdr_id <ID> | --sdr_name <NAME> | --sdr_ip <IP>  [--
sdr_port <PORT>]) [--sdr_ip_to_modify <IP>] --new_sdr_ip_role <ROLE>
```

Parameters

--sdr_id *<ID>*
SDR ID

--sdr_name *<NAME>*
SDR name

--sdr_ip *<IP>*
IP address assigned to the SDR

--sdr_port *<PORT>*
Port assigned to the SDR

--sdr_ip_to_modify *<IP>*
(Optional) The IP address of the SDR that you wish to modify

--new_sdr_ip_role *<ROLE>*
The new role assigned to the modified SDR IP address where *<ROLE>* can be one of the following:
application_only
Application only, for communicating with SDC
storage_only
Storage only, for communicating with SDS

external_only

External only, for communicating with remote SDR

application_and_storage

Application and storage, for communicating with SDC and SDS

application_and_external

Application and external, for communicating with SDC and remote SDR

storage_and_external

Storage and external, for communicating with SDS and remote SDR

all

Application, storage, and external, for communicating with SDC, SDS, and remote SDR

Example

```
scli --mdm_ip 192.168.1.200 --modify_sdr_ip_role --sdr_name SDR27  --sdr_ip_to_modify
10.76.30.14 --new_sdr_ip_role application_and_storage
```

modify_sdr_port

Modify the port assigned to the specified SDR.

Syntax

```
scli --modify_sdr_port (--sdr_id <ID> | --sdr_name <NAME> | --sdr_ip <IP> [--sdr_port
<PORT>]) --new_sdr_port <PORT>
```

Parameters

--sdr_id *<ID>*

SDR ID

--sdr_name *<NAME>*

SDR name

--sdr_ip *<IP>*

IP address of the SDR

--sdr_port *<PORT>*

Port assigned to the SDR

--new_sdr_port *<PORT>*

New port to be assigned to the SDR

Example

```
scli --mdm_ip 192.168.1.200 --modify_sdr_port --sdr_id ffa0be3900000000 --new_sdr_port
11088
```

pause_pair_initial_copy

Pause the initial copy process of the specified replication pair.

Syntax

```
scli --pause_pair_initial_copy (--replication_pair_id <ID> | --replication_pair_name
<NAME>)
```

Parameters

--replication_pair_id *<ID>*

Replication pair ID

--replication_pair_name *<NAME>*

Replication pair name

Example

```
scli --mdm_ip 192.168.1.200 --pause_pair_initial_copy --replication_pair_id
ca77eb5e00000000
```

pause_replication_consistency_group

Pause the replication consistency group (RCG). While in pause mode, new writes are collected at the source but are not transferred to the target.

Syntax

```
scli --pause_replication_consistency_group (--replication_consistency_group_id <ID> | --
replication_consistency_group_name <NAME>) --pause_mode <MODE>
```

Parameters

--replication_consistency_group_id *<ID>*

Replication Consistency Group ID

--replication_consistency_group_name *<NAME>*

Replication Consistency Group name

--pause_mode *<MODE>*

The pause mode. Options:

stop_data_transfer

Stop the data transmission

only_track_changes

Stop the data transmission and switch to FDRL mode

Example

```
scli --mdm_ip 192.168.1.200 --pause_replication_consistency_group --
replication_consistency_group_name rcg27 --pause_mode stop_data_transfer
```

print_recreate_replication_consistency_group_script

Display a list of SCLI commands required to recreate the specified Replication Consistency Group.

Syntax

```
scli --print_recreate_replication_consistency_group_script (--
replication_consistency_group_name <NAME>| --replication_consistency_group_id <ID>)
```

Parameters

--replication_consistency_group_name *<NAME>*

Replication Consistency Group name

--replication_consistency_group_id *<ID>*

Replication Consistency Group ID

Example

```
scli --mdm_ip 192.168.1.200 --print_recreate_replication_consistency_group_script --
replication_consistency_group_name rcg27
```

Sample output

```
scli --add_replication_consistency_group --replication_consistency_group_name RCG1 --
destination_system_object_id c876517d00000000 --protection_domain_id 036a797000000000
--remote_protection_domain_id 073b95a000000000 --rpo 3600 --target_volume_access_mode
no_access
scli --add_replication_pair --replication_consistency_group_name RCG1 --source_volume_id
6730b3a800000003 --destination_volume_id 2d3d6cd500000003 --copy_type full
```

query_all_replicated_volumes

Retrieve information about all replicated volumes. The query result can be optionally filtered by Protection Domain or by Storage Pool.

Syntax

```
scli --query_all_replicated_volumes [(((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --storage_pool_name <NAME>) | --storage_pool_id <ID>)]
```

Parameters

--protection_domain_id *<ID>*
Protection Domain ID

--protection_domain_name *<NAME>*
Protection Domain name

--storage_pool_name *<NAME>*
Storage Pool name

--storage_pool_id *<ID>*
Storage Pool ID

Example

```
scli --mdm_ip 192.168.1.200 --query_all_replicated_volumes
```

Sample output

```
Protection Domain f11d784d00000000 Name: pd1
Storage Pool bf83a76700000000 Name: sp1
 Volume ID: c3a62eb200000003 Name: vol1 Size: 16.0 GB (16384 MB)
   Pair ID: ef199ea900000000 (CG ID: 4607a08e00000000)

Protection Domain f11d784e00000001 Name: pd2
Storage Pool bf83a76800000001 Name: sp2
 <No volumes defined>

Protection Domain f11d9f5d00000002 Name: dfg
 <No volumes defined>
Query-all-replicated-volumes returned 1 volumes
```

query_all_replication_consistency_groups

Retrieve information about all the replication consistency groups (RCG).

Syntax

```
scli --query_all_replication_consistency_groups
```

Parameters

None.

Example

```
scli --mdm_ip 192.168.1.200 --query_all_replication_consistency_groups
```

Sample output

```
Protection Domain fcc9af5500000000 Name: pd1
 ID: 399c609c00000000 Name: rcg1 Activity: INACTIVE Peer System: peer_to_118 RPO: 1m
Compliance: 0% Status: Stopped by User

Protection Domain fcc9af5600000001 Name: pd2

Protection Domain fcc9d66500000002 Name: rrr

query_all_replication_consistency_groups returned 1 Replication Consistency Groups.
```

query_all_replication_journal_capacity

Retrieve information about all replication journal volumes. The results can be filtered by Protection Domain or by Storage Pool.

Syntax

```
scli --query_all_replication_journal_capacity [(((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --storage_pool_name <NAME>) | --storage_pool_id <ID>)]
```

Parameters

--protection_domain_id *<ID>*
: Protection Domain ID

--protection_domain_name *<NAME>*
: Protection Domain name

--storage_pool_name *<NAME>*
: Storage Pool name

--storage_pool_id *<ID>*
: Storage Pool ID

Example

```
scli --mdm_ip 192.168.1.200 --query_all_replication_journal_capacity --storage_pool_name
SP1
```

Sample output

```
Protection Domain a86cde9d00000000 Name: pd1
Storage Pool 6d1751fe00000000 Name: sp1
 Volume ID: f7d341ef00000000 Name: N/A Size: 8.0 GB (8192 MB) SDR ID: 3179c35600000000
Thin-provisioned  SdrSessionIndex 0
 Volume ID: f7d341f100000002 Name: N/A Size: 8.0 GB (8192 MB) SDR ID: 3179c35800000002
Thin-provisioned  SdrSessionIndex 2
 Volume ID: f7d341f400000001 Name: N/A Size: 8.0 GB (8192 MB) SDR ID: 3179c35a00000001
Thin-provisioned  SdrSessionIndex 1

Journal Capacity - Allocated: 24.0 GB (24576 MB) Max Allowed: 26.0 GB (26624 MB) (10% of
total)

Query-all-volumes returned 3 volumes
```

query_all_replication_pairs

Retrieve information about all the replication pairs in the system.

Syntax

```
scli --query_all_replication_pairs
```

Parameters

None.

Example

```
scli --mdm_ip 192.168.1.200 --query_all_replication_pairs
```

Sample output

```
CG Id: 399c609c00000000 CG Name: rcg1
Peer System Name: peer_to_118 Direction: LOCAL_TO_REMOTE
    Local ID: bee0361f00000000 Remote ID: c557f02500000000 Copy Type: IDENTICAL Initial
Copy State: DONE Name: (null)
    Local Volume ID f6351e3700000003 Remote Volume ID: dbf2341e00000003 Local Volume
Name: vol1_114 Remote Volume Name: vol1_118
    Initial Copy progress: 100% Local Activity State: INACTIVE
query_all_replication_pairs returned 1 Replication Pairs.
```

query_all_replication_peer_system

Retrieve information about all the replication peer systems.

Syntax

```
scli --query_all_replication_peer_system
```

Parameters

None.

Example

```
scli --mdm_ip 192.168.1.200 --query_all_replication_peer_system
```

Sample output

```
Query-all-Replication Peer System returned 1 Replication Peer System nodes.

Replication Peer System ID: 7c3e2613207b310f Replication Peer System
internal ID: 3ea1845400000000 Name: peer_to_118 State: Joined, SUCCESS IP:
10.55.137.118,10.55.137.120 Port: 7611 Version: 3.6.0

SDR-SDR connectivity status: All connected
```

query_all_sdr

Retrieve information about all SDRs.

Syntax

```
scli --query_all_sdr
```

Parameters

None.

Example

```
scli --mdm_ip 192.168.1.200 --query_all_sdr
```

Sample output

```
Query-all-SDR returned 3 SDR nodes.

Protection Domain ID: 036a797000000000
SDR ID: 5747c3f800000000 Name: SDR2 State: Connected, Joined IP:
10.55.105.40,10.76.30.11,192.168.177.28 Port: 11088 Version: 3.5.0
SDR ID: 5747c3f900000001 Name: SDR3 State: Connected, Joined IP:
10.55.105.41,10.76.30.14 Port: 11088 Version: 3.5.0
SDR ID: 5747c3fa00000002 Name: SDR1 State: Connected, Joined IP:
10.55.105.39,10.76.30.10 Port: 11088 Version: 3.5.0
```

query_replication_consistency_group

Retrieve information about the specified Replication Consistency Group.

Syntax

```
scli --query_replication_consistency_group (--replication_consistency_group_id <ID> | --
replication_consistency_group_name <NAME>)
```

Parameters

--replication_consistency_group_id *<ID>*

Replication Consistency Group ID

--replication_consistency_group_name *<NAME>*

Replication Consistency Group name

Example

```
scli --mdm_ip 192.168.1.200 --query_replication_consistency_group --
replication_consistency_group_id 399c609c00000000
```

Sample output

```
Replication Consistency Group ID: 399c609c00000000 Name: rcg1
  Protection Domain ID: fcc9af5500000000 Name: pd1

Configuration:
  RPO: 1m
  Target volume access mode: NO_ACCESS
  Number of volume pairs: 1, Capacity: 16.0 GB (16384 MB)

Remote info:
  Peer System ID: 7c3e2613207b310f Name: peer_to_118
  Protection Domain ID: e6bce87900000000 Name: pd1
  Remote CG Id: 2215ba5800000000

State:
  Lifetime State: NORMAL
  Transmit State: NOT_PAUSED
  Freeze State: UNFROZEN
  Consistency Mode: PARTIALLY_CONSISTENT
  Direction: LOCAL_TO_REMOTE
  Disaster recovery flow: NONE
  Disaster recovery state: None

Consistency state:
  RPO Compliance: 0%
  Lag Received: 0s (at 10:51:55)
  Lag Journalled: 0s (at 10:51:55)
  Lag Applied: 0s (at 10:51:55)
  Last Barrier: 0s (at 10:51:55)
  Next Planned Barrier: 0s (at 10:51:55)

Bandwidth:
  User Read: 479 IOPS 239.2 MB (244941 KB) per-second
  User Write: 514 IOPS 256.8 MB (262960 KB) per-second
  Transmit: 0 IOPS 0 Bytes per-second
  Apply: 0 IOPS 0 Bytes per-second

Latency:
  User Read: 240 IOPS 20.917 ms average latency
  User Write: 257 IOPS 221.345 ms average latency
  Transmit: 0 IOPS 0 us average latency
  Apply: 0 IOPS 0 us average latency
  Journaler Write: 0 IOPS 0 us average latency
  Journaler Read: 0 IOPS 0 us average latency

Journalled data:
  Journalled on Source: 0 Bytes
  Ready For Transmit: 0 Bytes
  Journalled on Target: 0 Bytes
  Ready For Apply: 0 Bytes

Activity State:
```

```
  Local Activity State: ACTIVE
  Remote Activity State: ACTIVE
```

query_replication_pair

Retrieve information about the specified replication pair.

Syntax

```
scli --query_replication_pair --replication_pair_id <ID> --replication_pair_name <NAME>
```

Parameters

--replication_pair_id *<ID>*

Replication pair ID

--replication_pair_name *<NAME>*

Replication pair name

Example

```
scli --mdm_ip 192.168.1.200 --query_replication_pair --replication_pair_id
bee0361f00000000
```

Sample output

```
Local ID: bee0361f00000000 Remote ID: c557f02500000000 Name: N/A Initial Copy State:
DONE Initial Copy Priority: -1 Initial Copy paused by user: FALSE
Peer System Name: peer_to_118 Direction: LOCAL_TO_REMOTE CG ID: 399c609c00000000 CG
Name: rcg1
Local Volume ID: f6351e3700000003 Remote Volume ID: dbf2341e00000003 Local Volume Name:
vol1_114 Remote Volume Name: vol1_118 Remote Volume Capacity (MB): 16384 Copy Type:
IDENTICAL Lifetime State: NORMAL
Initial Copy progress: 100%
Local Activity State: INACTIVE Remote Activity State: INACTIVE Last EnableRpl Failure
Reason: INVALID
```

query_replication_peer_system

Retrieve information about the specified replication peer system.

Syntax

```
scli --query_replication_peer_system (--peer_system_id <ID> | --peer_system_name <NAME>
| --peer_system_ip <IP> [--peer_system_port <PORT>] | --peer_system_object_id <ID>)
```

Parameters

--peer_system_id *<ID>*

Replication peer system ID

--peer_system_name *<NAME>*

Replication peer system name

--peer_system_ip *<IP>*

Replication peer system IP address

--peer_system_port *<PORT>*

Port assigned to the replication peer system port

--peer_system_object_id *<ID>*

Replication peer system internal object ID

Example

```
scli --mdm_ip 192.168.1.200 --query_replication_peer_system --peer_system_id
7c3e2613207b310f
```

Sample output

```
Replication Peer System 7c3e2613207b310f Internal ID 3ea1845400000000 Name: peer_to_118
Version: 3.6.0
IP information (total 2 IP addresses):
         1: 10.55.137.118
         2: 10.55.137.120
         Port: 7611
Membership-state: Joined; Coupling_rc: SUCCESS;
```

query_replication_peer_system_destination_volume_candidates

Retrieve information about destination volume candidates in the replication peer system. This command returns available remote volumes in the specified peer system's Protection Domain with the specified capacity.

Syntax

```
scli --query_replication_peer_system_destination_volume_candidates (--peer_system_id
<ID> | --peer_system_name <NAME> | --peer_system_ip <IP> [--peer_system_port <PORT>]
| --peer_system_object_id <ID>) (--protection_domain_id <ID> | --protection_domain_name
<NAME>) --size_gb <SIZE>
```

Parameters

--peer_system_id *<ID>*

Replication peer system ID

--peer_system_name *<NAME>*

Replication peer system name

--peer_system_ip *<IP>*

Replication peer system IP address

--peer_system_port *<PORT>*

Port assigned to the replication peer system

--peer_system_object_id *<ID>*

Replication Peer System internal object ID

--protection_domain_id *<ID>*

Protection Domain ID

--protection_domain_name *<NAME>*

Protection Domain name

--size_gb *<SIZE>*

Size of volume in GBs

Example

```
scli --mdm_ip 192.168.1.200 --
query_replication_peer_system_destination_volume_candidates --peer_system_id
1820896a6b53a70f --protection_domain_name pd1 --size_gb 540
```

query_replication_peer_system_protection_domains

Retrieve information about the available Protection Domains in the target replication peer system.

Syntax

```
Usage: scli --query_replication_peer_system_protection_domains (--peer_system_id <ID>
| --peer_system_name <NAME> | --peer_system_ip <IP> [--peer_system_port <PORT>] | --
peer_system_object_id <ID>)
```

Parameters

--peer_system_id *<ID>*

Replication peer system ID

--peer_system_name *<NAME>*

Replication peer system name

--peer_system_ip *<IP>*

IP address assigned to the replication peer system

--peer_system_port *<PORT>*

Port assigned to the replication peer system

--peer_system_object_id *<ID>*

Replication peer system internal object ID

Example

```
scli --mdm_ip 192.168.1.200 --query_replication_peer_system_protection_domains --
peer_system_id 1820896a6b53a70f
```

Sample output

```
Received 1 PDs
PD Id 382dca8200000000
PD Name pd1
```

query_sdr

Retrieve information about the specified SDR.

Syntax

```
scli --query_sdr (--sdr_id <ID> | --sdr_name <NAME> | --sdr_ip <IP> [--sdr_port <PORT>])
```

Parameters

--sdr_id *<ID>*
SDR ID

--sdr_name *<NAME>*
SDR name

--sdr_ip *<IP>*
IP address assigned to the SDR

--sdr_port *<PORT>*
Port assigned to the SDR

Example

```
scli --query_sdr --sdr_id 82dc5ccd00000000
```

Sample output

```
SDR 82dc5ccd00000000 Name: SDR3 Version: 3.6.0
 Protection Domain ID: fcc9af5500000000
IP information (total 1 IP addresses):
         1: 10.55.137.116    Role: all (application, storage, external)
         Port: 11088
Membership-state: Joined; Connection-state: Connected; Maintenance-state: Normal; SDR-
state: Normal;
```

query_sync_now_replication_consistency_group

Retrieve the status of a synchronization process for the specified replication consistency group (RCG).

Syntax

```
scli --query_sync_now_replication_consistency_group (--replication_consistency_group_id
<ID> | --replication_consistency_group_name <NAME>) --sync_now_key <SYNC_NOW_KEY>
```

Parameters

--replication_consistency_group_id *<ID>*

Replication consistency group ID

--replication_consistency_group_name *<NAME>*

Replication consistency group name

--sync_now_key *<SYNC_NOW_KEY>*

The key returned from running the `--sync_now_replication_consistency_group` command.

Example

```
scli --mdm_ip 192.168.1.200 --query_sync_now_replication_consistency_group --
replication_consistency_group_name rcg27 --sync_now_key 667432
```

Sample output

```
Successfully queried SyncNowKey 190531 of Replication Consistency Group RCG1: The data
was applied to the destination volumes
```

Return codes

The following are the possible return codes for this command:

Table 3. Replication Consistency Group Return Codes

Return code	Message
MOS_RC__SUCCESS	Success.
MOS_RC__RPL_CG_INVALID_ID	Invalid replication consistency group ID.
MOS_RC__RPL_CG_INVALID_KEY	Invalid replication consistency group key.
MOS_RC__RPL_CG_NOT_FOUND	Replication consistency group not found.
MOS_RC__RPL_CG_IS_INACTIVE	The RCG is not active due to deactivated protection domain.
MOS_RC__RPL_CG_COMMAND_IS_NOT_ALLOWED_ON_SYSTEM_RCG	A user operation is not allowed on a system RCG.
MOS_RC__RPL_CG_COMMAND_IS_NOT_ALLOWED_DURING_DISASTER_RECOVERY	The command is not allowed during disaster recovery.
MOS_RC__RPL_CG_IS_NOT_IN_NORMAL_STATE	Replication consistency group is not in normal state.

remove_replication_consistency_group

Initiate removal of a Replication Consistency Group.

Notes

Removal of a replication consistency group is a lengthy process and it may complete some time after the command completion.

Syntax

```
Usage: scli --remove_replication_consistency_group (--replication_consistency_group_id
<ID> | --replication_consistency_group_name <NAME>)
```

Parameters

--replication_consistency_group_id *<ID>*
Replication Consistency Group ID

--replication_consistency_group_name *<NAME>*
Replication Consistency Group Name

Example

```
scli --mdm_ip 192.168.1.200 --remove_replication_consistency_group --
replication_consistency_group_name rcg27
```

remove_replication_pair

Remove the specified replication pair.

NOTE: The target volume image may not be consistent after the pair removal. To obtain a consistent image on the target volume: Run `freeze_apply_replication_consistency_group` on the the replication consistency group, remove the replication pair, and then run `unfreeze_apply_replication_consistency_group` on the replication consistency group.

Syntax

```
scli --remove_replication_pair (--replication_pair_id <ID> | --replication_pair_name
<NAME>) [--i_am_sure]
```

Parameters

--replication_pair_id *<ID>*
Replication pair ID

--replication_pair_name *<NAME>*
Replication pair name

--i_am_sure
Preemptive approval

Example

```
scli --mdm_ip 192.168.1.200 --remove_replication_pair --replication_pair_name rp1 --
i_am_sure
```

remove_replication_peer_system

Remove a replication peer system.

Syntax

```
scli --remove_replication_peer_system (--peer_system_id <ID> | --peer_system_name <NAME>
| --peer_system_ip <IP> [--peer_system_port <PORT>] | --peer_system_object_id <ID>)
```

Parameters

--peer_system_id *<ID>*
Peer system ID

--peer_system_name *<NAME>*
Peer system name

--peer_system_ip *<IP>*
A comma-separated list of IP addresses assigned to the peer system

--peer_system_port *<PORT>*
Port assigned to the Replication peer system

--peer_system_object_id *<ID>*
Replication peer system internal object ID

Example

```
scli --mdm_ip 192.168.1.200 --remove_replication_peer_system --peer_system_name RPS1
```

remove_sdr

Initiate removal of the specified SDR.

Syntax

```
scli --remove_sdr (--sdr_id <ID> | --sdr_name <NAME> | --sdr_ip <IP> [--sdr_port <PORT>])
```

Parameters

--sdr_id *<ID>*
SDR ID

--sdr_name *<NAME>*

SDR name

--sdr_ip *<IP>*

SDR IP address

--sdr_port *<PORT>*

Port assigned to the SDR

Example

```
scli --mdm_ip 192.168.1.200 --remove_sdr --sdr_name SDR2
```

remove_sdr_ip

Remove an IP address from an SDR.

Syntax

```
scli --remove_sdr_ip (--sdr_id <ID> | --sdr_name <NAME> | --sdr_ip <IP> [--sdr_port
<PORT>]) --sdr_ip_to_remove <IP>
```

Parameters

--sdr_id *<ID>*

SDR ID

--sdr_name *<NAME>*

SDR name

--sdr_ip *<IP>*

IP address assigned to the SDR

--sdr_port *<PORT>*

Port assigned to the SDR

--sdr_ip_to_remove *<IP>*

IP address to remove from the SDR

Example

```
scli --mdm_ip 192.168.1.200 --remove_sdr_ip --sdr_name SDR27 --sdr_ip_to_remove
10.76.30.14
```

rename_replication_consistency_group

Name or rename the specified Replication Consistency Group.

Syntax

```
scli --rename_replication_consistency_group (--replication_consistency_group_id <ID> | --
replication_consistency_group_name <NAME>) (--new_name <NAME> | --remove_name)
```

Parameters

--replication_consistency_group_id *<ID>*

Replication Consistency Group ID

--replication_consistency_group_name *<NAME>*

Replication Consistency Group name

--new_name *<NAME>*

New name to be assigned to the Replication Consistency Group

--remove_name

Use this flag to delete the Replication Consistency Group name without giving it a new name.

Example

```
scli --mdm_ip 192.168.1.200 --rename_replication_consistency_group --
replication_consistency_group_name rcg27 --new_name RCG27
```

rename_replication_peer_system

Name or rename the replication peer system. The name should be unique.

Syntax

```
scli --rename_replication_peer_system (--peer_system_id <ID> | --peer_system_name <NAME>
| --peer_system_ip <IP> [--peer_system_port <PORT>] | --peer_system_object_id <ID>) --
new_peer_system_name <NAME>
```

Parameters

--peer_system_id *<ID>*

Replication peer system ID

--peer_system_name *<NAME>*

Replication peer system name

--peer_system_ip *<IP>*

IP address assigned to the replication peer system

--peer_system_port *<PORT>*

Port assigned to the replication peer system

--peer_system_object_id *<ID>*

Replication peer system internal object ID

--new_peer_system_name *<NAME>*

A new unique name to be assigned to the peer system.

Example

```
scli --mdm_ip 192.168.1.200 --rename_replication_peer_system --peer_system_id
1820896a6b53a70f --new_peer_system_name RPS1
```

rename_sdr

Name or rename an SDR.

Syntax

```
scli --rename_sdr (--sdr_id <ID> | --sdr_name <NAME> | --sdr_ip <IP> [--sdr_port
<PORT>]) --new_name <NAME>
```

Parameters

--sdr_id *<ID>*
SDR ID

--sdr_name *<NAME>*
SDR name

--sdr_ip *<IP>*
SDR IP address

--sdr_port *<PORT>*
Port assigned to the SDR

--new_name *<NAME>*
New name to be assigned to the SDR

Example

```
scli --mdm_ip 192.168.1.200 --rename_sdr --sdr_name SDR2 --new_name SDR27
```

resume_pair_initial_copy

Resume the paused initial copy process of the specified replication pair.

Syntax

```
scli --resume_pair_initial_copy (--replication_pair_id <ID> | --replication_pair_name
<NAME>)
```

Parameters

--replication_pair_id <ID>
Replication pair ID

--replication_pair_name <NAME>
Replication pair name

Example

```
scli --mdm_ip 192.168.1.200 --resume_pair_initial_copy --replication_pair_name RP1
```

resume_replication_consistency_group

Resume a paused replication consistency group (RCG). Once the RCG is resumed, the source will resume transmission of the collected write data to the target.

Syntax

```
scli --resume_replication_consistency_group (--replication_consistency_group_id <ID> | --
replication_consistency_group_name <NAME>)
```

Parameters

--replication_consistency_group_id *<ID>*

Replication Consistency Group ID

--replication_consistency_group_name *<NAME>*

Replication Consistency Group name

Example

```
scli --mdm_ip 192.168.1.200 --resume_replication_consistency_group --
replication_consistency_group_name rcg27
```

restore_replication_consistency_group

Once the replication consistency group is in failover or switchover mode, you can reverse or restore the replication. Restoring replication maintains the replication direction from the original source and overwrites all data at the target.

Notes

During a restore operation, data movement from the source to the target is carried out without maintaining consistency of the data during the process. Consistency is only achieved at the target once the source and target synchronized.

Prior to executing the restore operation, you consider creating a snapshot on the new target volumes in order to preserve the consistent image during the resynchronization process.

Syntax

```
scli --restore_replication_consistency_group (--replication_consistency_group_id <ID> |
--replication_consistency_group_name <NAME>)
```

Parameters

--replication_consistency_group_id *<ID>*

Replication Consistency Group ID

--replication_consistency_group_name *<NAME>*

Replication Consistency Group name

Example

```
scli --mdm_ip 192.168.1.200 --restore_replication_consistency_group --
replication_consistency_group_name rcg27
```

reverse_replication_consistency_group

Once the replication consistency group is in failover or switchover mode, you can reverse or restore the replication. Reversing replication changes the direction so that the original target becomes the source. All data at the original source is overwritten by the data at the target.

Notes

During a reverse operation, data movement from the new source to the new target is carried out without maintaining consistency of the data during the process. Consistency is only achieved at the new target once the new source and the new target synchronized.

Prior to executing the reverse operation, you consider creating a snapshot on the new target volumes in order to preserve the consistent image during the resynchronization process.

Syntax

```
scli --reverse_replication_consistency_group (--replication_consistency_group_id <ID> |
--replication_consistency_group_name <NAME>)
```

Parameters

--replication_consistency_group_id *<ID>*

Replication Consistency Group ID

--replication_consistency_group_name *<NAME>*

Replication Consistency Group name

Example

```
scli --mdm_ip 192.168.1.200 --reverse_replication_consistency_group --
replication_consistency_group_name rcg27
```

set_pair_initial_copy_queue_position

Set the priority of the initial copy process of the specified replication pair to be the highest or lowest priority in the initial copy queue.

NOTE: This command cannot be performed when the specified RCG is inactive.

Syntax

```
scli --set_pair_initial_copy_queue_position (--replication_pair_id <ID> | --
replication_pair_name <NAME>) (--initial_copy_priority <TYPE>)
```

Parameters

--replication_pair_id *<ID>*

Replication pair ID

--replication_pair_name *<NAME>*

Replication pair name

--initial_copy_priority *<TYPE>*

Desired initial copy priority. Can be one of the following:

highest

Move the initial copy process to the front of the queue

lowest

Move the initial copy process to the end of the queue

Example

```
scli --mdm_ip 192.168.1.200 --set_pair_initial_copy_queue_position --
replication_pair_name RP1 --initial_copy_priority highest
```

set_replication_consistency_group_consistent

Set the specified replication consistency group (RCG) as consistent. While in consistent mode, the journal at the target applies its content to the target volumes in a way that guarantees that a consistent image of the target volumes can be achieved.

This command is applicable when one of the two replication peer systems is running a version prior to v3.6.x.

NOTE: This command cannot be performed when the specified RCG is inactive. It cannot be performed when in failover or switchover mode. Similarly, it cannot be performed on a disabled Protection Domain or during freeze apply.

Syntax

```
scli --set_replication_consistency_group_consistent (--replication_consistency_group_id
<ID> | --replication_consistency_group_name <NAME>)
```

Parameters

--replication_consistency_group_id *<ID>*

Replication consistency group ID

--replication_consistency_group_name *<NAME>*

Replication consistency group name

Example

```
scli --mdm_ip 192.168.1.200 --set_replication_consistency_group_consistent --
replication_consistency_group_name rcg27
```

set_replication_consistency_group_inconsistent

Set the specified replication consistency group (RCG) as inconsistent. While in inconsistent mode, the contents of the target journal are applied to the target volumes without guarantee of consistency. This means that the target volume image may not be consistent during the time that the RCG is inconsistent. The RCG will automatically regain consistency when possible.

It is recommended to create a snapshot of the application volumes at the target before running this command. The snapshot will capture the last consistent image before the RCG is inconsistent.

NOTE: This command cannot be performed when the specified RCG is inactive or while in failover or switchover mode. It also cannot be performed on a disabled Protection Domain, during freeze apply, while creating snapshots, or during failover test.

Syntax

```
scli --set_replication_consistency_group_inconsistent (--
replication_consistency_group_id <ID> | --replication_consistency_group_name <NAME>)
```

Parameters

--replication_consistency_group_id *<ID>*

Replication Consistency Group ID

--replication_consistency_group_name *<NAME>*

Replication Consistency Group name

Example

```
scli --mdm_ip 192.168.1.200 --set_replication_consistency_group_inconsistent --
replication_consistency_group_name rcg27
```

set_replication_journal_capacity

Set the replication journal capacity ratio for the specified Storage Pool. This defines the maximum ratio of a storage pool capacity that can be used by the replication system for journaling.

Notes

Journal capacity is defined as a percentage of the total storage capacity in the Storage Pool and must equal at least 28 GB per SDR. In general, journal capacity should be at least 5% of replicated usable capacity in the Protection Domain, including volumes used as source and targets. It is important to assign enough storage capacity for the replication journal. For detailed instructions on calculating journal capacity, see "SDR journal capacity" in the *Getting to Know Dell PowerFlex Guide*.

Syntax

```
scli --set_replication_journal_capacity (((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --storage_pool_name <NAME>) | --storage_pool_id <ID>) --
replication_journal_capacity_max_ratio <RATIO>)
```

Parameters

--protection_domain_id *<ID>*

Protection Domain ID

--protection_domain_name *<NAME>*

Protection Domain name

--storage_pool_name *<NAME>*

Storage Pool name

--storage_pool_id *<ID>*

Storage Pool ID

--replication_journal_capacity_max_ratio *<RATIO>*

Maximum capacity percentage to be allocated for journal capacity (from 0 to 100).

Example

```
scli --mdm_ip 192.168.1.200 --set_replication_journal_capacity --protection_domain_name
PD1 --storage_pool_name SP1 --replication_journal_capacity_max_ratio 25
```

switchover_replication_consistency_group

Switch over the specified replication consistency group (RCG). Switchover is a planned failover. The application I/Os are stopped at the source and the data is synchronized so that the target volume is consistent with the source volume. The access mode of the original source volumes is changed to unavailable or read-only. The access mode of the original target volumes is changed so that the hosts can read and write to the volumes.

NOTE: This command cannot be performed when the specified RCG is inactive.

Syntax

```
scli --switchover_replication_consistency_group (--replication_consistency_group_id <ID>
| --replication_consistency_group_name <NAME>)[--i_am_sure]
```

Parameters

--replication_consistency_group_id *<ID>*

Replication Consistency Group ID

--replication_consistency_group_name *<NAME>*

Replication Consistency Group name

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

Example

```
scli --mdm_ip 192.168.1.200 --switchover_replication_consistency_group --
replication_consistency_group_name rcg27 --i_am_sure
```

sync_now_replication_consistency_group

Synchronize the source and target to ensure that all new data that has been written to the source is also written to the target volumes. A sync key is returned that can be used to query the status of the synchronization process.

Notes

In general, the disaster recovery processes of failover, switchover, recover, and restore perform all the required operations, including synchronizing the source and target volumes. This command is useful in situations when you choose to perform each of the individual operations directly.

Syntax

```
scli --sync_now_replication_consistency_group (--replication_consistency_group_id <ID> |
--replication_consistency_group_name <NAME>)
```

Parameters

--replication_consistency_group_id *<ID>*

Replication Consistency Group ID

--replication_consistency_group_name *<NAME>*

Replication Consistency Group name

Example

```
scli --mdm_ip 192.168.1.200 --sync_now_replication_consistency_group --
replication_consistency_group_name rcg27
```

terminate_replication_consistency_group

Terminate the specified replication consistency group (RCG).

Notes

Terminating the RCG moves it into an inactive state. Replication is stopped and the journal capacity is released. From this point on, the I/Os are handled by the SDS and not the SDR and no journal capacity is consumed. All RCG configurations are maintained, including source and target roles, pair configuration, and volume access modes.

Terminating an RCG does not require connectivity between the sites. If the RCG is terminated on a peer when there is no connectivity, then the other peer will terminate once connectivity is restored.

NOTE: Activating a terminated RCG requires full initialization. The initial copy process may take a long time.

Syntax

```
scli --terminate_replication_consistency_group (--replication_consistency_group_id <ID>
| --replication_consistency_group_name <NAME>)
```

Parameters

--replication_consistency_group_id *<ID>*

Replication Consistency Group ID

--replication_consistency_group_name *<NAME>*

Replication Consistency Group name

Example

```
scli --mdm_ip 192.168.1.200 --terminate_replication_consistency_group --
replication_consistency_group_name rcg27
```

unfreeze_apply_replication_consistency_group

Unfreeze the replication consistency group (RCG) and apply the I/Os stored in the target journal to the target volumes.

Syntax

```
scli --unfreeze_apply_replication_consistency_group (--replication_consistency_group_id
<ID> | --replication_consistency_group_name <NAME>)
```

Parameters

--replication_consistency_group_id *<ID>*

Replication Consistency Group ID

--replication_consistency_group_name *<NAME>*

Replication Consistency Group name

Example

```
scli --mdm_ip 192.168.1.200 --unfreeze_apply_replication_consistency_group --
replication_consistency_group_name rcg27
```

9

RFcache Commands

This section contains commands for managing RFcache.

disable_rfcache

Disable the RFcache feature on a specific SDS or in an entire Protection Domain.

Syntax

```
scli --disable_rfcache ((--protection_domain_id <ID> | --protection_domain_name <NAME>)
| (((--protection_domain_id <ID> | --protection_domain_name <NAME>) --storage_pool_name
<NAME>) | --storage_pool_id <ID>) | (--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP>
[--sds_port <PORT>]))
```

Parameters

--protection_domain_id *<ID>*
Protection Domain ID

--protection_domain_name *<NAME>*
Protection Domain name

--storage_pool_id *<ID>*
Storage Pool ID

--storage_pool_name *<NAME>*
Storage Pool name

--sds_id *<ID>*
SDS ID

--sds_name *<NAME>*
SDS name

--sds_ip *<IP>*
SDS IP address

--sds_port *<PORT>*
Port assigned to the SDS

Example

```
scli --disable_rfcache --protection_domain_name pd02
```

enable_rfcache

Enable RFcache on a specific SDS, a Storage Pool, or in an entire Protection Domain. The RFcache device must be associated with an Acceleration Pool defined specifically for the RFcache feature (or was created with the `--used_for_rfcache` flag).

Syntax

```
scli --enable_rfcache ((--protection_domain_id <ID> | --protection_domain_name <NAME>)
| (((--protection_domain_id <ID> | --protection_domain_name <NAME>) --storage_pool_name
<NAME>) | --storage_pool_id <ID>) | (--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP>
[--sds_port <PORT>]))
```

Parameters

`--protection_domain_id <ID>`
Protection Domain ID

`--protection_domain_name <NAME>`
Protection Domain name

`--storage_pool_name <NAME>`
Storage Pool name

`--storage_pool_id <ID>`
Storage Pool ID

`--sds_id <ID>`
SDS ID

`--sds_name <NAME>`
SDS name

`--sds_ip <IP>`
SDS IP address

`--sds_port <PORT>`
Port assigned to the SDS

Example

```
scli --enable_rfcache --protection_domain_name pd25 --sds_name sds25_10
```

set_rfcache_parameters

Set RFcache parameters per Protection Domain.

Syntax

```
scli --set_rfcache_parameters (--protection_domain_id <ID> | --protection_domain_name
<NAME>)
  [--page_size_kb <SIZE>]
  [--max_io_size_kb <SIZE>]
  [--rfcache_pass_through_mode <MODE>]
```

Parameters

NOTE: Adjusting these parameters can significantly impact system performance. It is recommended to use the defaults.

--protection_domain_id *<ID>*

Protection Domain ID

--protection_domain_name *<NAME>*

Protection Domain name

--page_size_kb *<SIZE>*

Cache page size in KB. Valid range: 4-64. (Default: 64)

--max_io_size_kb *<SIZE>*

Cache maximum I/O size in KB. Valid range: 32-256. (Default: 128)

--rfcache_pass_through_mode *<MODE>*

Pass-through mode. One of:

- pass_through_none
- pass_through_read
- pass_through_write
- pass_through_read_and_write
- pass_through_write_miss (default)

Example

```
scli --set_rfcache_parameters --protection_domain_name pd25 --page_size_kb 32 --
max_io_size_kb 128 --rfcache_pass_through_mode pass_through_write
```

10

SDC Commands

This section contains commands for managing SDCs.

add_sdc

Approve an SDC to map volumes and allow an SDC to access the MDM.

Syntax

```
scli --add_sdc --sdc_ip <ip> | --sdc_guid <guid>
  [--sdc_name <name>]
```

Parameters

--sdc_ip *<ip>*
SDC IP address

--sdc_guid *<guid>*
SDC GUID

--sdc_name *<name>*
Name to assign to the SDC

NOTE: You can find the SDC's GUID using of these methods:

- From the command line, run:

  ```
  scli --query_all_sdc
  ```

- View the SDC's properties in the PowerFlex GUI.
- Locally in `/opt/emc/scaleio/sdc/bin/drv_cfg`, with root access permission, run:

  ```
  drv_cfg --query_guid
  ```

Example

```
scli --add_sdc --sdc_ip 10.125.120.53 --sdc_name sdc53
```

check_sdc_authentication_status

Print the SDC authentication password status for all SDCs.

Syntax

```
scli --check_sdc_authentication_status [--run_test] [--filename <FILENAME>]
```

Parameters

--run_test

Run a connectivity test to check whether the SDCs can successfully authenticate using CHAP.

--filename *<FILENAME>*

The filename and filepath where detailed SDC authentication information will be stored. This parameter is mandatory when using the `--run_test` option.

Example

```
scli --mdm_ip 192.168.1.200 --check_sdc_authentication_status --run_test --filename
report.txt
```

generate_sdc_password

Generate a new SDC authentication password for the specified SDC.

Syntax

```
scli --generate_sdc_password (--sdc_id <ID> | --sdc_name <NAME> | --sdc_guid <GUID> |
--sdc_ip <IP>) [--reason <REASON>]
```

Parameters

--sdc_id *<ID>*

SDC ID

--sdc_name *<NAME>*

SDC name

--sdc_guid *<GUID>*

SDC GUID

--sdc_ip *<IP>*

SDC IP address

--reason *<REASON>*

Free text to describe the reason for the SDC password reset

Example

```
scli --mdm_ip 192.168.1.200 --generate_sdc_password --sdc_name SDC1 --reason
MonthlyUpdate
```

query_all_approved_sdc

Query the system when it is operating in restricted SDC mode for all SDCs that have been approved by the system.

NOTE: This query is useful when restricted SDC mode is enabled and you want to determine which SDCs can be used for volume mapping.

Syntax

```
scli  --query_all_approved_sdc
```

Parameters

None.

Example

```
scli --query_all_approved_sdc
```

query_all_sdc

Retrieve information about all SDCs in the system.

Syntax

```
scli --query_all_sdc
```

Parameters

None.

Example

```
scli --mdm_ip 192.168.1.200 --query_all_sdc
```

Sample output

```
MDM restricted SDC mode: NONE
Query all SDC returned 5 SDC nodes.
SDC ID: 30b5c18600000000 Name: N/A IP: 10.55.137.119 State: Connected GUID:
96CCE42A-6ADB-45EA-8E3F-B71C1F67808C  OS Type: LINUX Loaded Version: 3.6.0 Installed
Version: 3.6.0
    Reads:  0 IOPS 0 Bytes per second
    Writes: 0 IOPS 0 Bytes per second
SDC ID: 30b5c18700000001 Name: N/A IP: 10.55.137.114 State: Connected GUID:
C08C87B5-7E28-421A-A82C-C9C5B3070FF2  OS Type: LINUX Loaded Version: 3.6.0 Installed
Version: 3.6.0
    Reads:  0 IOPS 0 Bytes per second
    Writes: 0 IOPS 0 Bytes per second
SDC ID: 30b5c18800000002 Name: N/A IP: 10.55.137.115 State: Connected GUID:
89947FF3-6E84-4DF9-B869-481BCB8E3B3D  OS Type: LINUX Loaded Version: 3.6.0 Installed
Version: 3.6.0
    Reads:  0 IOPS 0 Bytes per second
    Writes: 0 IOPS 0 Bytes per second
SDC ID: 30b5c18900000003 Name: N/A IP: 10.55.137.116 State: Connected GUID: 7F5BC40E-
A1FB-44EC-A8B0-76C63A7B1123  OS Type: LINUX Loaded Version: 3.6.0 Installed Version:
3.6.0
    Reads:  0 IOPS 0 Bytes per second
    Writes: 0 IOPS 0 Bytes per second
```

```
SDC ID: 30b5c18a00000004 Name: N/A IP: 10.55.137.117 State: Connected GUID:
34EE9476-65D3-44C1-83DC-3CF6912CCB86  OS Type: LINUX Loaded Version: 3.6.0 Installed
Version: 3.6.0
    Reads:  0 IOPS 0 Bytes per second
    Writes: 0 IOPS 0 Bytes per second
```

query_restricted_sdc_mode

Query whether restricted SDC mode is enabled or disabled in the PowerFlex system.

If restricted SDC mode is enabled, you must register to the system each SDC to which you want to map volumes using the `--add_sdc` command.

Syntax

```
scli --query_restricted_sdc_mode
```

Parameters

None.

Example

```
scli --query_restricted_sdc_mode
```

query_sdc

Retrieve information about the specified SDC.

Syntax

```
scli --query_sdc --sdc_id <ID> | --sdc_name <NAME> | --sdc_guid <GUID> | --sdc_ip <IP>
  [--show_oscillating_failures]
```

Parameters

--sdc_id *<ID>*
SDC ID

--sdc_name *<NAME>*
SDC name

--sdc_guid *<GUID>*
SDC Global Unique Identifier

--sdc_ip *<IP>*
SDC IP address

--show_oscillating_failures
Show extended oscillating failure information if there are one or more failures.

Example

```
scli --mdm_ip 192.168.1.200 --query_sdc --sdc_id c27ab52600000001
```

Sample output

```
SDC ID: c27ab52600000001 Name: N/A IP: 10.55.137.119 State: Connected GUID:
C6533636-224B-44E3-A9EB-3CD651116CF5  OS Type: LINUX Loaded Version: 3.6.0 Installed
Version: 3.6.0
    Reads:  525 IOPS 262.4 MB (268697 KB) per-second
    Writes: 503 IOPS 251.2 MB (257222 KB) per-second

Volume information (total 2 volumes):
        1: Volume ID: f6351e3900000005 Name: vol3_pol1 Provisioning: Thin Size: 8.0 GB
(8192 MB) Access mode: READ_WRITE
        2: Volume ID: f6351e3700000003 Name: vol1_114 Provisioning: Thin Size: 16.0 GB
(16384 MB) Access mode: READ_WRITE
```

query_sdc_volume_limits

Retrieve the IOP and bandwidth limits that one SDC generates for the specified volume.

Syntax

```
scli --query_sdc_volume_limits (--volume_id <ID> | --volume_name <NAME>) (--sdc_id <ID>
| --sdc_name <NAME> | --sdc_guid <GUID> | --sdc_ip <IP>)
```

Parameters

--volume_id *<ID>*
: Volume ID

--volume_name *<NAME>*
: Volume name

--sdc_id *<ID>*
: SDC ID

--sdc_name *<NAME>*
: SDC name

--sdc_guid *<GUID>*
: SDC Global Unique Identifier

--sdc_ip *<IP>*
: SDC IP address

Example

```
scli --mdm_ip 192.168.1.200 --query_sdc_volume_limits --volume_id f6351e3700000003 --
sdc_id c27ab52600000001 --mdm_port 6611
```

Sample output

```
Successfully queried limits from SDC with ID c27ab52600000001 to volume with ID
f6351e3700000003: IOPS limit is unlimited. Bandwidth limit is unlimited.
```

remove_sdc

Remove an SDC from PowerFlex.

If your system is using Restricted SDC mode, this command also prevents an SDC's access to the MDM. (Its approved status is removed.) As a result, volumes cannot be mapped to the SDC.

If the SDC is currently connected to the MDM, the command will fail, and the connection will be maintained. The command will also fail if the SDC has volume mapping.

Syntax

```
scli --remove_sdc (--sdc_id <ID> | --sdc_name <NAME> | --sdc_guid <GUID> | --sdc_ip <IP>)
```

Parameters

--sdc_id *<ID>*

SDC ID. To obtain the ID, query the SDC using the `query_all_sdc`, or view the SDC's properties in the PowerFlex GUI.

--sdc_name *<NAME>*

SDC name

--sdc_guid *<GUID>*

SDC Global Unique Identifier. To obtain the GUID, query the SDC using the `query_all_sdc`, view the SDC's properties in the PowerFlex GUI, or run the `drv_cfg --query_guid` command locally from `/opt/emc/scaleio/sdc/bin` (root access permission required).

--sdc_ip *<IP>*

SDC IP address

Example

```
scli --remove_sdc --sdc_ip 10.125.120.53
```

rename_sdc

Assign a new name to the specified SDC.

Syntax

```
scli --rename_sdc (--sdc_id <ID> | --sdc_name <NAME> | --sdc_guid <GUID> | --sdc_ip
<IP>) --new_name <NAME>
```

Parameters

--sdc_id *<ID>*
SDC ID

--sdc_name *<NAME>*
SDC name

--sdc_guid *<GUID>*
SDC Global Unique Identifier

--sdc_ip *<IP>*
SDC IP address

--new_name *<NAME>*
New name to assign to the specified SDC

Example

```
scli --rename_sdc --sdc_id c8a300bd00000000 --new_name sdc-flash-21
```

reset_sdc_sds_connections

Reset SDC-SDS connections.

Syntax

```
scli --reset_sdc_sds_connections --i_am_sure
```

Parameters

--i_am_sure
Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

Example

```
scli --mdm_ip 192.168.1.200 --reset_sdc_sds_connections --i_am_sure
```

set_restricted_sdc_mode

Enable or disable restricted SDC mode. When enabled, SDCs must be approved (by GUID or IP address) before volumes can be mapped to them.

NOTE: In a system that is upgraded and already has volumes mapped to SDCs, if you want to enable restricted SDC mode, you must first approve the SDCs and only then enable restricted SDC mode. Use the `add_sdc` command to approve SDCs. After approving SDCs, add all the SDCs to the approved list before enabling the restricted mode.

Syntax

```
scli --set_restricted_sdc_mode --restricted_sdc_mode {approved_ip | guid | none}
```

Parameters

--restricted_sdc_mode {approved_ip | guid | none

Set the restricted SDC mode to one of the following:

- `approved_ip` — Volumes can be mapped only to approved SDCs that are also approved by IP address.
- `guid` — Volumes can be mapped only to approved SDCs.
- `none` — Restricted SDC mode is disabled and volumes can be mapped to any SDC.

Example

```
scli --add_sdc --sdc_ip approved_ip | guid | none
```

```
scli --set_approved_sdc_ips --sdc_id --sdc_ips approved_ip | guid | none
```

```
scli --set_restricted_sdc_mode --restricted_sdc_mode approved_ip | guid | none
```

set_sdc_authentication

Enable or disable SDC authentication. This feature ensures security by enforcing password-based authentication of the SDC to the MDM for access to mapped volumes. This prevents the SDC from accessing unauthorized volumes.

Syntax

```
scli --set_sdc_authentication (--enable | --disable) [--allow_old_sdc] [--
allow_no_sdc_password]
```

Parameters

--enable

Enable SDC authentication and authorization

--disable

Disable SDC authentication and authorization

--allow_old_sdc

Allow enabling SDC authentication and authorization when there is an old SDC. All old SDCs will be disconnected

--allow_no_sdc_password

Allow enabling SDC authentication and authorization when there is an SDC without a password. All SDCs without a password will be disconnected.

Example

```
scli --mdm_ip 192.168.1.200 --set_sdc_authentication --enable --allow_old_sdc
```

set_sdc_volume_limits

Set limits to the IOPS and bandwidth that one SDC generates for the specified volume. This enables you to control the quality of service (QoS).

Syntax

```
scli --set_sdc_volume_limits (--volume_id <ID> | --volume_name <NAME>) (--sdc_id <ID> |
--sdc_name <NAME> | --sdc_guid <GUID> | --sdc_ip <IP>) (Options)
  [--i_am_sure]
```

Parameters

--volume_id *<ID>*

Volume ID

--volume_name *<NAME>*

Volume name

--sdc_id *<ID>*

SDC ID

--sdc_name *<NAME>*

SDC name

--sdc_guid *<GUID>*

SDC Global Unique Identifier

--sdc_ip *<IP>*

SDC IP address

Options

Choose at least one:

--limit_iops *<NUMBER>*

Limit the volume IOPS. The number of IOPS must be larger than 10. 0 is unlimited.

--limit_bandwidth *<NUMBER>*

Limits the volume network bandwidth. The bandwidth is in MB/s. 0 is unlimited

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

Example

```
scli --mdm_ip 192.168.1.200 --set_sdc_volume_limit
--volume_name vol1 --sdc_ip 192.168.1.3 --limit_iops 100
```

set_approved_sdc_ips

Set a list of up to four IP addresses that are approved for the given SDC.

Syntax

```
scli --set_approved_sdc_ips (--sdc_id <ID> | --sdc_name <NAME> | --sdc_guid <GUID> | --
sdc_ip <IP>) --sdc_ips <IPs>
```

Parameters

--sdc_id *<ID>*

SDC ID

--sdc_name *<NAME>*

SDC name

--sdc_guid *<GUID>*

SDC GUID

--sdc_ip *<IP>*

IP address of an SDC

--sdc_ips *<IPS>*

A comma-separated list of SDC IP addresses. Omit the space after each comma.

Example

```
scli --set_approved_sdc_ips --sdc_ips 10.76.60.10,10.76.60.11,10.76.60.12,10.76.60.13
```

11

SDS Commands

This section contains commands for managing SDSs.

abort_remove_sds

Abort the removal of an SDS from a PowerFlex system. If it is performed at too late a stage in the removal process, the command may fail and the SDS will be removed.

Syntax

```
scli --abort_remove_sds (--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP> [--sds_port
<PORT>])
```

Parameters

--sds_id *<ID>*
: The ID of the SDS whose removal you want to stop

--sds_name *<NAME>*
: The name of the SDS whose removal you want to stop

--sds_ip *<IP>*
: The IP address of the SDS whose removal you want to stop

--sds_port *<PORT>*
: The port associated with the SDS

Example

```
scli --mdm_ip 192.168.1.200 --abort_remove_sds --sds_ip 192.168.1.4
```

add_sds

Add an SDS to a Protection Domain.

This command creates an SDS component and can be used to apply many configuration options at the time of creation. It informs the SDS about which devices to use, and for what purpose to use them (storage or acceleration). By default, the command also performs tests on the devices it adds, and saves the results, except for RFcache SSD acceleration devices, which are not tested.

NOTE: During execution of the command, PowerFlex checks that the devices are clear before adding them. If a device is not clear, an error message is returned, and the command fails for that device. If you would like to overwrite existing data on the devices by forcing the command, use the `force_device_takeover` flag. Note that this flag destroys all existing data on the device!

Syntax

```
scli --add_sds --sds_ip <IP> [--sds_ip_role <ROLE> {sdc_only | sds_only
| all}] (--protection_domain_id <ID> | --protection_domain_name <NAME>) [(--
storage_pool_name <NAME> | --storage_pool_id <ID>) --device_path <PATHS> [--
device_name <NAMES>] [--media_type <TYPES> {HDD | SSD}]] [(--acceleration_pool_name
<NAME> | --acceleration_pool_id <ID>) --acceleration_device_path <PATHS> [--
acceleration_device_name <NAMES>]] [--sds_name <NAME>] [--fault_set_id <ID> | --
fault_set_name <NAME>]
   [--rmcache_size_mb <SIZE>]
   [{--enable_rmcache | --disable_rmcache}]
   [--sds_port <PORT>]
   [--force_clean]
   [--force_device_takeover]
   [--test_time <TIME>]
   [{--test_only | --no_test | <BLANK>}]
   [--profile <PROFILE>]
   [--i_am_sure]
```

Parameters

--sds_ip *<IP>*

Comma-separated list of one or more IP addresses associated with the SDS over which the data will be transferred. See "SDS IP addresses and roles" below. Omit the space after each comma.

--sds_ip_role {sdc_only | sds_only | all}

Comma-separated list of IP roles associated with the SDS IP addresses. Omit the space after each comma. See "SDS IP addresses and roles" below.

--protection_domain_id *<ID>*

Protection Domain ID

--protection_domain_name *<NAME>*

Protection Domain name

--storage_pool_name *<NAME>*

Comma-separated list of Storage Pool names to be associated with the storage devices, respectively. Omit the space after each comma. If only one Storage Pool is specified in the list, all devices are associated with it. Otherwise, each device is associated with the Storage Pool situated in the same position in the list.

--storage_pool_id *<ID>*

Comma-separated list of Storage Pool IDs to be associated with the storage devices, respectively. Omit the space after each comma. If only one Storage Pool is specified in the list, all devices are associated with it. Otherwise, each device is associated with the Storage Pool situated in the same position in the list.

--device_path *<PATHS>*

Comma-separated list of the full path to each devices to be added. Omit the space after each comma. A device can be a disk, an unmounted partition, or a file that represents free space on a mounted device. To force device take-over, use this command with the `force_device_takeover` flag, proceeding with caution.

--device_name *<NAMES>*

Comma-separated list of device names. Omit the space after each comma. When using this flag to assign names to the SDS devices. The number of names must equal the number of devices, and the names will be appropriated to the devices in the same order as they appear in the device_path list.

--media_type {HDD | SSD}

If one media type is specified in the list, all devices are assigned with this type. Otherwise, each device is assigned the media type located in the same position as the list.

--acceleration_pool_name *<NAME>*

Comma-separated list of Acceleration Pool names to be associated with the acceleration devices. Omit the space after each comma. If one Acceleration Pool is specified, all acceleration devices are assigned to it. Otherwise, each device is assigned to the Acceleration Pool located in the same position in the list.

--acceleration_pool_id *<ID>*

Comma-separated list of Acceleration Pool IDs to be associated with the acceleration devices. Omit the space after each comma. If one Acceleration Pool is specified, all acceleration devices are assigned to it. Otherwise, each device is assigned to the Acceleration Pool located in the same position in the list.

--acceleration_device_path *<PATHS>*

Comma-separated list of SDS acceleration device paths. Omit the space after each comma. A device can be a disk, an unmounted partition, or a file that represents free space on a mounted device. To force device take-over, use the command with the `force_device_takeover` flag, proceeding with caution.

--acceleration_device_name *<NAMES>*

Comma-separated list of acceleration device names. Omit the space after each comma.

--sds_name *<NAME>*

Name assigned to the SDS

--fault_set_id *<ID>*

Fault Set ID

--fault_set_name *<NAME>*

Fault Set name

--rmcache_size_mb *<SIZE>*

Size of the SDS RAM cache in MB. Range: 128 MB to 300GB

--enable_rmcache | --disable_rmcache

Enable/disable RAM cache on the SDS

--sds_port *<PORT>*

Port associated with the SDS. Default: 7072

--force_clean

Clean a previous SDS configuration. Use this if the SDS was previously part of a PowerFlex system.

--force_device_takeover

Force adding the devices, ignoring data from other SDSs that are stored on the devices.

WARNING: Use this flag with caution. All data saved on the devices will be destroyed.

--test_time *<TIME>*

The maximum test run-time in seconds. Default: 10. The test will stop when it reaches either this limit, or the time it takes to complete 128 MB of data read/write, whichever is first.

Test options (choose one):

By default, PowerFlex tests the performance of each device being added before before its capacity can be used. Two tests are performed on the given devices: random writes and random reads. When the tests are complete, the device capacity is added automatically to the Protection Domain storage used by the MDM. To modify this behavior, specify one of the test options:

--test_only

Devices will be tested, but not used. To start using their capacity, run the `--activate_sds_device` command.

--no_test

The device capacity will be used without any device testing.

<BLANK>

Read and write test will be run on devices before their capacity is used.

--profile {compact | high_performance}

Set the performance profile. The default is `high_performance`.

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

NOTE: If you use the `storage_pool_id` parameter in the command, you do not need to also specify the Protection Domain ID or name, and the Storage Pool name.

Example

```
scli --mdm_ip 192.168.1.200 --add_sds --sds_ip 192.168.1.9,192.168.1.29 -sds_ip_role
sds_only,sdc_only --protection_domain_name rack_1.1 --device_path /dev/cciss/c0d1p3,/dev/
cciss/c0d1p4 --device_name sds01,sds02 --sds_name sds_1
```

SDS names

Assign each SDS and SDS device a meaningful name in order to facilitate future object identification. This can be particularly helpful for SDS devices, because the defined name remains constant even if the path changes. When a name has not been defined, the system may display default system-defined names that use the SDS's first IP address.

Each name should conform to the following rules:

1. Contains fewer than 32 characters
2. Contains only alphanumeric and punctuation characters
3. Is unique within the object type

NOTE: PowerFlex objects are assigned a unique ID that can be used to identify the object in CLI commands. You can retrieve the ID via a query or through the object's property sheet in the PowerFlex GUI.

SDS IP addresses and roles

Configure up to eight IP addresses per SDS. Define each one with one of the following roles:

- sdc_only: Communications are enabled only between this SDS and SDCs. The MDM will not be aware of this IP; therefore, internal error-checking will not be performed. If an incorrect IP address is supplied, throughput may be affected.
- sds_only: Communications are enabled only between this SDS and other SDS and MDM components.
- all (default): Communications are enabled with all components:

You can define a role for every IP address, or use the default for all of them. A situation where all IP addresses are either sdc_only or sds_only is not valid. Each SDS must have one of the following:

- At least one IP address with the role all

  or

- Two IP addresses with the roles sdc_only and sds_only

IP roles can be configured when adding an SDS, by adding a new IP address to an SDS, or by editing a current IP role.

add_sds_ip

Add an IP address to an existing SDS.

Syntax

```
scli --add_sds_ip (--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP> [--sds_port
<PORT>]) --new_sds_ip <IP>
  [--sds_ip_role {sdc_only | sds_only | all}] [--force_add_ip]
```

Parameters

`--sds_id <ID>`

SDS ID

--sds_name *<NAME>*

SDS name

--sds_ip *<IP>*

SDS IP address

--sds_port *<PORT>*

Port number on the server to be associated with the SDS

--new_sds_ip *<IP>*

New IP address to be associated with the SDS

--sds_ip_role {sdc_only | sds_only | all}

The IP role assigned to each SDS IP address: SDC only, SDS only, or all (the default). When adding multiple IP addresses, use a comma-separated list for IP roles, with no space after the commas.

--force_add_ip

Force the addition of the new IP address even if it could not be verified.

Example

```
scli --mdm_ip 192.168.1.200 --add_sds_ip --sds_ip 192.168.1.4 --new_sds_ip 192.168.1.24
```

disable_sds_rmcache

Disable Read RAM Cache on the specified SDS.

Syntax

```
scli --disable_sds_rmcache ((--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP> [--
sds_port <PORT>]) | (--protection_domain_id <ID> | --protection_domain_name <NAME>))
  [--i_am_sure]
```

Parameters

--sds_id *<ID>*

SDS ID

--sds_name *<NAME>*

SDS name

--sds_ip *<IP>*

SDS IP address

--sds_port *<PORT>*

Port assigned to the SDS

--protection_domain_id *<ID>*

Protection Domain ID

--protection_domain_name *<NAME>*

Protection Domain name

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

Example

```
scli --disable_sds_rmcache --sds_name sds23 --protection_domain_name pd01
```

NOTE: To disable the Read RAM Cache for all SDSs in a Protection Domain using a single command, add the relevant Protection Domain name or ID to the `--disable_sds_rmcache` command. For example:

```
scli --disable_sds_rmcache --sds_ip 10.100.5.25 --protection_domain pd25
```

enable_sds_rmcache

Enable Read RAM Cache on the specified SDS.

For a read to be stored in the RAM of a specific SDS, the Read RAM Cache feature on that SDS must be enabled, and the relevant Storage Pool and the relevant volume must both be configured to use Read RAM Cache. Caching will only begin after one or more devices are added to the SDS.

Syntax

```
scli --enable_sds_rmcache ((--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP> [--
sds_port <PORT>]) | (--protection_domain_id <ID> | --protection_domain_name <NAME>))
  [--i_am_sure]
```

Parameters

`--sds_id <ID>`
SDS ID

`--sds_name <NAME>`
SDS name

`--sds_ip <IP>`
SDS IP address

`--sds_port <PORT>`
Port assigned to the SDS

`--protection_domain_id <ID>`
Protection Domain ID

`--protection_domain_name <NAME>`
Protection Domain name

`--i_am_sure`
Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

Example

```
scli --enable_sds_rmcache --sds_ip 10.100.5.25 --protection_domain_name pd01
```

NOTE: To enable Read RAM Cache for all SDSs in a Protection Domain using a single command, add the relevant Protection Domain name or ID to the `--enable_sds_rmcache` command. For example:

```
scli --enable_sds_rmcache --sds_ip 10.100.5.25 --protection_domain pd25
```

enter_maintenance_mode

Place one or more SDSs in maintenance mode.

Syntax

```
scli --enter_maintenance_mode ((--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP>
[--sds_port <PORT>]) | (--fault_set_id <ID> | ((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --fault_set_name <NAME>)))
  [--force_insufficient_spare_or_free_space]
```

Parameters

--sds_id *<ID>*
SDS ID

--sds_name *<NAME>*
SDS name

--sds_ip *<IP>*
SDS IP address

--sds_port *<PORT>*
Port assigned to the SDS

--fault_set_id *<ID>*
Fault Set ID

--protection_domain_id *<ID>*
Protection Domain ID

--protection_domain_name *<NAME>*
Protection Domain name

--fault_set_name *<NAME>*
Fault Set name

--force_insufficient_spare_or_free_space
Allow entry into maintenance mode, even without enough available capacity.

Example

```
scli --enter_maintenance_mode --sds_name sds10 --force_insufficient_spare_or_free_space
```

exit_maintenance_mode

Cancel maintenance mode for one or more SDSs.

NOTE: The system displays an error message if you attempt to remove an SDS in Maintenance Mode, ("The task failed since there is an SDS in the Protection Domain currently in Maintenance Mode"). However, you can force an SDS out of Maintenance Mode by using one of the flags, and then removing the SDS.

Syntax

```
scli --exit_maintenance_mode ((--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP>
[--sds_port <PORT>]) | (--fault_set_id <ID> | ((--protection_domain_id <ID> | --
```

```
protection_domain_name <NAME>) --fault_set_name <NAME>)))
  [--force_failed_device_id <IDs>]
  [--force_failed_sds]
```

Parameters

--sds_id *<ID>*
SDS ID

--sds_name *<NAME>*
SDS name

--sds_ip *<IP>*
SDS IP address

--sds_port *<PORT>*
Port assigned to the SDS

--fault_set_id *<ID>*
Fault Set ID

--protection_domain_id *<ID>*
Protection Domain ID

--protection_domain_name *<NAME>*
Protection Domain name

--fault_set_name *<NAME>*
Fault Set name

--force_failed_device_id *<IDs>*
Allow exit from maintenance mode, even when the specified devices are in an error state.

--force_failed_sds
Allow exit from maintenance mode, even when an SDS is down.

NOTE: If you attempt to remove an SDS in maintenance mode, the system displays an error message: `The task failed since there is an SDS in the Protection Domain currently in Maintenance Mode`. Using the `force_failed_device_id` or `force_failed_sds` flag forces the SDS out of maintenance mode.

Example

```
scli --exit_maintenance_mode --sds_ip 10.76.1.10 --sds_port 7072
```

enter_protected_maintenance_mode

When entering Protected Maintenance Mode (PMM), a third copy of the SDS is created, which ensures that if there is a node failure where the second copy of SDS is required, there is still a full backup of the SDS. Move one or more SDSs to protected maintenance mode.

Syntax

```
scli --enter_protected_maintenance_mode ((--sds_id <ID> | --sds_name <NAME> | --sds_ip
<IP> [--sds_port <PORT>]) | (--fault_set_id <ID> | ((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --fault_set_name <NAME>)))
```

Parameters

None.

--sds_id *<ID>*
SDS ID

--sds_name *<NAME>*
SDS name

--sds_ip *<IP>*
SDS IP address

--sds_port *<PORT>*
Port assigned to the SDS

--protection_domain_id *<ID>*
Protection Domain ID

--protection_domain_name *<NAME>*
Protection Domain name

--fault_set_id *<ID>*
Fault Set ID

--fault_set_name *<NAME>*
Fault Set name

Example

```
scli --enter_protected_maintenance_mode --mdm_ip 192.168.1.200 --sds_name sds120
```

exit_protected_maintenance_mode

Move one or more SDSs out of protected maintenance mode.

Syntax

```
scli --exit_protected_maintenance_mode ((--sds_id <ID> | --sds_name <NAME> | --sds_ip
<IP> [--sds_port <PORT>]) | (--fault_set_id <ID> | ((--protection_domain_id <ID> |
--protection_domain_name <NAME>) --fault_set_name <NAME>))) [--force_failed_device_id
<IDs>] [--force_failed_sds] [--abort_entering_maintenance_mode]
```

Parameters

None.

--sds_id *<ID>*
SDS ID

--sds_name *<NAME>*
SDS name

--sds_ip *<IP>*
SDS IP address

--sds_port *<PORT>*
Port assigned to the SDS

--protection_domain_id *<ID>*

Protection Domain ID

--protection_domain_name *<NAME>*

Protection Domain name

--fault_set_id *<ID>*

Fault Set ID

--fault_set_name *<NAME>*

Fault Set name

--force_failed_device_id *<IDs>*

Allow exiting protected maintenance mode when the specified devices are in an error state.

--force_failed_sds

Allow exiting protected maintenance mode when the SDS is down.

--abort_entering_maintenance_mode

Acknowledge in case the process of entering protected maintenance mode is not yet complete.

Example

```
scli --exit_protected_maintenance_mode --mdm_ip 192.168.1.200 --sds_name sds120 --
force_failed_sds
```

modify_sds_ip_role

Modify the role of an SDS's IP address. The IP address can be used for SDC only, SDS only, or both.

Syntax

```
scli --modify_sds_ip_role (--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP> [--sds_port
<PORT>]) [--sds_ip_to_modify <IP>] --new_sds_ip_role {sdc_only | sds_only | all} [--
force_role_modification]
```

Parameters

--sds_id *<ID>*

SDS ID

--sds_name *<NAME>*

SDS name

--sds_ip *<IP>*

SDS IP address

--sds_port *<PORT>*

Port assigned to the SDS

--sds_ip_to_modify *<IP>*

IP address of the SDS being modified

--new_sds_ip_role {sdc_only | sds_only | all}

New role to assign the SDS IP: SDC only, SDS only, or all

--force_role_modification

Force the modification of the SDS's role even if the IP address could not be verified.

Example

```
scli --modify_sds_ip_role --sds_name sds01 --sds_ip_to_modify 192.168.1.4 --
new_sds_ip_role all
```

modify_sds_port

Modify the port used by the SDS for communication purposes.

Syntax

```
scli --modify_sds_port (--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP> [--sds_port
<PORT>]) --new_sds_port <PORT>
```

Parameters

--sds_id *<ID>*

SDS ID

--sds_name *<NAME>*

SDS name

--sds_ip *<IP>*

SDS IP address

--sds_port *<PORT>*

Port associated with the SDS

--new_sds_port *<PORT>*

New port to be associated with the SDS

Example

```
scli --mdm_ip 192.168.1.200 --modify_sds_port --sds_ip 192.168.1.4 --new_sds_port 6801
```

query_all_sds

Retrieve information about all SDSs in the system.

Syntax

```
scli --query_all_sds
```

Parameters

None.

Example

```
scli --mdm_ip 192.168.1.200 --query_all_sds
```

Sample output

```
Query-all-SDS returned 5 SDS nodes.

Protection Domain f11d784d00000000 Name: pd1
SDS ID: 7a3cf71e00000004 Name: sds_idd_1 State: Connected, Joined IP: 10.55.137.114
Port: 7072 Version: 3.6.0
SDS ID: 7a3cf71c00000003 Name: sds_idd_2 State: Connected, Joined IP: 10.55.137.115
Port: 7072 Version: 3.6.0
SDS ID: 7a3cf71d00000002 Name: sds_idd_4 State: Connected, Joined IP: 10.55.137.116
Port: 7072 Version: 3.6.0
SDS ID: 7a3cf71b00000001 Name: sds_idd_5 State: Connected, Joined IP: 10.55.137.117
Port: 7072 Version: 3.6.0
Protection Domain f11d784e00000001 Name: pd2
SDS ID: 7a3cf71a00000000 Name: sds_idd_3 State: Connected, Joined IP: 10.55.137.119
Port: 7072 Version: 3.6.0
Protection Domain f11d9f5d00000002 Name: dfg
```

query_sds_connectivity_status

Retrieve information regarding the connectivity status of the SDSs in the specified Protection Domain.

Syntax

```
scli --query_sds_connectivity_status (--protection_domain_id <ID> | --
protection_domain_name <NAME>)
```

Parameters

--protection_domain_id *<ID>*

Protection Domain ID

--protection_domain_name *<NAME>*

Protection Domain name

Example

```
scli --query_sds_connectivity_status --protection_domain_name pd_18
```

Sample output

```
Successfully queried protection domain pd1
Got responses from 4 SDSs:
        ID: 06b108a100000001 <===> ID: 06b108a000000000  Connected Count 4
                10.55.137.117  : Connected
        ID: 06b108a100000001 <===> ID: 06b108a100000001  Connected Count 8
                10.55.137.116  : Connected
        ID: 06b108a300000003 <===> ID: 06b108a000000000  Connected Count 4
                10.55.137.117  : Connected
```

```
        ID: 06b108a300000003 <===> ID: 06b108a100000001  Connected Count 4
            10.55.137.116   : Connected
        ID: 06b108a300000003 <===> ID: 06b108a300000003  Connected Count 8
            10.55.137.115   : Connected
        ID: 06b108a300000003 <===> ID: 06b108a200000002  Connected Count 4
            10.55.137.119   : Connected
        ID: 06b108a000000000 <===> ID: 06b108a000000000  Connected Count 8
            10.55.137.117   : Connected
        ID: 06b108a400000004 <===> ID: 06b108a000000000  Connected Count 4
            10.55.137.117   : Connected
        ID: 06b108a400000004 <===> ID: 06b108a100000001  Connected Count 4
            10.55.137.116   : Connected
        ID: 06b108a400000004 <===> ID: 06b108a300000003  Connected Count 4
            10.55.137.115   : Connected
        ID: 06b108a400000004 <===> ID: 06b108a400000004  Connected Count 8
            10.55.137.114   : Connected
        ID: 06b108a400000004 <===> ID: 06b108a200000002  Connected Count 4
            10.55.137.119   : Connected
```

query_sds

Retrieve detailed information about the specified SDS.

Syntax

```
scli --query_sds (--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP> [--sds_port <PORT>])
[--show_oscillating_failures]
```

Parameters

--sds_id *<ID>*
SDS ID

--sds_name *<NAME>*
SDS name

--sds_ip *<IP>*
SDS IP address

--sds_port *<PORT>*
Port associated with the SDS

--show_oscillating_failures
Show extended oscillating failure information, if such failures have occurred

Example

```
scli --mdm_ip 192.168.1.200 --query_sds --sds_id 06b108a400000004
```

Sample output

```
SDS 06b108a400000004 Name: sds_idd_1 Version: 3.6.0
Protection Domain: fcc9af5500000000, Name: pd1
Fault Set: 36c9d76800000000, Name: fs1_mg
Medium granularity DRL mode: Volatile
Authentication error: NONE
IP information (total 1 IP Addresses):
         1: 10.55.137.114    Role: All (SDS and SDC)
```

```
        Port: 7072
I/O flow control is enabled
Fine granularity number of concurrent writes: 1000
RAM Read Cache information:
        128.0 MB (131072 KB) total size
        Cache is enabled
        RAM Read Cache memory allocation state is pending
FGL metadata cache information:
        FGL metadata cache is disabled
        FGL metadata cache size: 0 MB
RFcache enabled

Storage Pool information (total of 1 Storage Pools):
         1: Storage Pool: sp1  inflight requests factor: 115, inflight bandwidth factor
115

Device information (total 2 devices):
         1: Name: N/A  Path: /dev/sdb  Original-path: /dev/sdb  ID: c8bea95800040000
                Storage Pool: sp1, Capacity: 148 GB, State: Normal
                Scanned 936040 MB, Unresolved Read errors: 0/0, Unresolved Compare
errors: 0/0
         2: Name: N/A  Path: /dev/sdc  Original-path: /dev/sdc  ID: c8bea95900040001
                Storage Pool: sp1, Capacity: 148 GB, State: Normal
                Scanned 936134 MB, Unresolved Read errors: 0/0, Unresolved Compare
errors: 0/0

Acceleration device information (total 0 devices):

Membership-state: Joined, Connection-state: Connected, SDS-state: Normal, 0 devices with
error, 0 RFcache devices with errors

Physical data:
        297.9 GB (305000 MB) total capacity
        280.5 GB (287184 MB) unused capacity
        17.4 GB (17816 MB) in-use capacity
        0 Bytes unreachable-unused capacity
        0 active moving-in forward-rebuild jobs
        0 active moving-in backward-rebuild jobs
        0 active moving-in rebalance jobs
        0 active moving-in enter protected maintenance mode jobs
        0 active moving-in exit protected maintenance mode jobs
        0 active moving-out forward-rebuild jobs
        0 active moving-out backward-rebuild jobs
        0 active moving-out enter protected maintenance mode jobs
        0 active moving-out exit protected maintenance mode jobs
        0 active moving-out rebalance jobs
        0 pending moving-in forward-rebuild jobs
        0 pending moving-in backward-rebuild jobs
        0 pending moving-in rebalance jobs
        0 pending moving-in enter protected maintenance mode jobs
        0 pending moving-in exit protected maintenance mode jobs
        0 pending moving-out forward-rebuild jobs
        0 pending moving-out backward-rebuild jobs
        0 pending moving-out rebalance jobs
        0 pending moving-out enter protected maintenance mode jobs
        0 pending moving-out exit protected maintenance mode jobs
        0 Bytes decreased capacity

Bandwidth:
        Primary-reads                             114 IOPS 81.2 MB (83148 KB) per-second
        Primary-writes                            150 IOPS 75.6 MB (77450 KB) per-second
        Secondary-reads                           0 IOPS 0 Bytes per second
        Secondary-writes                          142 IOPS 69.8 MB (71516 KB) per-second
        Backward-rebuild-reads                    0 IOPS 0 Bytes per second
        Backward-rebuild-writes                   0 IOPS 0 Bytes per second
        Forward-rebuild-reads                     0 IOPS 0 Bytes per second
        Forward-rebuild-writes                    0 IOPS 0 Bytes per second
        Rebalance-reads                           0 IOPS 0 Bytes per second
        Rebalance-writes                          0 IOPS 0 Bytes per second
        Volume-migration-reads                    0 IOPS 0 Bytes per second
        Volume-migration-writes                   0 IOPS 0 Bytes per second
        Enter-protected-maintenance-mode-reads    0 IOPS 0 Bytes per second
        Enter-protected-maintenance-mode-writes   0 IOPS 0 Bytes per second
```

```
        Exit-protected-maintenance-mode-reads    0 IOPS 0 Bytes per second
        Exit-protected-maintenance-mode-writes   0 IOPS 0 Bytes per second
```

query_sds_connectivity_status

Retrieve information regarding the connectivity status of the SDSs in the specified Protection Domain.

Syntax

```
scli --query_sds_connectivity_status (--protection_domain_id <ID> | --
protection_domain_name <NAME>)
```

Parameters

--protection_domain_id *<ID>*

Protection Domain ID

--protection_domain_name *<NAME>*

Protection Domain name

Example

```
scli --query_sds_connectivity_status --protection_domain_name pd_18
```

Sample output

```
Successfully queried protection domain pd1
Got responses from 4 SDSs:
        ID: 06b108a100000001 <===> ID: 06b108a000000000  Connected Count 4
                10.55.137.117  : Connected
        ID: 06b108a100000001 <===> ID: 06b108a100000001  Connected Count 8
                10.55.137.116  : Connected
        ID: 06b108a300000003 <===> ID: 06b108a000000000  Connected Count 4
                10.55.137.117  : Connected
        ID: 06b108a300000003 <===> ID: 06b108a100000001  Connected Count 4
                10.55.137.116  : Connected
        ID: 06b108a300000003 <===> ID: 06b108a300000003  Connected Count 8
                10.55.137.115  : Connected
        ID: 06b108a300000003 <===> ID: 06b108a200000002  Connected Count 4
                10.55.137.119  : Connected
        ID: 06b108a000000000 <===> ID: 06b108a000000000  Connected Count 8
                10.55.137.117  : Connected
        ID: 06b108a400000004 <===> ID: 06b108a000000000  Connected Count 4
                10.55.137.117  : Connected
        ID: 06b108a400000004 <===> ID: 06b108a100000001  Connected Count 4
                10.55.137.116  : Connected
        ID: 06b108a400000004 <===> ID: 06b108a300000003  Connected Count 4
                10.55.137.115  : Connected
        ID: 06b108a400000004 <===> ID: 06b108a400000004  Connected Count 8
                10.55.137.114  : Connected
        ID: 06b108a400000004 <===> ID: 06b108a200000002  Connected Count 4
                10.55.137.119  : Connected
```

query_sds_network_test_results

Retrieve the results of the last network test performed between an SDS and its peers.

Syntax

```
scli --query_sds_network_test_results (--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP>
[--sds_port <PORT>])
```

Parameters

--sds_id *<ID>*
SDS ID

--sds_name *<NAME>*
SDS name

--sds_ip *<IP>*
SDS IP address

--sds_port *<PORT>*
Port associated with the SDS

Example

```
scli --mdm_ip 192.168.1.200 --query_sds_network_test_results --sds_ip 192.168.1.4
```

query_sdc_to_sds_disconnections

Retrieve details on any disconnections between SDCs and SDSs currently in the system.

Syntax

```
scli --query_sdc_to_sds_disconnections
```

Parameters

None.

Example

```
scli --query_sdc_to_sds_disconnections
```

remove_sds

Initiate the removal of an SDS. The command removes an SDS object and disconnects the SDS from the MDM.

You can remove an SDS at any time, with no downtime required. During execution of this command, the associated data is replicated to different nodes. Therefore, the removal process is asynchronous and might take a long time.

NOTE: If the capacity of this SDS is still used by volumes, and the capacity cannot be replaced due to lack of available free space, the command will fail.

Syntax

```
scli --remove_sds (--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP> [--sds_port <PORT>])
```

Parameters

--sds_id *<ID>*
ID of the SDS to remove

--sds_name *<NAME>*
Name of the SDS to remove

--sds_ip *<IP>*
IP address of the SDS to remove

--sds_port *<PORT>*
Port associated with the SDS

Example

```
scli --mdm_ip 192.168.1.200 --remove_sds --sds_ip 192.168.1.4
```

Related operations

The removal operation runs in the background. To track its progress, run the `query_all_sds` command periodically.

To abort the removal, run the `abort_remove_sds` command.

remove_sds_ip

Remove an IP address from an SDS.

Possible scenarios for using this command are:

- In PowerFlex setups that include an SDS installed on a server with more than one NIC (network interface card), you might want to divert the usage of one of the NICs to a different purpose.
- An IP range may have changed. For example, an SDS may be using an IP address in a certain range, but the IP range has been changed to a different one in order to free up the previously used addresses.

Syntax

```
scli --remove_sds_ip (--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP> [--sds_port
<PORT>]) --sds_ip_to_remove <IP>
```

Parameters

--sds_id *<ID>*
SDS ID

--sds_name *<NAME>*
SDS name

--sds_ip *<IP>*
SDS IP address

--sds_port *<PORT>*
Port associated with the SDS

--sds_ip_to_remove *<IP>*
IP address to remove from the SDS

Example

```
scli --mdm_ip 192.168.1.200 --remove_sds_ip
 --sds_ip 192.168.1.4 --sds_ip_to_remove 192.168.1.24
```

rename_sds

Assign a new name to the specified SDS.

Syntax

```
scli --rename_sds (--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP> [--sds_port
<PORT>]) --new_name <NAME>
```

Parameters

--sds_id *<ID>*
SDS ID

--sds_name *<NAME>*
SDS name

--sds_ip *<IP>*
SDS IP address

--sds_port *<PORT>*
Port associated with the SDS

--new_name *<NAME>*
New, unique name to assign to the specified SDS

Example

```
scli --mdm_ip 192.168.1.200 --rename_sds --sds_ip 192.168.1.4 --new_name sds_new_name
```

SDS names

Assign the SDS a meaningful name in order to facilitate future object identification. The name should conform to the following rules:

1. Contains fewer than 32 characters
2. Contains only alphanumeric and punctuation characters
3. Is unique within the object type

set_drl_properties

Set the properties of the dirty-region-logging used by one or all SDSs.

NOTE: The settings apply to medium granularity layout devices only.

Syntax

```
scli --set_drl_properties ((--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP> [--
sds_port <PORT>]) | --set_all) (--volatile | --hardened)
```

Parameters

--sds_id *<ID>*
SDS ID

--sds_name *<NAME>*
SDS name

--sds_ip *<IP>*
SDS IP address

--sds_port *<PORT>*
Port associated with the SDS

--set_all
Apply the properties to all SDSs. Overrides previous settings

--volatile
Dirty-region-logging information will be stored in memory only. This is the default.

--hardened
Dirty-region-logging information will be stored both in memory and on disk.

Example

```
scli --set_drl_properties --sds_ip 192.168.1.4 --hardened
```

set_fgl_metadata_cache_size

Configure the fine granularity metadata cache size for the specified SDS.

Syntax

```
scli --set_fgl_metadata_cache_size (--sds_id <ID> | --sds_name <NAME> |
--sds_ip <IP> [--sds_port <PORT>]) (--metadata_cache_size_mb <SIZE> | --
use_protection_domain_default_size)
```

Parameters

--sds_id *<ID>*
SDS ID

--sds_name *<NAME>*
SDS name

--sds_ip *<IP>*
SDS IP address

--sds_port *<PORT>*
Port assigned to the SDS

--metadata_cache_size_mb *<SIZE>*
Define the default size of the metadata cache for the SDS in MB

--use_protection_domain_default_size
Use the default size of the Protection Domain

Example

```
scli --mdm_ip 192.168.1.200 --set_fgl_metadata_cache_size --sds_name SDS1 --
metadata_cache_size_mb 10248
```

set_sds_network_limits

Limit the network bandwidth used by all the SDSs in the specified Protection Domain for various traffic types, including I/O.

SDS nodes transfer data among themselves. This data consists of user-data being replicated as part of the RAID protection, and data copied for internal rebalancing and recovery from failures.

You can modify the balance between these types of data loads by limiting the data copy bandwidth. This change affects all SDSs in the specified Protection Domain.

WARNING: Contact EMC Support before you modify this configuration.

Syntax

```
scli --set_sds_network_limits (--protection_domain_id <ID> | --protection_domain_name
<NAME>)
[--rebuild_limit <LIMIT>] [--rebalance_limit <LIMIT>] [--volume_migration_limit <LIMIT>]
[--protected_maintenance_mode_limit <LIMIT>] [--overall_limit <LIMIT>]
  [--i_am_sure]
```

Parameters

--protection_domain_id *<ID>*
Protection Domain ID

--protection_domain_name *<NAME>*

Protection Domain name

--rebuild_limit *<LIMIT>*

Limit the network bandwidth for rebuild in MB/s. 0 is unlimited

--rebalance_limit *<LIMIT>*

Limit the network bandwidth for rebalance in MB/s. 0 is unlimited

--volume_migration_limit *<LIMIT>*

Limit the network bandwidth for V-Tree migration in MB/s. 0 is unlimited

--protected_maintenance_mode_limit *<LIMIT>*

Limit the network bandwidth for protected maintenance mode in MB/s. 0 means unlimited bandwidth.

--overall_limit *<LIMIT>*

Limit the overall network bandwidth in MB/s. This includes the rebuild, rebalance, and application I/O bandwidth. 0 is unlimited

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

Example

```
scli --mdm_ip 192.68.1.200 --set_sds_network_limits --protection_domain_name pd1 --
rebuild_limit 100
```

set_sds_rmcache_size

Set the RMcache size of the specified SDS.

By default, RMcache size is set to 128 MB in all SDSs. The amount of RAM you can allocate for cache is limited by the amount of RAM on the SDS server:

- If the RAM is less than 32 GB, 50 percent of memory can be used for cache.
- If the RAM is more than 32 GB, 75 percent of memory can be used for cache.

The maximum amount of RAM cache is 128 GB.

NOTE: In cases of NUMA (Non-Uniform Memory Access), the total RAM refers to the first node only.

Syntax

```
scli --set_sds_rmcache_size ((--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP> [--
sds_port <PORT>]) | (--protection_domain_id <ID> | --protection_domain_name <NAME>)) --
rmcache_size_mb <SIZE>
  [--i_am_sure]
```

Parameters

--sds_id *<ID>*

SDS ID

--sds_name *<NAME>*

SDS name

--sds_ip *<IP>*

SDS IP address

--sds_port *<PORT>*

Port associated with the SDS

--protection_domain_id *<ID>*

Protection Domain ID

--protection_domain_name *<NAME>*

Protection Domain name

--rmcache_size_mb *<SIZE>*

New size of RMcache in MB. Range: 128 MB-300 GB.

NOTE: For additional information on the maximum limit, see the *Getting to Know Dell PowerFlex Guide*.

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

Example

```
scli --set_sds_rmcache_size --protection_domain_name pd01 --sds_name sds10 --
rmcache_size_mb 256
```

show_certificate

Display the certificate properties for the specified SDS or SDR.

Syntax

```
scli --show_certificate ((--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP> [--sds_port
<PORT>]) | --all_sds | (--sdr_id <ID> | --sdr_name <NAME> | --sdr_ip <IP> [--sdr_port
<PORT>]) | --all_sdr | (--protection_domain_id <ID> | --protection_domain_name <NAME>))
  [--days_until_expiration <DAYS>]
```

Parameters

--sds_id *<ID>*

SDS ID

--sds_name *<NAME>*

SDS name

--sds_ip *<IP>*

SDS IP address

--sds_port *<PORT>*

Port associated with the SDS

--all_sds

Show certificates for all SDSs.

--sdr_id *<ID>*

SDR ID

--sdr_name *<NAME>*

SDR name

--sdr_ip *<IP>*

SDR IP address

--sdr_port *<PORT>*

Port associated with the SDR

--all_sdr

Show certificates for all SDRs.

--protection_domain_id *<ID>*

Protection Domain ID

--protection_domain_name *<NAME>*

Protection Domain name

--days_until_expiration *<DAYS>*

Only show certificates expiring within a given number of days.

Example

```
scli --show_certificate --all_sds --days_until_expiration 7
```

Sample output

```
Protection Domain fcc9af5500000000 Name: pd1
Protection Domain fcc9af5600000001 Name: pd2
Protection Domain fcc9d66500000002 Name: rrr
```

start_sds_network_test

Initiate a network test between an SDS and all of its peers.

Syntax

```
scli --start_sds_network_test (--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP> [--
sds_port <PORT>])
  [--parallel_messages <NUMBER>]
  [--network_test_size_gb <SIZE>]
  [--network_test_length_secs <SECONDS>]
  [--test_associated_protection_domains]
```

Parameters

--sds_id *<ID>*

SDS ID

--sds_name *<NAME>*

SDS name

--sds_ip *<IP>*

SDS IP address

--sds_port *<PORT>*

Port associated with the SDS

--parallel_messages *<NUMBER>*

Number of parallel messages sent during the test. Range is 1 - 16. Default: 4.

--network_test_size_gb *<SIZE>*

Amount of data sent between the tested SDS and every other SDS. Default size: 1 GB.

--network_test_length_secs *<SECONDS>*

Maximum amount of time spent testing the SDS, regardless of the number of SDSs in the system. Default value: 0 (unlimited).

--test_associated_protection_domains

Test associated Protection Domains. Protection Domains can be connected during V-Tree migration between different Protection Domains.

Example

```
scli --mdm_ip 192.168.1.200 --start_sds_network_test
--sds_ip 192.168.1.4 --parallel_messages 8 --network_test_size_gb 2 --
network_test_length_secs 120
```

update_sds_original_paths

Change the original path configuration of all devices to the path currently assigned to the devices in the SDS.

Syntax

```
scli --update_sds_original_paths (--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP> [--
sds_port <PORT>]) [--force_failed_devices]
```

Parameters

--sds_id *<ID>*

SDS ID

--sds_name *<NAME>*

SDS name

--sds_ip *<IP>*

SDS IP address

--sds_port *<PORT>*

SDS port

--force_failed_devices

Force path update even for devices in error

Example

```
scli --update_sds_original_paths --sds_name sds1 --force_failed_devices
```

12

Snapshot Commands

This section contains commands for managing snapshots.

add_snapshot_policy

Create a new snapshot policy to enable you to define the number of snapshots taken at a given time for one or more volumes.

Snapshots are taken of the volumes according to the rule defined in the snapshot policy. Define the time interval in between two rounds of snapshots as well as the number of snapshots to retain, in a multi-level structure. For example take snapshots every x minutes/hours/days/weeks. There are one to six levels, with the first level having the most frequent snapshots.

Syntax strings

```
scli  --add_snapshot_policy [--snapshot_policy_name <NAME>] --snapshot_creation_cadence
<CADENCE> --number_of_snapshots_per_retention_level <LEVELS> [--paused] [--
read_only_snapshots] [--secure_snapshots [--i_am_sure]]
```

Parameters

--snapshot_policy_name *<NAME>*

Snapshot policy name

--snapshot_creation_cadence *<CADENCE>*

The interval in minutes between two snapshots in the policy

--number_of_snapshots_per_retention_level *<LEVELS>*

A comma-separated list which represents the number of snapshots per retention level. There are one to six levels, and the first level has the most frequent snapshots. There can be up to 60 snapshots in all levels totalled (note that two consecutive levels overlap by one).

--paused

Indicates that the snapshot policy should paused on creation.

--read_only_snapshots

Specify all mappings of the created auto snapshots to be limited to read-only access.

--secure_snapshots

Use this flag to define all auto snapshots created as secure. It will not be possible to edit or remove them before the retention period is complete.

--i_am_sure

Preemptive approval for the creating the snapshots as secure.

Example

```
scli --add_snapshot_policy --snapshot_policy_name S_Pol_1 --snapshot_creation_cadence 4
--number_of_snapshots_per_retention_level 6,4,3,2,2,1 --read_only_snapshots
```

add_source_volume_to_snapshot_policy

Add a source volume to the snapshot policy specified by ID or name.

You can add multiple source volumes to a snapshot policy, but a specific volume can only be the source volume of a single policy. Only one volume per V-Tree may be used as a source volume of a policy.

Syntax

```
scli --add_source_volume_to_snapshot_policy (--snapshot_policy_id <ID> | --
snapshot_policy_name <NAME>) (--source_vol_id <ID> | --source_vol_name <NAME>)
```

Parameters

--snapshot_policy_id *<ID>*

Snapshot policy ID

--snapshot_policy_name *<NAME>*

Name of the snapshot policy

--source_vol_id *<ID>*

Source volume ID

--source_vol_name *<NAME>*

Source volume name

Example

```
scli --add_source_volume_to_snapshot_policy --snapshot_policy_name snap_pol_1 --
source_vol_id a7e798d100000000
```

align_snapshot_policy_to_current_time

Align the specified snapshot policy's cadence cycles to current time.

Syntax

```
scli --align_snapshot_policy_to_current_time (--snapshot_policy_id <ID> | --
snapshot_policy_name <NAME>)
```

Parameters

--snapshot_policy_id *<ID>*

Snapshot policy ID

--snapshot_policy_name *<NAME>*

Snapshot policy name

Example

```
scli --mdm_ip 192.168.1.200 --align_snapshot_policy_to_current_time --
snapshot_policy_name SNP1
```

clear_snapshot_security

Clear the snapshot security for the specified volume to enable editing or removing the volume. This command can only be run with a signature file obtained from customer support. If you need to clear the snapshot security, contact customer support to obtain the signature file.

Syntax

```
scli --clear_snapshot_security (--volume_id <ID> | --volume_name <NAME>) --
signature_file <FILE> [--i_am_sure]
```

Parameters

--volume_id *<ID>*

Volume ID

--volume_name *<NAME>*

Volume name

--signature_file *<FILE>*

Signature file provided by customer support.

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

Example

```
scli --mdm_ip 192.168.1.200 --clear_snapshot_security --volume_name vol1 --
signature_file /opt/vol-dd6e34001.sig --i_am_sure
```

lock_auto_snapshot

Lock an auto snapshot to prevent removal by the snapshot policy.

Syntax

```
scli --lock_auto_snapshot (--auto_snapshot_id <ID> | --auto_snapshot_name <NAME>
```

Parameters

--auto_snapshot_id *<ID>*

Auto snapshot ID

--auto_snapshot_name *<NAME>*

Auto snapshot name

Example

```
scli --lock_auto_snapshot --auto_snapshot_name auto_snapshot_1
```

modify_snapshot_policy

Modify the number of snapshots taken per minute and/or the number of snapshots per retention level for a specific policy

Syntax

```
scli --modify_snapshot_policy (--snapshot_policy_id <ID> | --
snapshot_policy_name <NAME>) ((--snapshot_creation_cadence <CADENCE> --
number_of_snapshots_per_retention_level <LEVELS>) | ([--set_read_only_snapshots] [--
set_secure_snapshots] [--apply_to_existing_snapshots])) [--i_am_sure]
```

Parameters

--snapshot_policy_id *<ID>*

Snapshot policy ID

--snapshot_policy_name *<NAME>*

Name of the snapshot policy

--snapshot_creation_cadence *<CADENCE>*

The interval in minutes between two snapshots in the policy

--number_of_snapshots_per_retention_level *<LEVELS>*

A comma-separated list which represents the number of snapshots per retention level. There are six levels, and the first level has the most frequent snapshots. There can be up to 60 snapshots total in all levels (note that two consecutive levels overlap by one).

--set_read_only_snapshots

All mappings of the created auto snapshots will be limited to read-only access

--set_secure_snapshots

The auto snapshots will be created as secure. They cannot be edited or removed prior to their policy expiration time

--apply_to_existing_snapshots

Applies the `--set_read_only_snapshots` and `--set_secure_snapshots` flags to the policy's existing auto snapshots (excluding locked snapshots)

--i_am_sure

Skip the safety questions for command execution.

Example

```
scli --modify_snapshot_policy --snapshot_policy_name SP_pol1 --snapshot_creation_cadence
2 --numer of snapshots_per_retention_level 5 --set_read_only_snapshots --i_am_sure
```

pause_snapshot_policy

Pause a snapshot policy so that no auto snapshots will be created or removed.

Syntax

```
scli --pause_snapshot_policy (--snapshot_policy_id <ID> | --snapshot_policy_name <NAME>
```

Parameters

--snapshot_policy_id *<ID>*
Snapshot policy ID

--snapshot_policy_name *<NAME>*
Name of the snapshot policy

Example

```
scli --pause_snapshot_policy --snapshot_policy_name snap_pol_1
```

query_all_snapshot_policies

Retrieve information about all snapshot policies.

Syntax

```
scli --query_all_snapshot_policies
```

Parameters

None.

Example

```
scli --query_all_snapshot_policies
```

Sample output

```
Query all snapshot policies returned 3 policies
Snapshot Policy policy1 (Id: 36a39c6200000000)
        Cadence: 30 minutes
        Number of snapshots per retention level: 1 (read-only)
        State: Active
        Time of last auto snapshot: N/A
        Next run time: 2020-11-12 09:56:17
        Number of source volumes: 0
        Number of auto snapshots: 0
```

```
Snapshot Policy policy2_secure (Id: 36a3c37200000001)
        Cadence: 5 minutes
        Number of snapshots per retention level: 1 (secure, read-only)
        State: Active
        Time of last auto snapshot: 2020-11-12 09:47:18
        Next run time: 2020-11-12 09:52:17
        Number of source volumes: 1
        Number of auto snapshots: 1

Snapshot Policy policy_read_write (Id: 36a3ea8200000002)
        Cadence: 5 minutes
        Number of snapshots per retention level: 1
        State: Active
        Time of last auto snapshot: 2020-11-12 09:44:18
        Next run time: 2020-11-12 09:49:17
        Number of source volumes: 1
        Number of auto snapshots: 1
```

query_snapshot_policy

Retrieve information about a snapshot policy.

Syntax

```
scli --query_snapshot_policy (--snapshot_policy_id <ID> | --snapshot_policy_name <NAME>)
[--show_auto_snapshots]
```

Parameters

--snapshot_policy_id *<ID>*
: Snapshot policy ID

--snapshot_policy_name *<NAME>*
: Name of the snapshot policy

--show_auto_snapshots
: Indicate whether to display the full list of auto snapshots

Example

```
scli --query_snapshot_policy --snapshot_policy_name policy1 --show_auto_snapshots
```

Sample output

```
Snapshot Policy policy1 (Id: aaa4278700000000)
        Cadence: 5 minutes
        Number of snapshots per retention level: 1
        State: Active
        Time of last auto snapshot: 2021-02-10 13:39:13
        Next run time: 2021-02-10 13:44:13
        Number of auto snapshots: 2

Source volume information (total 2 volumes):
         1: Volume ID: f6351e3700000003 Name: vol1_114 Provisioning: Thin Size: 16.0 GB
(16384 MB)
         2: Volume ID: f6351e3900000005 Name: vol3_pol1 Provisioning: Thin Size: 8.0 GB
(8192 MB)
```

remove_snapshot_policy

Remove the snapshot policy specified by name or ID.

Syntax

```
scli --remove_snapshot_policy (--snapshot_policy_id <ID> | --snapshot_policy_name <NAME>)
```

Parameters

--snapshot_policy_id *<ID>*

Snapshot policy ID

--snapshot_policy_name *<NAME>*

Name of the snapshot policy

Example

```
scli --remove_snapshot_policy --snapshot_policy_name SP_pol_1
```

remove_snapshots_consistency_group

Remove all snapshots that belong to a specific consistency group. Depending on the size of the consistency group, this command may take a long time.

NOTE: You cannot remove the consistency group if it contains auto snapshots.

Syntax

```
scli --remove_snapshots_consistency_group --scg_id <ID>
  [--i_am_sure]
```

Parameters

--scg_id *<ID>*

The consistency group's ID number.

NOTE: The consistency group ID is displayed when you create the consistency group. To find this ID at a later time, use the `query_all_volumes` command.

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

Example

```
scli  --remove_snapshots_consistency_group
--scg_id 783c7f00000000 --i_am_sure
```

remove_source_volume_from_snapshot_policy

Remove a source volume from the specified snapshot policy.

When removing the source volume from the policy, you must choose how to handle the snapshots created by the policy (auto snapshots). Your selection depends on if there are locked auto snapshots.

- If the source volume has no auto snapshots, you don't need to use the `--detach_auto_snapshots` or `--remove_auto_snapshots` options.
- If the source volume has auto snapshots but none of them are locked, you can choose to detach all snapshots with the `--detach_auto_snapshots` flag. They become regular snapshots as if they were created manually. If you use the `--remove_auto_snapshots`, the snapshots are deleted.
- If the source volume has locked auto snapshots, you can use `--detach_auto_snapshots` to turn the auto snapshots into regular snapshots, as if they were created manually. If you use `--remove_auto_snapshots`, the auto snapshots that are not locked are removed, while the auto snapshots that are locked are detached.

Syntax

```
scli --remove_source_volume_from_snapshot_policy (--snapshot_policy_id <ID> | --
snapshot_policy_name <NAME>) (--source_vol_id <ID> | --source_vol_name <NAME>) (--
detach_auto_snapshots | (--remove_auto_snapshots [--detach_locked_auto_snapshots]))
```

Parameters

`--snapshot_policy_id <ID>`
: Snapshot policy ID

`--snapshot_policy_name <NAME>`
: Name of the snapshot policy

`--source_vol_id <ID>`
: Source volume ID

`--source_vol_name <NAME>`
: Source volume name

`--detach_auto_snapshots`
: Detach all associated auto snapshots

`--remove_auto_snapshots`
: Remove all associated auto snapshots

`-- detach_locked_auto_snapshots`
: When the `remove_policy_snapshots` option is specified, any locked snapshots will not be removed, but rather detached.

Example

```
scli --remove_source_volume_from_snapshot_policy --snapshot_policy_name snap_pol_1 --
source_vol_id a7e798d100000000 --remove_auto_snapshots --detach_locked_auto_snapshots
```

rename_snapshot_policy

Name or rename the snapshot policy specified by name or ID.

Syntax

```
scli --rename_snapshot_policy (--snapshot_policy_id <ID> | --snapshot_policy_name
<NAME>) --new_name <NAME>
```

Parameters

--snapshot_policy_id *<ID>*
Snapshot policy ID

--snapshot_policy_name *<NAME>*
Name of the snapshot policy

--new_name *<NAME>*
New name of the snapshot policy

Example

```
scli --rename_snapshot_policy --snapshot_policy_name SP_pol_1 --new_name snap_pol_1
```

resume_snapshot_policy

Resume a paused snapshot policy so that auto snapshots will once again be created and removed.

Syntax

```
scli --resume_snapshot_policy (--snapshot_policy_id <ID> | --snapshot_policy_name <NAME>)
```

Parameters

--snapshot_policy_id *<ID>*
Snapshot policy ID

--snapshot_policy_name *<NAME>*
Name of the snapshot policy

Example

```
scli --resume_snapshot_policy --snapshot_policy_name snap_pol_1
```

set_snapshot_security

Set the specified snapshot as secure. The snapshot will not be editable or removable for the specified retention period. If the specified snapshot is already secure, the security expiration date may be postponed, but not configured to an earlier point in time.

Syntax

```
scli --set_snapshot_security (--volume_id <ID> | --volume_name <NAME>) --
retention_period <PERIOD> [--i_am_sure]
```

Parameters

--volume_id *<ID>*

Volume ID

--volume_name *<NAME>*

Volume name

--retention_period *<PERIOD>*

The snapshots will not be editable or removable for the specified time period. To specify the time period, use `m | h | D | W | M | Y` respectively indicating minutes, hours, days, weeks, months, or years. Examples: 30m (thirty minutes), 5D (five days), 1M (one month, which equals 31 days).

Example

```
scli --mdm_ip 192.168.1.200 --set_snapshot_security --volume_id 022beb2500000006 --
retention_period 2Y --i_am_sure
```

snapshot_volume

Creates a snapshot of one or more volumes.

When specifying more than one volume (a list), a consistency group is generated and the ID associated with it is displayed. The snapshots under the consistency group are taken simultaneously for all listed volumes, thus ensuring their consistency.

Secure snapshots, created using the `--retention_period` parameter, prohibit anyone from deleting the snapshot until the expiration time that is defined.

NOTE: The consistency group is for convenience purposes only. There are no protective measures to conserve the consistency group, and you can delete members from it.

Syntax

```
scli --snapshot_volume (--volume_id <ID>| --volume_name <NAME> | --volume_id_from_file
<FILE> | --volume_name_from_file <FILE>)
  [--snapshot_name <NAME> | --snapshot_name_from_file <FILE>] [--read_only] [--
retention_period <PERIOD>] [--i_am_sure]
```

Parameters

--volume_id *<ID>*

Comma-separated list of Volume IDs. Omit the space after each comma.

--volume_name *<NAME>*

Comma-separated list of Volume names. Omit the space after each comma.

--volume_id_from_file *<FILE>*

File containing new, line-separated list of volume IDs

--volume_name_from_file *<FILE>*

File containing a new, line-separated list of volume names

--snapshot_name *<NAME>*

Comma-separated list of names to be associated with the created snapshots. The names in this list correspond to the names in the volume name or volume ID list. Omit the space after each comma.

--snapshot_name_from_file *<FILE>*

File containing a new, line-separated list of names to be associated with the created snapshots

--read_only

All mappings of the created volume will be limited to read-only access

--retention_period *<PERIOD>*

Use this parameter to create the snapshots as secure. The snapshots will not be editable or removable for the specified time period. To specify the time period, use `m|h|D|W|M|Y` respectively indicating minutes, hours, days, weeks, months, or years. Examples: 30m (thirty minutes), 5D (five days), 1M (one month, which equals 31 days)

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

Example

```
scli --mdm_ip 192.168.1.200 --snapshot_volume
--volume_name vol_1 --snapshot_name snap_1 --retention_period 2Y
```

Snapshot names

Assign the snapshot a meaningful name. When a name has not been defined, the system may display default system-defined names that use the volume’s ID.

Assign snapshot names according to the following rules:

1. Contains fewer than 32 characters
2. Contains only alphanumeric and punctuation characters
3. Is unique within the object type

NOTE: PowerFlex objects are assigned a unique ID that can be used to identify the object in CLI commands. You can retrieve the ID via a query or through the object’s property sheet in the PowerFlex GUI.

unlock_auto_snapshot

Unlock an auto snapshot to allow the snapshot policy to remove it, if and when required.

Syntax

```
scli --unlock_auto_snapshot (--auto_snapshot_id <ID> | --auto_snapshot_name <NAME>) [--
auto_snapshot_will_be_removed]
```

Parameters

--auto_snapshot_id *<ID>*

Auto snapshot ID

--auto_snapshot_name *<NAME>*

Auto snapshot name

--auto_snapshot_will_be_removed

Acknowledge that if the auto snapshot has expired while it was locked, unlocking it will cause it to be removed.

Example

```
scli --unlock_auto_snapshot --auto_snapshot_name auto_snap_1 --
auto_snapshot_will_be_removed
```

13

Storage Pool Commands

This section contains commands for managing Storage Pools.

add_storage_pool

Add a Storage Pool to a Protection Domain.

A Storage Pool is a group of storage devices within a Protection Domain. Each time that you add devices to the system, you must map them to either Storage Pools or Acceleration Pools. Create Storage Pools before you start adding SDSs and storage devices to the system. You can modify Storage Pools post-installation using all management clients, with the exception of OpenStack.

Syntax

```
scli --add_storage_pool (--protection_domain_id <ID> | --protection_domain_name <NAME>)
  [--storage_pool_name <NAME>]
  [--use_rmcache | --dont_use_rmcache]
  [--rmcache_write_handling_mode  {cached | passthrough}]
  [--media_type {HDD | SSD}]
  [--external_acceleration <TYPE>] {none | read | write | read_and_write}]
  [--data_layout <LAYOUT> {medium_granularity |fine_granularity}
  [--compression_method <METHOD> {none | normal}]
  (--fine_granularity_acceleration_pool_name <NAME> | --
fine_granularity_acceleration_pool_id <ID>)]]
  [--enable_inflight_checksum | --disable_inflight_checksum]
  [--persistent_checksum_enabled | --persistent_checksum_disabled]
```

Parameters

--protection_domain_id *<ID>*

Protection Domain ID

--protection_domain_name *<NAME>*

Protection Domain name

--storage_pool_name *<NAME>*

Name to be associated with the new Storage Pool

--use_rmcache

Use RMcache on all SDSs in the Storage Pool. You can change this setting later on specific SDSs (they can be individually configured not to use RMcache). For more information, see the *Configure and Customize PowerFlex Guide*.

--dont_use_rmcache

Do not use Read RAM Cache on any SDSs in the Storage Pool (default)

--rmcache_write_handling_mode *<MODE>* **{cached | passthrough}**

Determines whether the system stores the data of this Storage Pool's writes in the SDS Read RAM Cache, or not. The default is to store the write data in cache (`cached`). You can change this setting later, using the command `set_rmcache_write_handling_mode`.

NOTE: Only I/Os that are multiples of 4k bytes can be cached.

--media_type *<TYPE>* **{HDD | SSD}**

Media type of the devices in the Storage Pool, where the devices will all be HDD or SSD.

--external_acceleration *<TYPE>*

External acceleration for the device where *<TYPE>* can be one of the following:

none

All devices are not accelerated by a non-PowerFlex read or write cache (default).

read

All devices are accelerated by a non-PowerFlex read cache.

write

All devices are accelerated by a non-PowerFlex write cache.

read_and_write

All devices are accelerated by both non-PowerFlex read and write cache.

--data_layout *<LAYOUT>* **{medium_granularity | fine_granularity}**

Indicates the layout of the data for devices in the Storage Pool, where *<LAYOUT>* can be medium granularity or fine granularity. The default is medium granularity.

--compression_method *<METHOD>*

When using fine granularity data layout, this Indicates the compression method to be used. *<METHOD>* can be one of the following:

none

No compression applied.

normal

Normal compression applied.

--fine_granularity_acceleration_pool_name *<NAME>*

Name of the NVDIMM Acceleration Pool to be used for fine granularity.

--fine_granularity_acceleration_pool_id *<ID>*

ID of the NVDIMM Acceleration Pool to be used for fine granularity.

--enable_inflight_checksum

Enable inflight checksum protection.

--disable_inflight_checksum

Disable inflight checksum protection.

--persistent_checksum_enabled

Enable persistent checksum protection.

--persistent_checksum_disabled

Disable persistent checksum protection.

Example

```
scli --add_storage_pool --protection_domain_name pd1 --storage_pool_name sp2
--dont_use_rmcache --media_type ssd --external_acceleration read_and_write --
persistent_checksum_enabled
```

Storage Pool names

Assign each Storage Pool a meaningful name. When a name has not been defined, the system may display default system-defined names that use the Storage Pool's IDs.

Each volume name should conform to the following rules:

1. Contains fewer than 32 characters
2. Contains only alphanumeric and punctuation characters
3. Is unique within the object type

NOTE: PowerFlex objects are assigned a unique ID that can be used to identify the object in CLI commands. You can retrieve the ID via a query or through the object's property sheet in the PowerFlex GUI.

disable_background_device_scanner

Disable the background device scanner on the specified Storage Pool.

Syntax

```
scli --disable_background_device_scanner (((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --storage_pool_name <NAME>) | --storage_pool_id <ID>)
```

Parameters

--protection_domain_id *<ID>*
Protection Domain ID

--protection_domain_name *<NAME>*
Protection Domain name

--storage_pool_name *<NAME>*
Storage Pool name

--storage_pool_id *<ID>*
Storage Pool ID

NOTE: If you use the `storage_pool_id` parameter in the command, you do not need to also specify the Protection Domain ID or name, and the Storage Pool name.

Example

```
scli --disable_background_device_scanner --protection_domain_name pd120 --
storage_pool_name sp180
```

disable_persistent_checksum

Disable persistent checksum for the specified Storage Pool.

Syntax

```
scli --disable_persistent_checksum (((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --storage_pool_name <NAME>) | --storage_pool_id <ID>)
```

Parameters

--protection_domain_id *<ID>*
Protection Domain ID

--protection_domain_name *<NAME>*
Protection Domain name

--storage_pool_name *<NAME>*
Storage Pool name

--storage_pool_id *<ID>*
Storage Pool ID

Example

```
scli --mdm_ip 192.168.1.200 --disable_persistent_checksum --storage_pool_id
d089bdc100000000
```

enable_background_device_scanner

Enable the background device scanner on the devices in the specified Storage Pool to check for errors (not supported in Fine Granularity Storage Pools).

Syntax

```
scli --enable_background_device_scanner (((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --storage_pool_name <NAME>) | --storage_pool_id <ID>) --
read_error_action <ACTION>  --compare_error_action <ACTION>
  [--scanner_bandwidth_limit <LIMIT>]
```

Parameters

--protection_domain_id *<ID>*

Protection Domain ID

--protection_domain_name *<NAME>*

Protection Domain name

--storage_pool_name *<NAME>*

Storage Pool name

--storage_pool_id *<ID>*

Storage Pool ID

--read_error_action

Read error handling. Can be one of the following:

--report_and_fix

Perform read operations and automatically fix from peer on errors.

--report_only

Perform read operations and report errors without fixing them.

--compare_error_action

Compare error handling between primary and secondary copies of data.

NOTE: Zero padding must be enabled in order to set the background device scanner to data compare mode.

Can be one of the following:

--no_compare

Perform read operations without comparing data content with peer.

--report_and_fix

Perform read operations, compare data content with peer, and automatically fix.

--report_only

Perform read operations, compare data content with peer, and report mismatches without fixing them.

--scanner_bandwidth_limit *<LIMIT>*

Bandwidth limit in KB per second per device. The value should be in the range of 10 KB to 10 MB. Default: 3 MB.

NOTE: High bandwidth should be used very carefully for extreme cases only (such as an urgent need to check some devices), as it may create negative impact on the system performance. Setting the background device scanner bandwidth should take into account maximum bandwidth of the devices.

NOTE: If you use the `storage_pool_id` parameter in the command, you do not need to also specify the Protection Domain ID or name, and the Storage Pool name.

Example

```
scli --enable_background_device_scanner --protection_domain_name pd120 --
storage_pool_name sp18 --read_error_action --report_only --compare_error_action --
report_only --scanner_bandwidth_limit 2000
```

Errors

To check for errors detected by the background device scanner, query SDSs using the `--query_sds` command. Error information is provided in event reports. For more information about viewing events, see the *Monitor Dell PowerFlex*.

Bandwidth limits

High bandwidth may create negative impact on system performance and should be used carefully and in extreme cases only—for example, when there is an urgent need to check certain devices. When setting the background device scanner bandwidth, you should take into account the maximum bandwidth of the devices.

enable_persistent_checksum

Enable persistent checksum for the specified Storage Pool. Persistent checksum can be used to support the medium granularity layout in protecting the storage device from data corruption.

Syntax

```
scli --enable_persistent_checksum (((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --storage_pool_name <NAME>) | --storage_pool_id <ID>) [--
validate_on_read <MODE>] [--builder_bandwidth_limit <LIMIT>]
```

Parameters

--protection_domain_id *<ID>*
Protection Domain ID

--protection_domain_name *<NAME>*
Protection Domain name

--storage_pool_name *<NAME>*
Storage Pool name

--storage_pool_id *<ID>*
Storage Pool ID

--validate_on_read *<MODE>*
Validate checksum upon reading data: `enabled` or `disabled`

--builder_bandwidth_limit *<LIMIT>*

Bandwidth limit in KB/s for the checksum building process. Valid range: 1024-10240. Default: 3072

Example

```
scli --mdm_ip 192.168.1.200 --enable_persistent_checksum --storage_pool_id
d089bdc100000000 --validate_on_read enabled --builder_bandwidth_limit 6144
```

modify_compression_method

Modify the compression method of a Storage Pool or a V-Tree. Compression can only be used for fine granularity layouts.

NOTE: When fine granularity layouts are not used, use the `none` compression option.

Syntax

```
scli --modify_compression_method ( (((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --storage_pool_name <NAME>) | --storage_pool_id <ID>) | --
vtree_id <ID> | --volume_id <ID> | --volume_name <NAME> ) --compression_method <METHOD>
```

Parameters

--protection_domain_id *<ID>*

Protection Domain ID

--protection_domain_name *<NAME>*

Protection Domain name

--storage_pool_name *<NAME>*

--storage_pool_id *<ID>*

Storage Pool ID

--vtree_id *<ID>*

V-Tree ID

--volume_id *<ID>*

Volume ID

--volume_name *<NAME>*

Volume name

--compression_method *<NAME>*

Indicates the compression method to be used on the Storage Pool or V-Tree. Compression is only relevant for fine granularity data layouts. Set to `none` if fine granularity is not used. Options include:

`none`- No compression is applied

`normal`- Normal compression is applied

Example

```
scli --modify_compression_method --protection_domain_name pd2 --storage_pool_name sp21 --
compression_method normal
```

modify_persistent_checksum

Modify the persistent checksum configuration for the specified Storage Pool.

Syntax

```
scli --modify_persistent_checksum (((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --storage_pool_name <NAME>) | --storage_pool_id <ID>) [--
validate_on_read <MODE>] [--builder_bandwidth_limit <LIMIT>]
```

Parameters

--protection_domain_id *<ID>*
Protection Domain ID

--protection_domain_name *<NAME>*
Protection Domain name

--storage_pool_name *<NAME>*
Storage Pool name

--storage_pool_id *<ID>*
Storage Pool ID

--validate_on_read *<MODE>*
Validate checksum upon reading data: `enabled` or `disabled`

--builder_bandwidth_limit *<LIMIT>*
Bandwidth limit in KB/s for the checksum building process. Valid range: 1024-10240. Default: 3072

Example

```
scli --mdm_ip 192.168.1.200 --modify_persistent_checksum --storage_pool_id
d089bdc100000000 --validate_on_read disabled --builder_bandwidth_limit 1024
```

modify_zero_padding_policy

Modify the zero padding policy for the Storage Pool. Use the zero-padded policy when the Storage Pool data layout is Fine Granularity (FG). The zero-padded policy ensures that every read from an area previously not written to, returns zeros.

Syntax

```
scli --modify_zero_padding_policy (((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --storage_pool_name <NAME>) | --storage_pool_id <ID>) (--
enable_zero_padding | --disable_zero_padding)
```

Parameters

--protection_domain_id *<ID>*
Protection Domain ID

--protection_domain_name *<NAME>*
Protection Domain name

--storage_pool_name *<NAME>*

Storage Pool name

--storage_pool_id *<ID>*

Storage Pool ID

--enable_zero_padding

Enable zero padding in this Storage Pool

--disable_zero_padding

Disable zero padding in this Storage Pool

NOTE: If you use the `storage_pool_id` parameter in the command, you do not need to also specify the Protection Domain ID or name, and the Storage Pool name.

Example

```
scli --modify_zero_padding_policy --protection_domain_name pd10 --storage_pool_name sp1
--zero_padding_disabled
```

query_inaccessible_user_address_space

List all inaccessible address spaces for the specified Storage Pool.

Syntax

```
scli --query_inaccessible_user_address_space (((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --storage_pool_name <NAME>) | --storage_pool_id <ID>)
```

Parameters

--protection_domain_id *<ID>*

Protection Domain ID

--protection_domain_name *<NAME>*

Protection Domain name

--storage_pool_name *<NAME>*

Storage Pool name

--storage_pool_id <ID>

Storage Pool ID

Example

```
scli --mdm_ip 192.168.1.200 --query_inaccessible_user_address_space --storage_pool_id
d089bdc100000000
```

query_storage_pool

Retrieve aggregated information about all objects in the specified Storage Pool.

Syntax

```
scli --query_storage_pool (((--protection_domain_id <ID> | --protection_domain_name
<NAME>) --storage_pool_name <NAME>) |--storage_pool_id <ID>)
```

Parameters

--protection_domain_id *<ID>*

Protection Domain ID

--protection_domain_name *<NAME>*

Protection Domain name

--storage_pool_name *<NAME>*

Storage Pool name

--storage_pool_id *<ID>*

Storage Pool ID

NOTE: If you use the `storage_pool_id` parameter in the command, you do not need to also specify the Protection Domain ID or name, and the Storage Pool name.

Example

```
scli --mdm_ip 192.168.1.200 --query_storage_pool --protection_domain_name pd2 --
storage_pool_name sp2
```

Sample output

```
Storage Pool sp2 (Id: 3df1cd9700000001) has 0 volumes and 22.3 GB (22875 MB) free net
capacity. 0 Bytes volume allocation limit.
        The number of parallel rebuild/rebalance jobs: 2
        Rebuild is enabled and using Limit-Concurrent-IO policy with the following
parameters:
          Number of concurrent IOs per device: 1
        Rebalance is enabled and using Favor-Application-IO policy with the following
parameters:
          Number of concurrent IOs per device: 1, Bandwidth limit per device: 10240 KB
per second
        VTree Migration is enabled and using Favor-Application-IO policy with the
following parameters:
          Number of concurrent IOs per device: 1, Bandwidth limit per device: 10240 KB
per second
        Protected Maintenance Mode is enabled and using Limit-Concurrent-IO policy with
the following parameters:
          Number of concurrent IOs per device: 1
        Fragmentation: Enabled
        Background device scanner: Enabled, Read Error Action: report and fix, Compare
Error Action: report and fix, Bandwidth Limit 3072 KBps per device
        Zero padding: Enabled
        Spare policy: 70% out of total
        Inflight Checksum: Disabled
        Doesn't use RAM Read Cache
        Uses Read Flash Cache
        Capacity alert thresholds: High: 80, Critical: 90
```

```
        Media Type: SSD, External Acceleration Type: NONE, Inflight Requests Factor:
115, Inflight Bandwidth Factor: 115
        Data layout: Medium granularity
        Replication Capacity Max Allowed Ratio: 10%
        Replication Capacity Max Allowed: 3.0 GB (3072 MB)
        Replication Allocated Capacity: 0 Bytes
        Replication Used Capacity: 0 Bytes
        Number of devices with inaccessible capacity: 0
        Medium granularity persistent checksum:
                Persistent checksum is enabled
                State is PROTECTED
                Checksum builder limit is: 3072 KBps
                Validation on read is disabled


Physical data:
        148.9 GB (152500 MB) total capacity
                0 Bytes in-use capacity
                44.7 GB (45750 MB) unused capacity
                104.2 GB (106750 MB) spare capacity
                0 Bytes decreased capacity
        Capacity Health:
                0 Bytes protected capacity
                0 Bytes failed capacity
                0 Bytes degraded-failed capacity
                0 Bytes degraded-healthy capacity
                0 Bytes unreachable-unused capacity
                0 Bytes semi-protected capacity
                0 Bytes in-maintenance capacity
                0 Bytes temporary capacity
                0 Bytes inaccessible capacity

User data:
        0 Bytes total user data
        0 Bytes total thin user data
        Thin and snapshot saving: Not applicable
        Overall saving: Not applicable
        Data Movement:
                0 Bytes rebalance capacity
                0 Bytes forward-rebuild capacity
                0 Bytes backward-rebuild capacity
                0 Bytes enter protected maintenance mode capacity
                0 Bytes exit protected maintenance mode capacity
                0 Bytes at-rest capacity
                0 Bytes total moving capacity
                        0 Bytes active moving capacity
                                0 Bytes active rebalance capacity
                                0 Bytes active forward-rebuild capacity
                                0 Bytes active backward-rebuild capacity
                                0 Bytes active enter protected maintenance mode capacity
                                0 Bytes active exit protected maintenance mode capacity
                        0 Bytes pending moving capacity
                                0 Bytes pending rebalance capacity
                                0 Bytes pending forward-rebuild capacity
                                0 Bytes pending backward-rebuild capacity
                                0 Bytes pending enter protected maintenance mode capacity
                                0 Bytes pending exit protected maintenance mode capacity

Bandwidth:
        Primary-reads                             0 IOPS 0 Bytes per second
        Primary-writes                            0 IOPS 0 Bytes per second
        Secondary-reads                           0 IOPS 0 Bytes per second
        Secondary-writes                          0 IOPS 0 Bytes per second
        Backward-rebuild-reads                    0 IOPS 0 Bytes per second
        Backward-rebuild-writes                   0 IOPS 0 Bytes per second
        Forward-rebuild-reads                     0 IOPS 0 Bytes per second
        Forward-rebuild-writes                    0 IOPS 0 Bytes per second
        Rebalance-reads                           0 IOPS 0 Bytes per second
        Rebalance-writes                          0 IOPS 0 Bytes per second
        Volume-migration-reads                    0 IOPS 0 Bytes per second
        Volume-migration-writes                   0 IOPS 0 Bytes per second
        Enter-protected-maintenance-mode-reads    0 IOPS 0 Bytes per second
        Enter-protected-maintenance-mode-writes   0 IOPS 0 Bytes per second
```

```
        Exit-protected-maintenance-mode-reads    0 IOPS 0 Bytes per second
        Exit-protected-maintenance-mode-writes   0 IOPS 0 Bytes per second

Volumes summary:
        0 Bytes all volumes size
        0 Bytes base volumes size
        0 Volumes.
```

remove_storage_pool

Remove a Storage Pool from a Protection Domain.

NOTE: Each Protection Domain must contain at least one Storage Pool, which can be either the default or a user-defined Storage Pool. You can only remove all Storage Pools from a Protection Domain if they are empty and do not contain any devices.

Syntax

```
scli --remove_storage_pool (((--protection_domain_id <ID> | --protection_domain_name
<NAME>) --storage_pool_name <NAME>) |--storage_pool_id <ID>)
```

Parameters

--protection_domain_id *<ID>*
Protection Domain ID

--protection_domain_name *<NAME>*
Protection Domain name

--storage_pool_name *<NAME>*
Storage Pool name

--storage_pool_id *<ID>*
Storage Pool ID

NOTE: If you use the `storage_pool_id` parameter in the command, you do not need to also specify the Protection Domain ID or name, and the Storage Pool name.

Example

```
scli --mdm_ip 192.168.1.200 --remove_storage_pool
--protection_domain_name rack_1.1
--storage_pool_name sp_1.1_rack_1.1
```

rename_storage_pool

Name, or rename, a Storage Pool.

Syntax

```
scli --rename_storage_pool (((--protection_domain_id <ID> | --protection_domain_name
<NAME>) --storage_pool_name <NAME>) |--storage_pool_id <ID>) --new_name <NAME>
```

Parameters

--protection_domain_id *<ID>*
Protection Domain ID

--protection_domain_name *<NAME>*
Protection Domain name

--storage_pool_name *<NAME>*
Storage Pool name

--storage_pool_id *<ID>*
Storage Pool ID

--new_name *<NAME>*
New name to associate with the Storage Pool

NOTE: If you use the `storage_pool_id` parameter in the command, you do not need to also specify the Protection Domain ID or name, and the Storage Pool name.

Example

```
scli --mdm_ip 192.168.1.200 --rename_storage_pool --protection_domain_name rack_1.1 --
storage_pool_name sp_1.1_rack_1.1 --new_name sp_2.2_rack_1.1
```

Storage Pool names

Assign the Storage Pool a meaningful name in order to facilitate future object identification. The name should conform to the following rules:

1. Contains fewer than 32 characters
2. Contains only alphanumeric and punctuation characters
3. Is unique within the object type

reset_scanner_error_counters

Reset background device scanner error counters for the specified Storage Pool or device. You can use this command to reset counters for comparison errors, read errors, or both counter types.

Syntax

```
scli --reset_scanner_error_counters ((((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --storage_pool_name <NAME>) |--storage_pool_id <ID>) |
(--device_id <ID> | ((--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP> [--sds_port
<PORT>]) (--device_name <NAME> | --device_path <PATH>))) )
  [--reset_compare_errors]
  [--reset_read_errors]
```

Parameters

--protection_domain_id *<ID>*
Protection Domain ID

--protection_domain_name *<NAME>*
Protection Domain name

--storage_pool_name *<NAME>*

Storage Pool name

--sds_id *<ID>*

SDS ID

--sds_name *<NAME>*

SDS name

--sds_ip *<SDS IP ADDRESS>*

SDS IP address

--sds_port *<PORT>*

Port assigned to the SDS

--device_id *<ID>*

Device ID

--device_name *<NAME>*

Device name

--device_path *<PATH>*

SDS storage device path or file path

--storage_pool_id *<ID>*

Storage Pool ID

--reset_read_errors

Reset the read error counter.

--reset_compare_errors

Reset the compare error counter.

NOTE: If you use the `storage_pool_id` parameter in the command, you do not need to also specify the Protection Domain ID or name, and the Storage Pool name.

NOTE: If you use the `device_id` parameter in the command, you do not need to also supply the SDS ID or SDS name.

Example

```
scli --reset_scanner_error_counters --protection_domain_name pd120 --storage_pool_name
sp120  --sds_name sds13 --reset_compare_errors
```

set_capacity_alerts_threshold

Set the threshold for triggering capacity usage alerts.

Alerts thresholds are calculated from each Storage Pool capacity after deducting the defined amount of spare capacity. You can set thresholds for a single Storage Pool or for all Storage Pools.

Syntax

```
scli --set_capacity_alerts_threshold --capacity_high_threshold <PERCENT> --
capacity_critical_threshold <PERCENT> (--all_storage_pools | --system_default | (((--
protection_domain_id <ID> | --protection_domain_name <NAME>) --storage_pool_name <NAME>)
|--storage_pool_id <ID>))
```

Parameters

--capacity_high_threshold *<PERCENT>*

Threshold of the non-spare capacity of the Storage Pool that will trigger a high-priority alert, expressed as a percentage

--capacity_critical_threshold *<PERCENT>*

Threshold of the non-spare capacity of the Storage Pool that will trigger a critical-priority alert, expressed as a percentage

--protection_domain_id *<ID>*

Protection Domain ID

--protection_domain_name *<NAME>*

Protection Domain name

--storage_pool_name *<NAME>*

Storage Pool name

--storage_pool_id *<ID>*

Storage Pool ID

--all_storage_pools

Thresholds apply to all Storage Pools

--system_default

Thresholds apply to newly created Storage Pools

Example

```
scli --mdm_ip 192.168.1.200 --set_capacity_alerts_threshold --system_default --
capacity_high_threshold 80 --capacity_critical_threshold 90
```

set_rebalance_mode

Enable or disable rebalancing of the specified Storage Pool.

WARNING: Rebalancing is an essential part of the PowerFlex system and should only be disabled temporarily, in special circumstances. Disabling rebalance may cause the system to become unbalanced even if no capacity is added or removed—for example, during recovery from an SDS or device failure.

Syntax

```
scli --set_rebalance_mode (((--protection_domain_id <ID> | --protection_domain_name
<NAME>) --storage_pool_name <NAME>) |--storage_pool_id <ID>) (--enable_rebalance | --
disable_rebalance)
  [--i_am_sure]
```

Parameters

--protection_domain_id *<ID>*

Protection Domain ID

--protection_domain_name *<NAME>*

Protection Domain name

--storage_pool_name *<NAME>*

Storage Pool name

--storage_pool_id *<ID>*

Storage Pool ID

--enable_rebalance

Enable rebalancing of the specified Storage Pool.

--disable_rebalance

Disable rebalancing of the specified Storage Pool.

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

NOTE: If you use the `storage_pool_id` parameter in the command, you do not need to also specify the Protection Domain ID or name, and the Storage Pool name.

Example

```
scli --set_rebalance_mode --protection_domain_name pd10 --storage_pool_name sp20 --
enable_rebalance
```

set_rebalance_policy

Set the rebalance I/O priority policy for a Storage Pool. This feature affects system performance and should only be modified by advanced users.

Syntax

```
scli --set_rebalance_policy (((--protection_domain_id <ID> | --protection_domain_name
<NAME>) --storage_pool_name <NAME>) |--storage_pool_id <ID>) --policy <POLICY> (Options)
  [--i_am_sure]
```

Parameters

--protection_domain_id *<ID>*

Protection Domain ID

--protection_domain_name *<NAME>*

Protection Domain name

--storage_pool_name *<NAME>*

Storage Pool name

--storage_pool_id *<ID>*

Storage Pool ID

--policy *<POLICY>*

Policy to use for rebalance I/O priority. Can be one of the following:

no_limit

Rebalance I/Os are not limited.

limit_concurrent_io

Limit the number of allowed concurrent rebalance I/Os to the value defined for `concurrent_io_limit`.

favor_application_io

(Default) Limit the number of allowed concurrent rebalance I/Os to the values defined for `concurrent_io_limit` and `bandwidth_limit`, regardless of user I/Os.

Options

Choose one or several:

--concurrent_io_limit *<LIMIT>*

The maximum number of concurrent rebalance I/Os per device. Default: 1.

--bandwidth_limit <BANDWIDTH>

The maximum bandwidth of rebalance I/Os, in KB/s, per device. The valid range is 1024-1048576. Default: 10,240. This property will take effect only if the policy is set to `favor_application_io`.

--i_am_sure

Skip the safety questions for command execution. (For example: "This could damage the stored data. Are you sure?")

NOTE: If you use the `storage_pool_id` parameter in the command, you do not need to also specify the Protection Domain ID or name, and the Storage Pool name.

Example

```
scli --set_rebalance_policy --protection_domain_name pd1 --storage_pool_name sp1 --
policy favor_application_io --bandwidth_limit 5120
```

set_rebuild_mode

Enable or disable rebuilds in the specified Storage Pool.

WARNING: Rebuilds are an essential part of the PowerFlex system and should only be disabled temporarily, in special circumstances. If rebuilds are disabled, redundancy will not be restored after failures.

Syntax

```
scli --set_rebuild_mode (((--protection_domain_id <ID> | --protection_domain_name
<NAME>) --storage_pool_name <NAME>) |--storage_pool_id <ID>) (--enable_rebuild | --
disable_rebuild)
  [--i_am_sure]
```

Parameters

--protection_domain_id <ID>

Protection Domain ID

--protection_domain_name <NAME>

Protection Domain name

--storage_pool_name <NAME>

Storage Pool name

--storage_pool_id <ID>

Storage Pool ID

--enable_rebuild

Enable rebuilds in the specified Storage Pool.

--disable_rebuild

Disable rebuilds in the specified Storage Pool.

--i_am_sure

Skip the safety questions for command execution. (For example: "This could damage the stored data. Are you sure?")

NOTE: If you use the `storage_pool_id` parameter in the command, you do not need to also specify the Protection Domain ID or name, and the Storage Pool name.

Example

```
scli --set_rebuild_mode --protection_domain_name pd10 --storage_pool_name sp20 --
enable_rebuild
```

set_rebuild_policy

Set the rebuild I/O priority policy for a Storage Pool. This feature affects system performance and should only be modified by advanced users.

Syntax

```
scli --set_rebuild_policy (((--protection_domain_id <ID> | --protection_domain_name
<NAME>) --storage_pool_name <NAME>) |--storage_pool_id <ID>) --policy <POLICY> (Options)
  [--i_am_sure]
```

Parameters

--protection_domain_id *<ID>*

Protection Domain ID

--protection_domain_name *<NAME>*

Protection Domain name

--storage_pool_name *<NAME>*

Storage Pool name

--storage_pool_id *<ID>*

Storage Pool ID

--policy *<POLICY>*

Policy to use for rebuild I/O priority. Can be one of the following:

no_limit

Rebuild I/Os are not limited.

limit_concurrent_io

(Default) Limit the number of allowed concurrent rebuild I/Os to the value defined for `concurrent_io_limit`.

favor_application_io

Limit the number of allowed concurrent rebuild I/Os to the values defined for `concurrent_io_limit` and `bandwidth_limit`, regardless of user I/Os.

Options

Choose one or several:

--concurrent_io_limit *<LIMIT>*

The maximum number of concurrent rebuild I/Os per device. Default: 1.

--bandwidth_limit *<BANDWIDTH>*

The maximum bandwidth of rebuild I/Os, in KB/s, per device. The valid range is 1024-1048576. Default: 10,240. This property will take effect only if the policy is set to `favor_application_io`.

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

NOTE: If you use the `storage_pool_id` parameter in the command, you do not need to also specify the Protection Domain ID or name, and the Storage Pool name.

Example

```
scli --set_rebuild_policy --protection_domain_name pd1 --storage_pool_name sp1 --policy
favor_application_io --bandwidth_limit 5120
```

set_rebuild_rebalance_parallelism

Set the maximum number of concurrent rebuild and rebalance activities on all SDSs within the specified Storage Pool.

Syntax

```
scli --set_rebuild_rebalance_parallelism (((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --storage_pool_name <NAME>) |--storage_pool_id <ID>) (--
limit <LIMIT>)
```

Parameters

--protection_domain_id *<ID>*
Protection Domain ID

--protection_domain_name *<NAME>*
Protection Domain name

--storage_pool_name *<NAME>*
Storage Pool name

--storage_pool_id *<ID>*
Storage Pool ID

--limit *<LIMIT>*
The maximum number of concurrent rebuild and rebalance activities (1..10) on SDSs in the Storage Pool.

NOTE: If you use the `storage_pool_id` parameter in the command, you do not need to also specify the Protection Domain ID or name, and the Storage Pool name.

Example

```
scli --set_rebuild_rebalance_parallelism --protection_domain_name pd_18 --
storage_pool_name flash_18 --limit 5
```

set_rmcache_usage

Control the use of Read RAM Cache in the specified Storage Pool.

When using Read RAM Cache, you must also enable caching in each SDS in the Storage Pool using the `enable_sds_rmcache` command. Caching only begins once devices have been added to the SDSs.

NOTE: You can configure Read RAM Cache for a Storage Pool and then disable caching on one or more SDSs individually.

Syntax

```
scli --set_rmcache_usage (((--protection_domain_id <ID> | --protection_domain_name
<NAME>) --storage_pool_name <NAME>) |--storage_pool_id <ID>) (--use_rmcache | --
```

```
dont_use_rmcache)
  [--i_am_sure]
```

Parameters

--protection_domain_id *<ID>*

Protection Domain ID

--protection_domain_name *<NAME>*

Protection Domain name

--storage_pool_name *<NAME>*

Storage Pool name

--storage_pool_id *<ID>*

Storage Pool ID

--use_rmcache

Use Read RAM Cache in the Storage Pool.

--dont_use_rmcache

Do not use Read RAM Cache in the Storage Pool (the default).

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

NOTE: If you use the `storage_pool_id` parameter in the command, you do not need to also specify the Protection Domain ID or name, and the Storage Pool name.

Example

```
scli --set_rmcache_usage --protection_domain_name pd10 --storage_pool_name spflash --
use_rmcache
```

set_rmcache_write_handling_mode

Set the RMcache write handling mode of the specified Storage Pool. Writes typically skip the cache, but in certain circumstances you may want to cache writes, as well as reads.

Syntax

```
scli --set_rmcache_write_handling_mode (((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --storage_pool_name <NAME>) |--storage_pool_id <ID>) --
rmcache_write_handling_mode {passthrough | cached}
[--i_am_sure]
```

Parameters

--protection_domain_id *<ID>*

Protection Domain ID

--protection_domain_name *<NAME>*

Protection Domain name

--storage_pool_name *<NAME>*

Storage Pool name

--storage_pool_id *<ID>*

Storage Pool ID

--rmcache_write_handling_mode {passthrough | cached}

The write-handling mode used by the RMcache. One of:

passthrough

Writes skip the cache and are stored in storage only.

cached

Writes are stored in both cache and storage (the default).

--i_am_sure

Skip the safety questions for command execution. (For example: "This could damage the stored data. Are you sure?")

NOTE: If you use the `storage_pool_id` parameter in the command, you do not need to also specify the Protection Domain ID or name, and the Storage Pool name.

Example

```
scli --set_rmcache_write_handling_mode --protection_domain_name pd1 --storage_pool_name
spflash --rmcache_write_handling_mode cached
```

set_storage_pool_external_acceleration_type

Set the external acceleration type for devices in the Storage Pool.

Syntax

```
scli --set_storage_pool_external_acceleration_type ((((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --storage_pool_name <NAME>) |--storage_pool_id <ID>) --
external_acceleration <TYPE>
  [--override_sds_device_specific_external_acceleration | --
keep_sds_device_specific_external_acceleration]
```

Parameters

--protection_domain_id *<ID>*

Protection Domain ID

--protection_domain_name *<NAME>*

Protection Domain name

--storage_pool_name *<NAME>*

Storage Pool name

--storage_pool_id *<ID>*

Storage Pool ID

--external_acceleration *<TYPE>*

External acceleration for the device, where TYPE is one of the following:

none

All devices are not accelerated by a non-PowerFlex read or write cache.

read

All devices are accelerated by a non-PowerFlex read cache.

write

All devices are accelerated by a non-PowerFlex write cache.

read_and_write

All devices are accelerated by both non-PowerFlex read and write cache.

--override_sds_device_specific_external_acceleration

Override SDS device-specific external acceleration.

NOTE: The device's external acceleration type will be overridden according to the external acceleration type set for the `--external_acceleration` parameter. If there is a mismatch between the Storage Pool's external acceleration type and the device's external acceleration type, the command will fail with informative error message.

--keep_sds_device_specific_external_acceleration

Keep SDS device-specific external acceleration.

Example

```
scli --set_storage_pool_external_acceleration_type --protection_domain_name
pd1 --storage_pool_name sp2 --external_acceleration read_and_write --
override_sds_device_specific_external_acceleration
```

set_storage_pool_media_type

Set the media type of the devices in the Storage Pool.

Syntax

```
scli --set_storage_pool_media_type (((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --storage_pool_name <NAME>) |--storage_pool_id <ID>) --
media_type <TYPE>
  [--override_sds_device_media_type]
```

Parameters

--protection_domain_id *<ID>*

Protection Domain ID

--protection_domain_name *<NAME>*

Protection Domain name

--storage_pool_name *<NAME>*

Storage Pool name

--storage_pool_id *<ID>*

Storage Pool ID

--media_type *<TYPE>*

Media type of the device where TYPE is one of the following:

HDD

All devices in this Storage Pool will be HDDs.

SSD

All devices in this Storage Pool will be SSDs.

transitional

Media type will be defined per device

--override_sds_device_media_type

Override SDS device media type

Example

```
scli --set_storage_pool_media_type --protection_domain_name pd2 --storage_pool_name sp2
--media_type ssd --override_sds_device_media_type
```

set_checksum_mode

Inflight checksum protection mode can be used to validate data reads and writes in Storage Pools, in order to protect data from data corruption .Modify the checksum protection policy of a Storage Pool

Syntax

```
scli --set_checksum_mode (((--protection_domain_id <ID> | --protection_domain_name
<NAME>) --storage_pool_name <NAME>) | --storage_pool_id <ID>) (--
enable_inflight_checksum | --disable_inflight_checksum)
```

Parameters

--protection_domain_id *<ID>*
Protection Domain ID

--protection_domain_name *<NAME>*
Protection Domain name

--storage_pool_name *<NAME>*
Storage Pool name

--storage_pool_id *<ID>*
Storage Pool ID

--enable_inflight_checksum
Enable inflight checksum for the Storage Pool.

--disable_inflight_checksum
Disable inflight checksum for the Storage Pool.

Example

```
scli --set_checksum_mode --protection_domain_name PD_1 --storage_pool_name SP_2 --
enable_inflight_checksum
```

set_protected_maintenance_mode_policy

Set the protected maintenance mode I/O priority policy for a Storage Pool. This feature affects system performance and should only be modified by advanced users.

Syntax

```
scli --set_protected_maintenance_mode_policy (((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --storage_pool_name <NAME>) |--storage_pool_id <ID>) --
```

```
policy <POLICY> (Options)
  [--i_am_sure]
```

Parameters

--protection_domain_id *<ID>*

Protection Domain ID

--protection_domain_name *<NAME>*

Protection Domain name

--storage_pool_name *<NAME>*

Storage Pool name

--storage_pool_id *<ID>*

Storage Pool ID

--policy *<POLICY>*

Policy to use for protected maintenance mode I/O priority. Can be one of the following:

no_limit

Protected maintenance mode I/Os are not limited.

limit_concurrent_io

(Default) Limit the number of allowed concurrent protected maintenance mode I/Os to the value defined for `concurrent_io_limit`.

favor_application_io

Limit the number of allowed concurrent protected maintenance mode I/Os to the values defined for `concurrent_io_limit` and `bandwidth_limit`, regardless of user I/Os.

Options

Choose one or several:

--concurrent_io_limit *<LIMIT>*

The maximum number of concurrent protected maintenance mode I/Os per device. Default: 1.

--bandwidth_limit *<BANDWIDTH>*

The maximum bandwidth of protected maintenance mode I/Os, in KB/s, per device. The valid range is 1024-1048576. Default: 10,240. This property will take effect only if the policy is set to `favor_application_io`.

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

NOTE: If you use the `storage_pool_id` parameter in the command, you do not need to also specify the Protection Domain ID or name, and the Storage Pool name.

Example

```
scli --set_protected_maintenance_mode_policy --protection_domain_name pd1 --
storage_pool_name sp1 --policy favor_application_io --bandwidth_limit 5120
```

test_inaccessible_data

After identifying where there is inaccessible data using the `--query_inaccessible_user_address_space` command, manually clear the inaccessible data from the Storage Pool. You can then test the inaccessible data, which will trigger a rebuild operation, and the data will be accessible again.

Syntax

```
scli --test_inaccessible_data (--device_id <ID> | ((--sds_id <ID> | --sds_name <NAME> |
--sds_ip <IP> [--sds_port <PORT>]) (--device_name <NAME> | --device_path <PATH>)))
```

Parameters

--sds_id *<ID>*
SDS ID

--sds_name *<NAME>*
SDS name

--sds_ip *<IP>*
SDS IP address

--sds_port *<PORT>*
Port assigned to the SDS

--device_id *<ID>*
Device ID

--device_name *<NAME>*
Device name

--device_path *<PATH>*
SDS storage device path or filepath

Example

```
scli --mdm_ip 192.168.1.200 --test_inaccessible_data --device_id fceff00700040001
```

14

System Commands

This section contains commands for managing the system.

abort_upgrade

Stop an upgrade that is in process.

Syntax

```
scli --abort_upgrade
```

Parameters

None.

Example

```
scli --abort_upgrade
```

add_certificate

Add a Certificate Authority (CA) signed certificate to the CLI's Trusted Certificates list. All applied certificates signed by the CA will be trusted automatically by this CLI.

Syntax

```
scli ---add_certificate --certificate_file <FILE>
```

Parameters

--certificate_file *<FILE>*
File name of the Root or Trusted Certificate

Example

```
scli --add_certificate --certificate_file CAcert-cacert.pem
```

add_standby_mdm

Add a standby MDM to the system. This MDM can later be added to the cluster.

Syntax

```
scli --add_standby_mdm --mdm_role {manager    | tb} --new_mdm_ip <IP>
  [--new_mdm_port <PORT>]
  [--new_mdm_management_ip <IP>]
  [--new_mdm_virtual_ip_interface <INTF>]
  [--new_mdm_name <NAME>]
  [--allow_asymmetric_ips]
  [--force_clean]
  [--i_am_sure]
```

Designate the standby MDM's role, either manager or tiebreaker (default). You can give the standby a unique name.

Parameters

--mdm_role {manager | tb}

Designates whether the MDM role is a manager (primary MDM of the cluster) or will only serve as a tiebreaker in the cluster primary MDM election process

--new_mdm_ip *<IP>*

Comma-separated list of up to four IP addresses assigned to the new MDM. Can contain both IPv4 and IPv6 addresses. Omit the space after each comma.

--new_mdm_port *<PORT>*

Port of the new MDM. Default: 9011.

--new_mdm_management_ip *<IP>*

Comma-separated list of up to four IP addresses used to manage the MDM. Can contain both IPv4 and IPv6 addresses. Omit the space after each comma.

--new_mdm_virtual_ip_interface *<INTF>*

Comma-separated list of interface names to be used for the MDM virtual IP addresses. The order of interfaces must match the order of virtual IP addresses assigned to the cluster. Omit the space after each comma.

--new_mdm_name *<NAME>*

Name of the new MDM

--allow_asymmetric_ips

Allow the added node to have a different number of IP addresses from the primary node.

--force_clean

Clean the previous configuration of the MDM.

--i_am_sure

Skip the safety questions for command execution, and give preemptive approval.

Example

```
scli --add_standby_mdm --mdm_role manager --new_mdm_ip 192.168.1.153 --
new_mdm_management_ip 10.103.110.153 --new_mdm_virtual_ip_interface eth4
```

allow_commands_during_upgrade

Allows cluster commands to be sent during an upgrade for this session.

Syntax

```
scli --allow_commands_during_upgrade
```

Parameters

None.

Example

```
scli --allow_commands_during_upgrade
```

approve_all_mdm_certificates

Approve all MDM cluster certificates.

The command displays all MDM cluster certificates to be approved, one at a time. If there is a connection problem with a node, or a certificate is not approved, the process stops.

Syntax

```
scli --approve_all_mdm_certificates
```

Example

```
scli --approve_all_mdm_certificates
```

create_mdm_cluster

Create an MDM cluster from the selected primary MDM.

This command does not require the user to be logged in before running it. When the command is executed, the MDM cluster operates in single mode.

Syntax

```
scli --create_mdm_cluster --master_mdm_ip <IP>
  [--master_mdm_management_ip <IP>]
  [--cluster_virtual_ip <IP>]
  [--master_mdm_virtual_ip_interface <INTF>]
  [--master_mdm_name <NAME>]
  [--accept_license]
  [--disable_client_secure_communication]
```

```
  [--approve_certificate_once]
  [--approve_certificate]
```

Parameters

--master_mdm_ip *<IP>*

Comma- separated list of IP addresses assigned to the primary MDM, to be used for MDM cluster internal control communications. Omit the space after each comma.

--master_mdm_management_ip *<IP>*

Comma-separated list of IP addresses, used to manage the MDM. Omit the space after each comma.

--cluster_virtual_ip *<IP>*

Comma-separated list of virtual IP addresses to be used for the cluster. Omit the space after each comma.

--master_mdm_virtual_ip_interface *<INTF>*

Comma-separated list of interface names to be used for the MDM virtual IP addresses. The order of interfaces must match the order of the virtual IP addresses assigned to the cluster. Omit the space after each comma.

--master_mdm_name *<NAME>*

MDM name. Each MDM name must be unique. Ensure that each server on which the MDM is installed has a unique hostname.

--accept_license

Accept the license agreement.

--disable_client_secure_communication

Create the cluster without management client secure communications.

--approve_certificate_once

One-time approval of the MDM's certificate (without adding the certificate to the truststore).

--approve_certificate

Automatic approval of the MDM's certificate for the next commands (adds the certificate to the truststore).

Example

```
scli --create_mdm_cluster --master_mdm_ip 192.168.1.152 --cluster_virtual_ip
192.168.100.152 --master_mdm_virtual_ip_interface eth4
```

finalize_upgrade

Finalize the upgrade process.

Syntax

```
scli --finalize_upgrade
```

Parameters

None.

Example

```
scli --finalize_upgrade
```

generate_certificate

Generate a new certificate and private key for the SDS or SDR. The certificate and key are signed by the system.

Syntax

```
scli --generate_certificate ((--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP>[--
sds_port <PORT>]) | (--sdr_id <ID> | --sdr_name <NAME> | --sdr_ip <IP> [--sdr_port
<PORT>]))
```

Parameters

--sds_id *<ID>*
SDS ID

--sds_name *<NAME>*
SDS name

--sds_ip *<IP>*
SDS IP address

--sds_port *<PORT>*
Port assigned to the SDS

--sdr_id *<ID>*
SDR ID

--sdr_name *<NAME>*
SDR name

--sdr_ip *<IP>*
SDR IP address

--sdr_port <PORT>
Port assigned to the SDR

Example

```
scli --generate_certificate --sdr_id ffa0be3900000000
```

generate_mdm_certificate

Create a new self-signed certificate and private key for the specfied MDM.

NOTE: Running this command on the primary MDM causes the MDM to restart. This may cause the MDM to switch ownership.

Syntax

```
scli --generate_mdm_certificate (--target_mdm_id <ID> | --target_mdm_ip <IP> [--
target_mdm_port <PORT>] | --target_mdm_name <NAME>)
```

Parameters

--target_mdm_id *<ID>*
ID of the MDM for which the certificate and private key are required

--target_mdm_ip *<IP>*
IP address of the MDM for which the certificate and private key are required

--target_mdm_port *<PORT>*
Port of the MDM for which the certificate and private key are required

--target_mdm_name *<NAME>*
Name of the MDM for which the certificate and private key are required

Example

```
scli --generate_mdm_certificate --target_mdm_ip 198.168.1.2 --target_mdm_port 9011
```

generate_mdm_csr_file

Create a CSR (Certificate Signing Request) file for the specified MDM.

The file is used to obtain security certificates from a Certificate Authority and is saved in the specified MDM's directory:

- Linux: `/opt/emc/scaleio/mdm/cfg`
- Windows: `C:\Program Files\emc\scaleio\mdm\cfg`

Syntax

```
scli --generate_mdm_csr_file (--target_mdm_id <ID> | --target_mdm_ip <IP> [--
target_mdm_port <PORT>] | --target_mdm_name <NAME>)
  [OPTIONS]
```

Parameters

--target_mdm_id *<ID>*
ID of the MDM for which the certificate and private key are required

--target_mdm_ip *<IP>*
IP address of the MDM for which the certificate and private key are required

--target_mdm_port *<PORT>*
Port of the MDM for which the certificate and private key are required

--target_mdm_name *<NAME>*
Name of the MDM for which the certificate and private key are required

--common_name *<NAME>*
Common name to appear in the CSR. The default is the machine's host name.

Options

NOTE: If you use any of these options, the default values will be disabled for all of them.

--country *<NAME>*

Country to appear in the CSR

--state *<NAME>*

State to appear in the CSR

--location *<NAME>*

Location to appear in the CSR

--organization *<NAME>*

Organization to appear in the CSR

--organizational_unit *<NAME>*

Organizational unit to appear in the CSR

--email_address *<EMAIL>*

Email address to appear in the CSR

Example

```
scli --generate_mdm_csr_file --target_mdm_ip 192.168.1.2 --target_mdm_port 9011 --
country australia --state victoria --location melbourne --organization example_pty_ltd
--organizational_unit sales --email_address sales@example.com.au
```

modify_cluster_virtual_ips

Set or modify the virtual IP address of an MDM cluster.

NOTE: After changing or adding a virtual IP address, you are required to update the SDC with the new virtual IP address. See "Update the SDC parameters in VMware based HCI or Compute node" in the *Configure and Customize Dell PowerFlex*

Notes

For data networks using IPv6, if you plan to implement a floating virtual IP address for the MDM (when the MDM is installed on a Linux server), disable the IPv6 DAD setting on the server's relevant interfaces, using the command: `sysctl net.ipv6.conf.<interface_name>.dad_transmits=0`

Syntax

```
scli --modify_cluster_virtual_ips (--cluster_virtual_ip <IP> | --clear_all)
```

Parameters

--cluster_virtual_ip *<IP>*

Comma-separated list of IP addresses to be used for the cluster. Omit the space after each comma.

--clear_all

Clear all virtual IP addresses.

Example

```
scli --modify_cluster_virtual_ips --cluster_virtual_ip 192.168.100.152,192.168.100.153
```

modify_management_ip

Modify an MDM node's management IP addresses.

If you are using this command to add additional management IP addresses, you must enter all of the existing IP addresses in the command, and add the new ones at the end. Up to four IP addresses are supported.

Syntax

```
scli --modify_management_ip (--new_mdm_management_ip <IP> | --clear_all) (--
target_mdm_id <ID> | --target_mdm_ip <IP> [--target_mdm_port <PORT>] | --target_mdm_name
<NAME>)
  [--allow_duplicate_management_ips]
  [--i_am_sure]
```

Parameters

--new_mdm_management_ip *<IP>*

Comma-separated list of IP address to be used to manage the MDM. Omit the space after each comma.

--clear_all

Clear all management IP addresses.

--target_mdm_id *<ID>*

MDM ID

--target_mdm_ip *<IP>*

MDM IP address

--target_mdm_port *<PORT>*

MDM port

--target_mdm_name *<NAME>*

MDM name

--allow_duplicate_management_ips

Allow duplicate management IP addresses.

--i_am_sure

Skip the safety questions for command execution. (For example: "This could damage the stored data. Are you sure?")

Example

```
scli --modify_management_ip --target_mdm_id 0x34e8f1df6c84a410 --new_mdm_management_ip
fd00::192:168:1:17,10.103.110.17
```

Implementing the changes

After modifying MDM IP addresses, ensure that you implement the change in all components and interfaces that are mapped to the MDM:

1. PowerFlex GUI: In the **Login** window, log in using the new IP address, then clear the old IP address from the history if it is now obsolete.
2. REST Gateway and OpenStack: In the REST Gateway, modify the mapping to the new management IP address.
3. On each SDC, perform the following locally, in command line with root access permissions:

- If for any reason, the management IP address is not updated in the SDC, run:

```
/opt/emc/scaleio/sdc/bin/drv_cfg --mod_mdm_ip
```

- If the SDC is mapped to more than one PowerFlex system, run:

```
/opt/emc/scaleio/sdc/bin/drv_cfg --add_mdm
```

4. vSphere: Using the vSphere web plug-in interface, unregister PowerFlex and reregister it with the new IP address.

modify_spare_policy

Modify the current spare capacity reservation policy.

To ensure data protection during server failures, PowerFlex reserves 10 percent of its capacity by default, instead of allowing this capacity to be used for volume allocation. To ensure full system protection in the event of a node failure, the spare capacity must be at least equal to the amount of capacity in the node containing the maximum capacity or the maximum Fault Set capacity.

If all nodes contain equal capacity, you should set the capacity value to at least 1/N of the total capacity (where N is the number of SDS nodes).

Syntax

```
scli --modify_spare_policy (((--protection_domain_id <ID> | --protection_domain_name
<NAME>) --storage_pool_name <NAME>) | --storage_pool_id <ID>) --spare_percentage
<PERCENT>
  [--i_am_sure]
```

Parameters

--protection_domain_id *<ID>*
Protection Domain ID

--protection_domain_name *<NAME>*
Protection Domain name

--storage_pool_name *<NAME>*
Storage Pool name

--storage_pool_id *<ID>*
Storage Pool ID

--spare_percentage *<PERCENT>*
Percentage of the total capacity to be set aside as spare

--i_am_sure
Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

NOTE: If you use the `storage_pool_id` parameter in the command, you do not need to also specify the Protection Domain ID or name, and the Storage Pool name.

Example

```
scli --mdm_ip 192.168.1.200 --modify_spare_policy
--protection_domain_name rack_1.1 --storage_pool_name --spare_percentage 10
```

modify_virtual_ip_interfaces

Set the virtual IP interfaces of a node in the MDM cluster.

Syntax

```
scli --modify_virtual_ip_interfaces (--new_mdm_virtual_ip_interface <INTF> | --
clear_all) (--target_mdm_id <ID> | --target_mdm_ip <IP> [--target_mdm_port <PORT>] | --
target_mdm_name <NAME>)
```

Parameters

--new_mdm_virtual_ip_interface *<INTF>*

Comma-separated list of interface names to be used for the MDM virtual IP interfaces. Omit the space after each comma. The order of interfaces must match the order of virtual IPs assigned to the cluster.

--clear_all

Clear all virtual IP interfaces.

--target_mdm_id *<ID>*

MDM ID

--target_mdm_ip *<IP>*

MDM IP address

--target_mdm_port *<PORT>*

MDM port

--target_mdm_name *<NAME>*

MDM name

Example

```
scli --modify_virtual_ip_interfaces --target_mdm_ip 192.168.1.152 --
new_mdm_virtual_ip_interface eth4
```

query_all

Retrieve aggregated information about all objects in all Protection Domains.

Syntax

```
scli --query_all
```

Parameters

None.

Example

```
scli --mdm_ip 192.168.1.200 --query_all
```

Sample output

```
# scli --query_all
System Info:
        Product: DellEMC PowerFlex Version: R3_6.0.270
        ID: 1df6301000964d0f
        Manager ID: 0000000000000000

License info:
        Installation ID: 1e14a1273652dda9
        SWID:
        Maximum capacity: Unlimited
        Usage time left: 84 days *** Non-Production License ***
        Enterprise features: Enabled
        The system was activated 6 days ago

System settings:
        Capacity alert thresholds: High: 80, Critical: 90
        MDM restricted SDC mode: NONE
        Management Clients secure communication: Enabled
        TLS version: TLSv1.2
        CLI preemptive login banner acceptance via shell: Enabled
        CLI login banner acceptance by password: Enabled
        User authentication method: Native
        MDM connection authentication: Enabled
        Defragmentation: Enabled

SDC-SDS connectivity status: All connected

SDC-SDR connectivity status: All connected

SDR-SDS connectivity status: All connected

Query all returned 2 Protection Domains:
Protection Domain pd1 (Id: fcc9af5500000000) has 1 storage pools, 0 acceleration pools,
1 Fault Sets, 4 SDS nodes and 7 volumes
        Free net capacity: 231.5 GB (237022 MB), Volume allocation limit: 2.5 TB (2560
GB)
        Operational state is Active
        RFcache enabled, Mode: Write miss, Page Size 64 KB, Max IO size 128 KB,
Acceleration Pool ID: N/A
        I/O flow control is enabled
        Fine granularity default number of concurrent writes: 1000
        FGL metadata cache is disabled
        FGL default metadata cache size: 0 MB
        Replication capacity utilization is in Normal level

Storage Pool sp1 (Id: 3df1cd9600000000) has 7 volumes and 231.5 GB (237022 MB) free net
capacity. 2.5 TB (2560 GB) volume allocation limit.
        The number of parallel rebuild/rebalance jobs: 2
        Rebuild is enabled and using Limit-Concurrent-IO policy with the following
parameters:
          Number of concurrent IOs per device: 1
        Rebalance is enabled and using Favor-Application-IO policy with the following
parameters:
          Number of concurrent IOs per device: 1, Bandwidth limit per device: 10240 KB
per second
        VTree Migration is enabled and using Favor-Application-IO policy with the
following parameters:
          Number of concurrent IOs per device: 1, Bandwidth limit per device: 10240 KB
per second
        Protected Maintenance Mode is enabled and using Limit-Concurrent-IO policy with
the following parameters:
          Number of concurrent IOs per device: 1
        Fragmentation: Enabled
        Background device scanner: Enabled, Read Error Action: report and fix, Compare
Error Action: report and fix, Bandwidth Limit 3072 KBps per device
        Zero padding: Enabled
        Spare policy: 56% out of total
        Inflight Checksum: Disabled
        Doesn't use RAM Read Cache
```

```
        Doesn't use Read Flash Cache
        Capacity alert thresholds: High: 80, Critical: 90
        Media Type: HDD, External Acceleration Type: NONE, Inflight Requests Factor:
115, Inflight Bandwidth Factor: 115
        Data layout: Medium granularity
        Replication Capacity Max Allowed Ratio: 10%
        Replication Capacity Max Allowed: 27.0 GB (27648 MB)
        Replication Allocated Capacity: 24.0 GB (24576 MB)
        Replication Used Capacity: 5.8 GB (5926 MB)
        Number of devices with inaccessible capacity: 0
        Medium granularity persistent checksum:
                Persistent checksum is enabled
                State is PROTECTED
                Checksum builder limit is: 3072 KBps
                Validation on read is disabled

SDR-SDS connectivity status: All connected

Protection Domain pd2 (Id: fcc9af5600000001) has 1 storage pools, 1 acceleration pools,
0 Fault Sets, 1 SDS nodes and 0 volumes
        Free net capacity: 22.3 GB (22875 MB), Volume allocation limit: 0 Bytes
        Operational state is Active
        RFcache enabled, Mode: Write miss, Page Size 64 KB, Max IO size 128 KB,
Acceleration Pool ID: 6794008e00000000
        I/O flow control is enabled
        Fine granularity default number of concurrent writes: 1000
        FGL metadata cache is disabled
        FGL default metadata cache size: 0 MB
        Replication capacity utilization is in Normal level

Storage Pool sp2 (Id: 3df1cd9700000001) has 0 volumes and 22.3 GB (22875 MB) free net
capacity. 0 Bytes volume allocation limit.
        The number of parallel rebuild/rebalance jobs: 2
        Rebuild is enabled and using Limit-Concurrent-IO policy with the following
parameters:
          Number of concurrent IOs per device: 1
        Rebalance is enabled and using Favor-Application-IO policy with the following
parameters:
          Number of concurrent IOs per device: 1, Bandwidth limit per device: 10240 KB
per second
        VTree Migration is enabled and using Favor-Application-IO policy with the
following parameters:
          Number of concurrent IOs per device: 1, Bandwidth limit per device: 10240 KB
per second
        Protected Maintenance Mode is enabled and using Limit-Concurrent-IO policy with
the following parameters:
          Number of concurrent IOs per device: 1
        Fragmentation: Enabled
        Background device scanner: Enabled, Read Error Action: report and fix, Compare
Error Action: report and fix, Bandwidth Limit 3072 KBps per device
        Zero padding: Enabled
        Spare policy: 70% out of total
        Inflight Checksum: Disabled
        Doesn't use RAM Read Cache
        Uses Read Flash Cache
        Capacity alert thresholds: High: 80, Critical: 90
        Media Type: SSD, External Acceleration Type: NONE, Inflight Requests Factor:
115, Inflight Bandwidth Factor: 115
        Data layout: Medium granularity
        Replication Capacity Max Allowed Ratio: 10%
        Replication Capacity Max Allowed: 3.0 GB (3072 MB)
        Replication Allocated Capacity: 0 Bytes
        Replication Used Capacity: 0 Bytes
        Number of devices with inaccessible capacity: 0
        Medium granularity persistent checksum:
                Persistent checksum is enabled
                State is PROTECTED
                Checksum builder limit is: 3072 KBps
                Validation on read is disabled

Acceleration Pool RFcache_Pool (Id: 6794008e00000000)
        Media Type: SSD
        Used for RFcache: yes
```

```
SDR-SDS connectivity status: All connected

SDS Summary:
        Total 5 SDS Nodes
        5 SDS nodes have membership state 'Joined'
        5 SDS nodes have connection state 'Connected'

Physical data:
        1.3 TB (1340 GB) total capacity
                61.3 GB (62756 MB) in-use capacity
                        61.3 GB (62756 MB) user data
                        0 Bytes metadata
                        0 Bytes fine granularity spares
                507.6 GB (519794 MB) unused capacity
                771.4 GB (789950 MB) spare capacity
                0 Bytes decreased capacity
        Capacity Health:
                61.3 GB (62756 MB) protected capacity
                0 Bytes failed capacity
                0 Bytes degraded-failed capacity
                0 Bytes degraded-healthy capacity
                0 Bytes unreachable-unused capacity
                0 Bytes semi-protected capacity
                0 Bytes in-maintenance capacity
                0 Bytes temporary capacity

User data:
        30.6 GB (31378 MB) total user data
                30.6 GB (31378 MB) medium granularity user data
                0 Bytes fine granularity user data
                        0 Bytes compressed data (Compression ratio: Not applicable)
                        0 Bytes uncompressed data
        0 Bytes trimmed capacity
        30.6 GB (31378 MB) user data without trimmed capacity
        1.0:1 total compression ratio
        2.1:1 thin and snapshot saving
        2.1:1 overall saving
        Data Movement:
                0 Bytes rebalance capacity
                0 Bytes forward-rebuild capacity
                0 Bytes backward-rebuild capacity
                0 Bytes enter protected maintenance mode capacity
                0 Bytes exit protected maintenance mode capacity
                30.6 GB (31378 MB) at-rest capacity
                0 Bytes total moving capacity
                        0 Bytes active moving capacity
                                0 Bytes active rebalance capacity
                                0 Bytes active forward-rebuild capacity
                                0 Bytes active backward-rebuild capacity
                                0 Bytes active enter protected maintenance mode capacity
                                0 Bytes active exit protected maintenance mode capacity
                        0 Bytes pending moving capacity
                                0 Bytes pending rebalance capacity
                                0 Bytes pending forward-rebuild capacity
                                0 Bytes pending backward-rebuild capacity
                                0 Bytes pending enter protected maintenance mode capacity
                                0 Bytes pending exit protected maintenance mode capacity

Bandwidth:
        Primary-reads                             348 IOPS 182.9 MB (187289 KB) per-second
        Primary-writes                            556 IOPS 273.2 MB (279737 KB) per-second
        Secondary-reads                           0 IOPS 0 Bytes per second
        Secondary-writes                          557 IOPS 271.8 MB (278283 KB) per-second
        Backward-rebuild-reads                    0 IOPS 0 Bytes per second
        Backward-rebuild-writes                   0 IOPS 0 Bytes per second
        Forward-rebuild-reads                     0 IOPS 0 Bytes per second
        Forward-rebuild-writes                    0 IOPS 0 Bytes per second
        Rebalance-reads                           0 IOPS 0 Bytes per second
        Rebalance-writes                          0 IOPS 0 Bytes per second
        Volume-migration-reads                    0 IOPS 0 Bytes per second
        Volume-migration-writes                   0 IOPS 0 Bytes per second
        Enter-protected-maintenance-mode-reads    0 IOPS 0 Bytes per second
```

```
        Enter-protected-maintenance-mode-writes  0 IOPS 0 Bytes per second
        Exit-protected-maintenance-mode-reads    0 IOPS 0 Bytes per second
        Exit-protected-maintenance-mode-writes   0 IOPS 0 Bytes per second

Volumes summary:
        6 thin-provisioned volumes. Total user data size: 30.6 GB (31378 MB)
        1 snapshot. Total user data size: 0 Bytes
        64.0 GB (65536 MB) all volumes size
        56.0 GB (57344 MB) base volumes size
        6 volumes are not mapped
        1 snapshot policy
        1 volume are replicated
        3 replication journal volumes
        3 user volumes

SDR Summary:
        Total 3 SDR Nodes
        3 SDR nodes have membership state 'Joined'
        3 SDR nodes have connection state 'Connected'

Replication Peer System Summary:
        Total 1 Replication Peer System Nodes
        1 Replication Peer System nodes have membership state 'Joined'
```

Output description

The following tables explain the output for `query_all`.

Table 4. System information

System Info	Description
Product	The PowerFlex version of the system
ID	The system ID number
Manager ID	The ID of the management application that controls the MDM

Table 5. License information

License Info	Description
Installation ID	The Installation ID number, which is required for licensing purposes
SWID	The SWID number, which is required for licensing purposes
Maximum capacity	The maximum capacity permitted by the system's license
Usage time left	The amount of time left until the license expires
Enterprise features	Indicates whether enterprise features are enabled
The system was activated *n* days ago	The number of days since the system was activated

Table 6. System settings

System Settings	Description
Volumes are/are not obfuscated	The obfuscation setting for all volumes in the system. Default: not obfuscated.
Capacity alert thresholds	The alert thresholds, as percentages, currently configured for capacity usage
Thick volume reservation	The amount of capacity, as a percentage, reserved for thick volumes,
MDM restricted SDC mode	Indicates whether users on remote clients are restricted to read-only access:

Table 6. System settings (continued)

System Settings	Description
	• enabled: read-only access • disabled: full read-write access
Management Clients secure communication	The status of the enabled/disabled
TLS version	
User authentication method	
SDS connection authentication	The status of the ..... : enabled/disabled

Table 7. Protection domain information

Protection Domain	Description
Protection Domain summary	The following items are displayed separately for each Protection Domain: • Name • ID • Number of Storage Pools in the Protection Domain • Number of Fault Sets in the Protection Domain • Number of SDS nodes in the Protection Domain • Number of volumes associated with the Protection Domain • Amount of capacity in the Protection Domain that is available for volume allocation
Operational state	Indicates the Protection Domain's current state, such as active, inactive, and so on.
RFcache status	Indicates whether RFcache is enabled/disabled in the Protection Domain. If RFcache is enabled, the following information is displayed: • Mode: pass-through mode used by the caching feature • Page size: cache page size in KB • Maximum I/O size: cache maximum I/O size in KB

Table 8. Storage pool information

Storage Pool	Description
Storage Pool summary	Lists the following information for each Storage Pool: • Name • ID • Number of volumes associated with the Storage Pool • Amount of capacity in the Storage Pool that is available for volume allocation
The number of parallel rebuild/rebalance jobs	The number of parallel Rebuild and Rebalance jobs currently existing in the system
Rebuild is enabled/disabled	Indicates whether the Rebuild feature is enabled or disabled. (In normal production conditions, Rebuild should be enabled.) When enabled, the Rebuild policy configured in the system is displayed, along with the relevant parameters.
Rebalance is enabled/disabled	Indicates whether the Rebalance feature is enabled or disabled. (In normal production conditions, Rebalance should be enabled.) When enabled, the Rebalance policy configured in the system is displayed, along with the relevant parameters.
Background device scanner	Displays the mode and the bandwidth limit per device
Zero padding is enabled/disabled	Indicates whether zero padding is enabled or disabled

Table 8. Storage pool information (continued)

Storage Pool	Description
Spare policy	The percentage of total capacity reserved as Spare
Read RAM Cache	Indicates whether the Read RAM Cache feature is used in the Storage Pool. If the Read RAM Cache is used, it displays the write handling mode currently in use (passthrough or cached)
Read Flash Cache	Indicates whether Read Flash Cache is used in the Storage Pool
Capacity alert thresholds	The alert thresholds, as percentages, configured for capacity usage. Thresholds may be configured for the entire system, or per Storage Pool.

Table 9. SDS summary and capacity information

SDS Summary	Description
Total	The total number of SDS nodes
n SDS nodes have membership state 'Joined'	The number of SDS nodes that are connected to the cluster, can receive I/O, and hold primary and secondary copies of data. It takes some time from actual disconnection until the SDS is disconnected from the cluster, or the reverse. There are also up-pending and down-pending states.
n SDS nodes have connection state 'Connected'	The number of SDS nodes currently connected to the MDM
x TB (*y* GB) total capacity	The total amount of available raw storage This does not represent the total capacity available for volume allocation.
x TB (*y* GB) unused capacity	The quantity of raw capacity in the system that can be earmarked for specific purposes, such as Spare, or used for new volume creation
x Bytes snapshots capacity	The quantity of capacity used for storing snapshots
x TB (*y* GB) in-use capacity	The total quantity of healthy, degraded, and failed capacity
x Bytes thin capacity	The quantity of capacity currently needed for storage purposes
x TB (*y* GB) protected capacity	The quantity of capacity that is fully protected (primary and secondary copies of the data exist)
x Bytes failed capacity	The quantity of capacity that is not available at all (neither primary, nor secondary copies)
x Bytes degraded-failed capacity	The quantity of degraded-failed capacity. When an SDS fails, all of its capacity is defined as degraded-failed, and the secondary copies (which are spread across the Protection Domain) are defined as degraded-healthy.
x Bytes degraded-healthy capacity	The quantity of degraded-healthy capacity. When an SDS fails, all of its capacity is defined as degraded-failed, and the secondary copies (which are spread across the Protection Domain) are defined as degraded-healthy.
x Bytes unreachable-unused capacity	The quantity of capacity not configured for any use type in the system that is currently unavailable
x Bytes active rebalance capacity	The quantity of capacity that is currently being migrated to a different location for load balancing purposes

Table 9. SDS summary and capacity information (continued)

SDS Summary	Description
x Bytes pending rebalance capacity	The quantity of capacity that is waiting in the job queue for migration to a different location for load balancing purposes
x Bytes active forward-rebuild capacity	The quantity of capacity for which one copy of data exists and a second copy is currently being created
x Bytes pending forward-rebuild capacity	The quantity of capacity for which one copy of data exists, and a job for the creation of a second copy is waiting in the job queue
x Bytes active backward-rebuild capacity	The quantity of capacity for which one copy of data went offline and came back online, and changes are currently being synchronized in that copy
x Bytes pending backward-rebuild capacity	The quantity of capacity for which one copy of data went offline and came back online, and changes are waiting in the job queue to be synchronized in that copy
x Bytes rebalance capacity	The total quantity of capacity that is either currently rebalancing or is pending Rebalance
x Bytes forward-rebuild capacity	The total quantity of degraded capacity that is either currently in Forward Rebuild state or is pending Forward Rebuild
x Bytes backward-rebuild capacity	The total quantity of degraded capacity that is either currently in Backward Rebuild state or is pending Backward Rebuild
x Bytes active moving capacity	The quantity of capacity that is currently being migrated from one location to another
x Bytes pending moving capacity	The quantity of capacity that is waiting in the job queue for migration from one location to another
x Bytes total moving capacity	The total quantity of active and pending moving (migrating) capacity
x TB (*y* GB) spare capacity	The quantity of capacity that is reserved for system use when recovery from failure is required. This capacity cannot be used for storage purposes.
x TB (*y* GB) at-rest capacity	The quantity of capacity that is fully protected and not in a Rebuild or Rebalance state
x Bytes semi-protected capacity	
x Bytes in-maintenance capacity	
x Bytes decreased capacity	The quantity of Decreased capacity that was deducted from devices (using the **Set Device Capacity Limit** GUI command or `modify_sds_device_capacity` CLI command) and cannot be used for any purpose
Primary-reads *x* IOPS *y* Bytes per second	Number of primary data copy read IOPS, and read bandwidth
Primary-writes *x* IOPS *y* Bytes per second	Number of primary data copy write IOPS, and write bandwidth
Secondary-reads *x* IOPS *y* Bytes per second	Number of secondary data copy (protection) read IOPS, and read bandwidth
Secondary-writes *x* IOPS *y* Bytes per second	Number of secondary copy (protection) write IOPS, and write bandwidth
Backward-rebuild-reads *x* IOPS *y* Bytes per second	Number of Backward Rebuild read IOPS, and read bandwidth
Backward-rebuild-writes *x* IOPS *y* Bytes per second	Number of Backward Rebuild write IOPS, and write bandwidth
Forward-rebuild-reads *x* IOPS *y* Bytes per second	Number of Forward Rebuild read IOPS, and read bandwidth
Forward-rebuild-writes *x* IOPS *y* Bytes per second	Number of Forward Rebuild write IOPS, and write bandwidth

Table 9. SDS summary and capacity information (continued)

SDS Summary	Description
Rebalance-reads *x* IOPS *y* Bytes per second	Number of Rebalance read IOPS, and read bandwidth
Rebalance-writes *x* IOPS *y* Bytes per second	Number of Rebalance write IOPS, and write bandwidth

Table 10. Volumes summary

Volumes summary	Description
n thick-provisioned volumes	The number of thick-provisioned volumes, and the total size in TB and GB
n volumes mapped to all SDC nodes	The number of volumes mapped to SDC nodes

query_cluster

Retrieve MDM cluster information.

Syntax

```
scli --query_cluster [--show_certificate_info]
```

Parameters

--show_certificate_info

Display certificates for all the MDMs

Example

```
scli --mdm_ip 192.168.1.200 --query_cluster --show_certificate_info
```

Sample output

The following is sample output for a five-node cluster:

```
scli --query_cluster
Cluster:
    Mode: 5_node, State: Normal, Active: 5/5, Replicas: 3/3
    Virtual IP Addresses: N/A
Master MDM:
    Name: MDM_116, ID: 0x155c8db3025d5f01
        IP Addresses: 10.55.137.116, Port: 9011, Virtual IP interfaces: N/A
        Certificate Info:
            Subject:    /GN=MDM/CN=vm-centos8u1-3.sio.lab.emc.com/L=Hopkinton/
ST=Massachusetts/C=US/O=EMC/OU=ASD
            Issuer:     /GN=MDM/CN=vm-centos8u1-3.sio.lab.emc.com/L=Hopkinton/
ST=Massachusetts/C=US/O=EMC/OU=ASD
            Valid From: Jan 30 09:34:16 2021 GMT
            Valid To:   Jan 29 10:34:16 2031 GMT
            Thumbprint: 2E:68:A4:48:A3:BC:DF:77:D9:8B:1C:77:01:99:04:5B:1D:F9:77:B6
Slave MDMs:
    Name: MDM_114, ID: 0x75c68cb264723f00
        IP Addresses: 10.55.137.114, Port: 9011, Virtual IP interfaces: N/A
        Certificate Info:
            Subject:    /GN=MDM/CN=vm-centos8u1-1.sio.lab.emc.com/L=Hopkinton/
ST=Massachusetts/C=US/O=EMC/OU=ASD
```

```
            Issuer:     /GN=MDM/CN=vm-centos8u1-1.sio.lab.emc.com/L=Hopkinton/
ST=Massachusetts/C=US/O=EMC/OU=ASD
            Valid From: Jan 30 09:34:16 2021 GMT
            Valid To:   Jan 29 10:34:16 2031 GMT
            Thumbprint: F7:B6:37:3C:45:EF:98:5B:64:57:5E:D3:7D:A1:B7:AD:E4:7B:5A:EA
    Name: MDM_115, ID: 0x7386c66b551fdf02
        IP Addresses: 10.55.137.115, Port: 9011, Virtual IP interfaces: N/A
        Certificate Info:
            Subject:    /GN=MDM/CN=localhost/L=Hopkinton/ST=Massachusetts/C=US/O=EMC/
OU=ASD
            Issuer:     /GN=MDM/CN=localhost/L=Hopkinton/ST=Massachusetts/C=US/O=EMC/
OU=ASD
            Valid From: Jan 30 09:34:18 2021 GMT
            Valid To:   Jan 29 10:34:18 2031 GMT
            Thumbprint: A8:FB:1E:A2:6C:92:32:6C:42:FB:A0:A3:50:1D:6F:90:06:E2:1B:A0
```

Possible values for items in the sample output

Table 11. Possible values for query_cluster output items

Item	Possible values
Mode	Cluster mode: • 1_node • 3_node • 5_node
State	State of the cluster: • Normal • Degraded (one or more cluster members are down or inactive)
Active	The number of cluster members that are active. If all members are in normal state, this number equals the number of cluster members.
Replicas	The number of MDM repository holders (the primary MDM, plus the number of secondary MDMs in normal state)
Secondary MDM status	• Normal • Error • Disconnected • Not synchronized
Tiebreaker status	• Normal • Error • Disconnected
Standby MDM	Role (Manager or tiebreaker)

query_performance_parameters

Retrieve performance parameters. The output displays the configurations of performance-related parameters for the performance profile that is currently active. When no parameters are added to the command, the name of the active profile is shown.

Syntax

```
scli --query_performance_parameters
  [--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP> [--sds_ip <IP>] | --all_sds]
[--sdc_id <ID> | --sdc_name <NAME> | --sdc_guid <GUID> | --sdc_name <NAME> | --
all_sdc] [--peer_system_id <ID> | --peer_system_name <NAME> | --peer_system_ip <IP> [--
peer_system_port <PORT>] | --peer_system_object_id <ID> | --all_peer_system] [--sdr_id
<ID> | --sdr_name <NAME> | --sdr_ip <IP> [--sdr_port <PORT>] | --all_sdr] [--print_all]
```

Parameters

--sds_id *<ID>*
SDS ID

--sds_name *<NAME>*
SDS name

--sds_ip *<IP>*
SDS IP address

--sds_port *<PORT>*
Port associated with the SDS

--all_sds
Query SDS performance parameters from all SDSs.

--sdc_id *<ID>*
SDC ID

--sdc_name *<NAME>*
SDC name

--sdc_guid *<GUID>*
SDC Global Unique Identifier

--sdc_ip *<IP>*
SDC IP address

--all_sdc
Query SDC performance parameters from all SDCs.

--peer_system_id *<ID>*
Replication Peer System ID

--peer_system_name *<NAME>*
Replication Peer System Name

--peer_system_ip *<IP>*
Replication Peer System IP address

--peer_system_port *<PORT>*
Port assigned to the Replication Peer System

--peer_system_object_id *<ID>*
Replication Peer System internal object ID

--all_peer_system
Query Replication Peer System performance parameters from all Replication Peer Systems.

--sdr_id *<ID>*
SDR ID

--sdr_name *<NAME>*
SDR name

--sdr_ip *<IP>*
SDR IP address

--sdr_port *<PORT>*
Port assigned to the SDR

--all_sdr
Query SDR performance parameters from all SDRs.

--print_all
Print all performance parameters, including those with active profile default values.

Example

```
scli --query_performance_parameters --print_all
```

Sample output

```
MDM Performance Parameters:
        Active profile: high_performance
                                                                  Current Value
Profile Value
        MDM_NUMBER_SDC_RECEIVE_UMT                                        5
5
        MDM_NUMBER_SDS_RECEIVE_UMT                                       20
20
        MDM_NUMBER_SDS_SEND_UMT                                          10
10
        MDM_NUMBER_SDS_KEEPALIVE_RECEIVE_UMT                             10
10
        MDM_SDS_CAPACITY_COUNTERS_UPDATE_INTERVAL                         1
1
        MDM_SDS_CAPACITY_COUNTERS_POLLING_INTERVAL                        5
5
        MDM_SDS_VOLUME_SIZE_POLLING_INTERVAL                             15
15
        MDM_SDS_VOLUME_SIZE_POLLING_RETRY_INTERVAL                        5
5
        MDM_NUMBER_SDS_TASKS_UMT                                       1536
1536
        MDM_INITIAL_SDS_SNAPSHOT_CAPACITY                              1024
1024
        MDM_SDS_SNAPSHOT_CAPACITY_CHUNK_SIZE                           5120
5120
        MDM_SDS_SNAPSHOT_USED_CAPACITY_THRESHOLD                         50
50
        MDM_SDS_SNAPSHOT_FREE_CAPACITY_THRESHOLD                        200
200
        MDM_NET_ALLOC_RCV_BUFFER_WAIT_MS                                500
500
        MDM_NET_BREAK_DO_IO_LOOP                                          5
5
        MDM_FGL_INITIAL_SNAPSHOT_CAPACITY                              5120
5120
        MDM_FGL_SNAPSHOT_CAPACITY_CHUNK_SIZE                           5120
5120
        MDM_FGL_SNAPSHOT_USED_CAPACITY_THRESHOLD                         50
50
        MDM_FGL_SNAPSHOT_FREE_CAPACITY_THRESHOLD                        200
200
        MDM_RENAME_DISABLE_MODE_ON_SNAP_DELETION                          0
0
        MDM_PORT_FLAP_FEATURE_ENABLED                                     1
1
        MDM_DISQUALIFICATION_EPOCH_SEC                                  120
120
        MDM_DISCONNECTIONS_ERT                                            3
3
        MDM_DISCONNECTIONS_ERTF_MS                                    60000
60000
        MDM_HIGH_LATENCY_ERT                                             50
50
        MDM_HIGH_LATENCY_ERTF_MS                                      60000
60000
        MDM_PACKET_LOSS_ERT                                              50
50
        MDM_PACKET_LOSS_ERTF_MS                                       60000
60000
        RPL_MDM_PORT_FLAP_FEATURE_ENABLED                                 0
0
```

```
        RPL_MDM_DISQUALIFICATION_EPOCH_SEC                          120
120
        RPL_MDM_DISCONNECTIONS_ERT                                    3
3
        RPL_MDM_DISCONNECTIONS_ERTF_MS                            60000
60000
        RPL_MDM_HIGH_LATENCY_ERT                                     50
50
        RPL_MDM_HIGH_LATENCY_ERTF_MS                              60000
60000
        RPL_MDM_PACKET_LOSS_ERT                                      50
50
        RPL_MDM_PACKET_LOSS_ERTF_MS                               60000
60000
        RPL_MDM_CAP_MGR_JOURNAL_VOLUME_ALLOC_SIZE_GB                  8
8
        RPL_MDM_SLEEP_TIME_FOR_CE_ANALYZER_MS                       500
500
        RPL_MDM_SYNCED_OPERATION_TIMEOUT_SECONDS                     30
30
        RPL_MDM_CAP_MGR_MONITOR_TIMEOUT_MS                        1000
1000
        RPL_MDM_NUM_CG_TASKS_UMT                                      1
1
        RPL_MDM_NUM_PAIR_TASKS_UMT                                    1
1
        RPL_MDM_NUM_SDR_RECEIVE_UMT                                  10
10
        RPL_MDM_NUM_SDR_SEND_UMT                                     10
10
        RPL_MDM_NUM_SDR_KEEPALIVE_RECEIVE_UMT                        10
10
        RPL_MDM_SDR_COUNTERS_POLLING_INTERVAL_SECONDS                 1
1
        RPL_MDM_CAP_MGR_INCREASE_THRESHOLD_SIZE_GB                    4
4
        RPL_MDM_CAP_MGR_DECREASE_CAP_TIMEOUT_MS                    2000
2000
        RPL_MDM_RPO_DRIFT_UNTIL_BARRIER                              64
64
        RPL_MDM_NUM_SDR_TASKS_UMT                                   512
512
        RPL_MDM_NUM_PEER_MDM_TASKS_UMT                              126
126
        RPL_MDM_NUM_PEER_MDM_RECEIVE_UMT                             10
10
        RPL_MDM_NUM_PEER_MDM_SEND_UMT                                10
10
        RPL_MDM_CURRENT_BARRIER_MAX_NUM_IOS_THRESHOLD          1999000000
1999000000
        RPL_MDM_CURRENT_BARRIER_MAX_SIZE_MB_THRESHOLD             65536
65536
        RPL_MDM_SDR_SESSION_CAPACITY_LOW_WATERMARK_MB             20480
20480
        RPL_MDM_SDR_SESSION_CAPACITY_HIGH_WATERMARK_MB             4096
4096
        RPL_MDM_CAP_MGR_CRITICAL_CAPACITY_THRESHOLD_MB               32
32
        RPL_MDM_CAP_MGR_SWITCH_SESSIONS_INTERVAL_MS               120000
120000
        RPL_MDM_CAP_MGR_MINOR_UTILIZATION_THRESHOLD                  75
75
        RPL_MDM_CAP_MGR_MAJOR_UTILIZATION_THRESHOLD                  80
80
        RPL_MDM_CAP_MGR_VERY_HIGH_UTILIZATION_THRESHOLD              90
90
        RPL_MDM_CAP_MGR_CRITICAL_UTILIZATION_THRESHOLD               95
95
```

query_properties

Retrieve any set of properties, on any set of objects of the specified type.

The properties that can be queried depend on the type of the object that is being queried. For example, SDS has different attributes than Volume.

A number of preset combinations are available for use with this command. A preset is a predefined set of properties, usually with some logical relationship, which specifies a group of commonly queried properties. The available presets are also dependent on the object type. All object types have at least one preset called `all`, which queries all the properties for that type.

Syntax

```
scli --query_properties --object_type <TYPE> (--object_id <ID> | --all_objects) (--
properties <PROPS> | --preset <PRESET>)
  [--group_by_property]
  [--display_property_options]
  [--flat_list]
```

Parameters

`--object_type <TYPE>`

Object type to query. One of:

SYSTEM

PROTECTION_DOMAIN

STORAGE_POOL

FAULT_SET

SDS

Host

VOLUME

DEVICE

VTREE

MDM

SP_SDS

ACCELERATION_POOL

SNAPSHOT_POLICY

SDR

PEER_MDM

REPLICATION_PAIR

REPLICATION_CONSISTENCY_GROUP

SDT

NOTE: To view detailed presets and properties options for a type, enter `scli --query_properties --object_type <TYPE>`, where *<TYPE>* is one of the above options.

`--object_id <ID>`

Comma-separated list of object ID used for querying one or more specific objects. Omit the space after each comma.

`--all_objects`

Query all objects of the specified type.

`--properties <PROPS>`

Query a comma-separated list of properties. Omit the space after each comma. See the scli help output for a list of properties that can be queried per object type

--preset *<PRESET>*

Query a predefined set of properties. Omit the space after each comma. See the scli help output for the list of presets that can be used per object type.

--group_by_property

Group results by property type, rather than by object (the default).

--display_property_options

Display all available properties grouped by object type.

--flat_list

Print results as a flat list rather than by group (better for searching).

NOTE: Use `scli --query_properties --object_type <TYPE>` with one of the valid types above, to get detailed presets and properties options.

Example

```
scli --query_properties --object_type SDS --all_objects --properties
IPS,PORT,PROTECTION_DOMAIN_ID,RMCACHE_ENABLED
```

Sample output

```
# scli --query_properties --object_type SDS --all_objects --properties
IPS,PORT,PROTECTION_DOMAIN_ID,RMCACHE_ENABLED
SDS 5fcafa4300000001:
        IPS                                             10.55.105.40,10.76.30.11
        PORT                                            7072
        PROTECTION_DOMAIN_ID                            a080539500000000
        RMCACHE_ENABLED                                 Yes

SDS 5fcafa4600000004:
        IPS                                             10.55.105.43,10.76.30.15
        PORT                                            7072
        PROTECTION_DOMAIN_ID                            a080539500000000
        RMCACHE_ENABLED                                 Yes
```

Object presets

The following table provides a list of presets that can be used per object type.

Table 12. Object presets for query_properties

Object type	ALL	CAPACITY	RPL_JOURNAL_CAPACITY	RPL_BW_COUNTERS	IO	RAM_CACHE	PERFORMANCE_PARAMETERS	RF_CACHE
SYSTEM	x	x	N/A	N/A	x	x	x	x
PROTECTION_DOMAIN	x	x	N/A	N/A	x	x	N/A	x
STORAGE_POOL	x	x	x	N/A	x	x	x	x
FAULT_SET	x	x	N/A	N/A	x	x	N/A	x
SDS	x	x	N/A	N/A	x	x	x	x
SDC	x	N/A	N/A	N/A	x	N/A	x	N/A
VOLUME	x	N/A	N/A	N/A	x	x	N/A	N/A

Table 12. Object presets for query_properties (continued)

Object type	ALL	CAPACITY	RPL_JOURNAL_CAPACITY	RPL_BW_COUNTERS	IO	RAM_CACHE	PERFORMANCE_PARAMETERS	RF_CACHE
DEVICE	x	x	N/A	N/A	x	N/A	N/A	x
VTREE	x	x	N/A	N/A	N/A	N/A	N/A	N/A
MDM	x	N/A	N/A	N/A	N/A	N/A	N/A	N/A
SP_SDS	x	x	N/A	N/A	x	N/A	N/A	x
ACCELERATION_POOL	x	N/A	N/A	N/A	N/A	N/A	N/A	N/A
SNAPSHOT_POLICY	x	N/A	N/A	N/A	N/A	N/A	N/A	N/A
SDR	x	N/A	x	x	N/A	N/A	x	N/A
PEER_MDM	x	N/A	N/A	N/A	N/A	N/A	x	N/A
REPLICATION_PAIR	x	N/A	N/A	N/A	N/A	N/A	N/A	N/A
REPLICATION_CONSISTENCY_GROUP	x	N/A	N/A	N/A	N/A	N/A	N/A	N/A

query_remote_read_only_limit_state

Query the MDM's remote access restriction state. If the state is enabled, remote users may only issue read-only commands to the MDM. If the state is disabled, all command types may be issued to the MDM by remote clients.

Syntax

```
scli --query_remote_read_only_limit_state
```

Parameters

None.

Example

```
scli --query_remote_read_only_limit_state
```

query_system_limits

Retrieve information about the system limits.

Syntax

```
scli --query_system_limits
```

Parameters

None.

Example

```
scli --query_system_limits
```

Sample output

```
System limits (45 entries)
  Number of protections domains:  Max: 256
  Number of storage pools:  Max: 1024
  Number of fault sets:  Max: 16384
  Number of SDSs:  Max: 512
  Number of SDCs:  Max: 1024
  Number of volume trees:  Max: 131072
  Number of volumes:  Max: 131072
  Number of snapshots:  Max: 133120
  Number of system users:  Max: 256
  Number of LDAP services:  Max: 8
  Number of sessions:  Max: 128
  Number of rsyslog clients:  Max: 16
  Number of storage pools per protection domain:  Max: 64
  Number of SDSs per protection domain:  Max: 128
  Number of fault sets per protection domain:  Max: 64
  Number of SDSs per fault set:  Max: 64
  Number of devices per SDS:  Max: 64
  Number of RFcache devices per SDS:  Max: 8
  Number of devices per storage pool:  Max: 300
  Number of volumes per medium-granularity volume tree:  Max: 127
  Number of volumes per fine-granularity volume tree:  Max: 127
  Number of snapshots per medium-granularity volume tree:  Max: 254
  Number of snapshots per fine-granularity volume tree:  Max: 254
  Number of volumes mapped to a single SDC:  Max: 8192
  Number of volumes per protection domain:  Max: 32768
  Number of LDAP groups per LDAP service:  Max: 8
  Number of IPs per MDM:  Min: 1  Max: 4
  Number of IPs per SDS/SDC:  Min: 1  Max: 8
  Single volume size in GB:  Min: 8  Max: 1048576
  Total size of all volumes in Storage Pool in GB:  Max: 4194303
  Total size of all volumes in System (GB):  Max: 16777216
  Number of VTrees per protection domain:  Max: 32768
  Number of acceleration devices in acceleration pool:  Max: 512
  Number of acceleration pools per protections domain:  Max: 32
  Number of acceleration devices per acceleration pool that belong to the same SDS:
Max: 8
  Capacity per medium granularity device in GB:  Min: 90
  Capacity per fine granularity device in GB:  Min: 128
  Capacity per SDS in GB:  Max: 131072
  Number of Replication Consistency Groups:  Max: 1024
  Number of Replication Destination Systems:  Max: 1
```

```
   Total Number of SDRs per system:  Max: 128
   Total Number of Replication Pairs in system:  Max: 32768
   Total Number of concurrent Move Pairs operations:  Max: 512
   Single replication volume size in GB:  Min: 8  Max: 65536
   Number of mappings between volumes and SDCs:  Max: 262143
```

query_upgrade

Retrieve information about the upgrade process.

Syntax

```
scli --query_upgrade
   [--show_sdc_info]
```

Parameters

--show_sdc_info
Include SDC information

Example

```
scli --query_upgrade --show_sdc_info
```

refresh_mdm_cluster_capabilities

Refresh the exposed cluster node capabilities.

This command is usually used to instruct the cluster nodes to check whether LDAP and SSL libraries are installed.

Syntax

```
scli --refresh_mdm_cluster_capabilities
```

Parameters

Example

```
scli --refresh_mdm_cluster_capabilities --mdm_port 6611
```

remove_standby_mdm

Remove a standby MDM or standby tiebreaker, that is not part of the cluster, from the system.

Syntax

```
scli --remove_standby_mdm (--remove_mdm_id <ID> | --remove_mdm_ip <IP> [--
remove_mdm_port <PORT>] | --remove_mdm_name <NAME>)
```

Parameters

--remove_mdm_id *<ID>*
ID of the MDM to remove

--remove_mdm_ip *<IP>*
IP address of the MDM to remove

--remove_mdm_port *<PORT>*
Port associated with the MDM to remove

--remove_mdm_name *<NAME>*
Name of the MDM to remove

Example

```
scli --remove_standby_mdm --remove_mdm_name mdm153
```

rename_mdm

Rename an MDM Cluster node.

Syntax

```
scli --rename_mdm (--target_mdm_id <ID> | --target_mdm_ip <IP> [--target_mdm_port
<PORT>] | --target_mdm_name <NAME>) --new_mdm_name <NAME>
```

Parameters

--target_mdm_id *<ID>*
ID of the MDM to be renamed

--target_mdm_ip *<IP>*
IP address of the MDM to be renamed

--target_mdm_port *<PORT>*
Port of the MDM to be renamed

--target_mdm_name *<NAME>*
Current name of the MDM to be renamed

--new_mdm_name *<NAME>*
New MDM name. The MDM name must be unique. If a new name is not provided, the old name will be deleted.

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

Example

```
scli --rename_mdm --new_mdm_name mdm3 --target_mdm_name mdm2
```

rename_system

Name or rename the system.

To determine the current system name, use the `query_cluster` or `query_all` command.

Syntax

```
scli --rename_system --new_name <NAME>
```

Parameters

--new_name *<NAME>*

New name to associate with the system.

Example

```
scli --rename_system --new_name sio-prod-sys2
```

replace_cluster_mdm

Replace members of the MDM cluster. Up to half (rounded down) of the cluster members can be replaced at one time.

Syntax

```
scli --replace_cluster_mdm ((--add_slave_mdm_id <ID> | --add_slave_mdm_ip <IP> [--
add_slave_mdm_port <PORT>] | --add_slave_mdm_name <NAME>)| (--remove_slave_mdm_id <ID>
| --remove_slave_mdm_ip <IP> [--remove_slave_mdm_port <PORT>] | --remove_slave_mdm_name
<NAME>)) ((--add_tb_id <ID> | --add_tb_ip <IP> [--add_tb_port <PORT>] | --add_tb_name
<NAME>) | (--remove_tb_id <ID> | --remove_tb_ip <IP> [--remove_tb_port <PORT>] | --
remove_tb_name <NAME>))
  [--allow_leave_failed]
  [--i_am_sure]
```

Parameters

--add_slave_mdm_id *<ID>*

Comma-separated list of secondary MDM IDs that will be part of the cluster. A maximum of two IDs are allowed. Omit the space after the comma.

--add_slave_mdm_ip *<IP>*

Comma-separated list of secondary MDM IP addresses that will be part of the cluster. A maximum of two IP addresses are allowed. Omit the space after the comma.

--add_slave_mdm_port *<PORT>*

Comma-separated list of secondary MDM ports that will be part of the cluster. A maximum of two ports are allowed. Omit the space after the comma.

--add_slave_mdm_name *<NAME>*

Comma-separated list of secondary MDM names that will be part of the cluster. A maximum of two names are allowed. Omit the space after the comma.

--remove_slave_mdm_id *<ID>*

Comma-separated list of secondary MDM IDs that will be removed from the cluster. A maximum of two IDs are allowed. Omit the space after the comma.

--remove_slave_mdm_ip *<IP>*

Comma-separated list of secondary MDM IP addresses that will be removed from the cluster. A maximum of two IP addresses are allowed. Omit the space after the comma.

--remove_slave_mdm_port *<PORT>*

Comma-separated list of secondary MDM ports that will be removed from the cluster. A maximum of two ports are allowed. Omit the space after the comma.

--remove_slave_mdm_name *<NAME>*

Comma-separated list of secondary MDM names that will be removed from the cluster. A maximum of two names are allowed. Omit the space after the comma.

--add_tb_id *<ID>*

Comma-separated list of tiebreaker IDs that will be part of the cluster. A maximum of two IDs are allowed. Omit the space after the comma.

--add_tb_ip *<IP>*

Comma-separated list of tiebreaker IP addresses that will be part of the cluster. A maximum of two IP addresses are allowed. Omit the space after the comma.

--add_tb_port *<PORT>*

Comma-separated list of tiebreaker ports that will be part of the cluster. A maximum of two ports are allowed. Omit the space after the comma.

--add_tb_name *<NAME>*

Comma-separated list of tiebreaker names that will be part of the cluster. A maximum of two names are allowed. Omit the space after the comma.

--remove_tb_id *<ID>*

Comma-separated list of tiebreaker IDs that will be removed from the cluster. A maximum of two IDs are allowed. Omit the space after the comma.

--remove_tb_ip *<IP>*

Comma-separated list of tiebreaker IP addresses that will be removed from the cluster. A maximum of two IP addresses are allowed. Omit the space after the comma.

--remove_tb_port *<PORT>*

Comma-separated list of tiebreaker ports that will be removed from the cluster. A maximum of two ports are allowed. Omit the space after the comma.

--remove_tb_name *<NAME>*

Comma-separated list of tiebreaker names that will be removed from the cluster. A maximum of two names are allowed. Omit the space after the comma.

--allow_leave_failed

Allow leaving currently failed MDMs in the cluster.

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

Example

```
scli --replace_cluster_mdm --add_slave_mdm_name mdm33 --remove_slave_mdm_name mdm31 --
add_tb_name tb44 --remove_tb_name tb41
```

replace_mdm_security_files

Replace the MDM's security certificate and key. The replacement certificate can be either a self-generated and self-signed certificate, or a certificate generated and signed by an external Certificate Authority. Security permissions are required in order to execute this command.

NOTE: Running the command on the primary MDM causes the MDM to restart. It may also cause a brief, single point of failure period due to an MDM switch over. For more information on externally signed security certificates, and the entire workflow, see the *Configure and Customize Dell PowerFlex*.

Syntax

```
scli --replace_mdm_security_files (--target_mdm_id <ID> | --target_mdm_ip <IP> [--
target_mdm_port <PORT>] | --target_mdm_name <NAME>)
```

Parameters

--target_mdm_id *<ID>*
MDM ID

--target_mdm_ip *<IP>*
MDM IP address

--target_mdm_port *<PORT>*
MDM port

--target_mdm_name *<NAME>*
MDM name

Example

```
scli --replace_mdm_security_files --target_mdm_name mdm153
```

set_component_authentication_properties

Enable or disable component authentication properties.

Syntax

```
scli --set_component_authentication_properties (--use_authentication | --
dont_use_authentication)
```

Parameters

None.

--use_authentication

Use authentication between system components

--dont_use_authentication

Do not use authentication between system components

Example

```
scli --set_component_authentication_properties --use_authentication
```

set_management_client_communication

Enable or disable secure communications between the MDM and the management client.

Syntax

```
scli --set_management_client_communication (--enable_client_secure_communication | --
disable_client_secure_communication)
```

Parameters

--enable_client_secure_communication

Enable the use of security when communicating with management clients.

--disable_client_secure_communication

Disable the use of security when communicating with management clients.

Example

```
scli --set_management_client_communication --enable_client_secure_communication
```

set_performance_parameters

Apply performance profiles to system components. You can apply separate profiles for SDSs, SDCs, the MDM cluster, replication peer systems, and SDRs.

When a container is provided in the command (Protection Domain, Fault Set, all SDSs, all SDCs), the command configures all objects currently in the container. Objects that are added later will not be configured.

NOTE: If you change the performance profile of an SDS located on an SVM, you must afterwards perform manual memory allocation on the SVM.

Syntax

```
scli --set_performance_parameters ([--sds_id <ID> | --sds_name <NAME> | --sds_ip
<IP> [--sds_port <PORT>] | --protection_domain_id <ID> | --protection_domain_name
<NAME> | --fault_set_id <ID> | ((--protection_domain_id <ID> | --protection_domain_name
<NAME>) --fault_set_name <NAME>) | --all_sds] [--sdc_id <ID> | --sdc_name <NAME>
| --sdc_guid <GUID> | --sdc_ip <IP> | --all_sdc] [(((--protection_domain_id <ID>
| --protection_domain_name <NAME>) --storage_pool_name <NAME>) | --storage_pool_id
<ID>)] [--peer_system_id <ID> | --peer_system_name <NAME> | --peer_system_ip <IP>
[--peer_system_port <PORT>] | --peer_system_object_id <ID> | --all_peer_system] [--
```

```
sdr_id <ID> | --sdr_name <NAME> | --sdr_ip <IP> [--sdr_port <PORT>] | --all_sdr] [--
apply_to_mdm]) (--profile <PROFILE>)
```

Parameters

--sds_id *<ID>*
SDS ID

--sds_name *<NAME>*
SDS name

--sds_ip *<IP>*
SDS IP address

--sds_port *<PORT>*
Port associated with the SDS

--protection_domain_id *<ID>*
Protection Domain ID

--protection_domain_name *<NAME>*
Protection Domain name

--fault_set_id *<ID>*
Fault Set ID

--fault_set_name *<NAME>*
Name of the new Fault Set

--all_sds
Set SDS-related parameters on all SDSs.

--sdc_id *<ID>*
SDC ID

--sdc_name *<NAME>*
SDC name

--sdc_guid *<GUID>*
SDC Global Unique Identifier

--sdc_ip *<IP>*
SDC IP address

--all_sdc
Set SDS-related parameters on all SDCs.

--peer_system_id *<ID>*
Replication Peer System ID

--peer_system_name *<NAME>*
Replication Peer System Name

--peer_system_ip *<IP>*
Replication Peer System IP address

--peer_system_port *<PORT>*
Port assigned to the Replication Peer System

--peer_system_object_id *<ID>*
Replication Peer System internal object ID

--all_peer_system
Set Replication Peer System related parameters to all Replication Peer Systems.

--sdr_id *<ID>*
SDR ID

--sdr_name *<NAME>*
SDR name

--sdr_ip *<IP>*

SDR IP address

--sdr_port *<PORT>*

SDR port

--all_sdr

Set SDR-related parameters on all SDRs.

--apply_to_mdm

Apply the performance profile to the MDM. This flag is only applicable when setting the profile.

--profile {compact | high_performance}

Set the performance profile as compact or high_performance. The high performance profile configures a predefined set of parameters for very high performance use cases.

NOTE: For a complete list of parameters controlled by the profiles, refer to the *PowerFlex Performance Fine-Tuning Technical Notes*.

Example

```
scli --set_performance_parameters --all_sds --all_sdc --profile high_performance
```

start_upgrade

Start the upgrade process.

Syntax

```
scli --start_upgrade
```

Parameters

None.

Example

```
scli --start_upgrade
```

switch_cluster_mode

Switch the MDM cluster from one operating mode or configuration to another.

Syntax

```
scli --switch_cluster_mode --cluster_mode <MODE> ((--add_slave_mdm_id <ID> | --
add_slave_mdm_ip <IP> [--add_slave_mdm_port <PORT>] | --add_slave_mdm_name <NAME>)| (--
remove_slave_mdm_id <ID> | --remove_slave_mdm_ip <IP> [--remove_slave_mdm_port <PORT>]
| --remove_slave_mdm_name <NAME>)) ((--add_tb_id <ID> | --add_tb_ip <IP> [--add_tb_port
<PORT>] | --add_tb_name <NAME>) | (--remove_tb_id <ID> | --remove_tb_ip <IP> [--
remove_tb_port <PORT>] | --remove_tb_name <NAME>))
```

```
   [--allow_leave_failed]
   [--i_am_sure]
```

Parameters

--cluster_mode *<MODE>*

Cluster operation mode. Can be one of the following:

1_node

The MDM cluster will operate in single-node mode.

NOTE: Single-node mode should only be used temporarily during maintenance activities. Single-node mode does not provide failover protection for the MDM.

3_node

The MDM cluster will operate in 3-node mode.

5_node

The MDM cluster will operate in 5-node mode.

--add_slave_mdm_id *<ID>*

Comma-separated list of secondary MDM IDs that will be part of the cluster. A maximum of two IDs are allowed. Omit the space after the comma.

--add_slave_mdm_ip *<IP>*

Comma-separated list of secondary MDM IP addresses that will be part of the cluster. A maximum of two IP addresses are allowed. Omit the space after the comma.

--add_slave_mdm_port *<PORT>*

Comma-separated list of secondary MDM ports that will be part of the cluster. A maximum of two ports are allowed. Omit the space after the comma.

--add_slave_mdm_name *<NAME>*

Comma-separated list of secondary MDM names that will be part of the cluster. A maximum of two names are allowed. Omit the space after the comma.

--remove_slave_mdm_id *<ID>*

Comma-separated list of secondary MDM IDs that will be removed from the cluster. A maximum of two IDs are allowed. Omit the space after the comma.

--remove_slave_mdm_ip *<IP>*

Comma-separated list of secondary MDM IP addresses that will be removed from the cluster. A maximum of two IP addresses are allowed. Omit the space after the comma.

--remove_slave_mdm_port *<PORT>*

Comma-separated list of secondary MDM ports that will be removed from the cluster. A maximum of two ports are allowed. Omit the space after the comma.

--remove_slave_mdm_name *<NAME>*

Comma-separated list of secondary MDM names that will be removed from the cluster. A maximum of two names are allowed. Omit the space after the comma.

--add_tb_id *<ID>*

Comma-separated list of tiebreaker IDs that will be part of the cluster. A maximum of two IDs are allowed. Omit the space after the comma.

--add_tb_ip *<IP>*

Comma-separated list of tiebreaker IP addresses that will be part of the cluster. A maximum of two IP addresses are allowed. Omit the space after the comma.

--add_tb_port *<PORT>*

Comma-separated list of tiebreaker ports that will be part of the cluster. A maximum of two ports are allowed. Omit the space after the comma.

--add_tb_name *<NAME>*

Comma-separated list of tiebreaker names that will be part of the cluster. A maximum of two names are allowed. Omit the space after the comma.

--remove_tb_id *<ID>*

Comma-separated list of tiebreaker IDs that will be removed from the cluster. A maximum of two IDs are allowed. Omit the space after the comma.

--remove_tb_ip *<IP>*

Comma-separated list of tiebreaker IP addresses that will be removed from the cluster. A maximum of two IP addresses are allowed. Omit the space after the comma.

--remove_tb_port *<PORT>*

Comma-separated list of tiebreaker ports that will be removed from the cluster. A maximum of two ports are allowed. Omit the space after the comma.

--remove_tb_name *<NAME>*

Comma-separated list of tiebreaker names that will be removed from the cluster. A maximum of two names are allowed. Omit the space after the comma.

--allow_leave_failed

Allow leaving currently failed MDMs in the cluster.

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

Example

```
scli --switch_cluster_mode --cluster_mode 5_node --add_slave_mdm_ip 192.168.1.164 --
add_tb_ip 192.168.1.134
```

switch_mdm_ownership

Switch MDM cluster ownership from the current primary MDM to a different MDM.

Syntax

```
scli --switch_mdm_ownership (--new_master_mdm_id <ID> | --new_master_mdm_ip <IP> [--
new_master_mdm_port <PORT>]| --new_master_mdm_name <NAME>)
```

Parameters

--new_master_mdm_id *<ID>*

MDM ID. Must be an MDM with manager role.

--new_master_mdm_ip *<IP>*

MDM IP. Must be an MDM with manager role.

--new_master_mdm_port *<PORT>*

MDM port. Must be an MDM with manager role.

--new_master_mdm_name *<NAME>*

MDM name. Must be an MDM with manager role.

Example

```
scli --switch_mdm_ownership --new_master_mdm_ip 192.168.1.164
```

15

User Commands

This section contains commands for managing users.

add_user

Add a user to the system. A randomly generated password for the created user is returned.

This command is available only to administrator users.

Each user name should conform to the following rules:

1. Contains fewer than 32 characters
2. Contains only alphanumeric and punctuation characters (when punctuation characters are being used, you may need to use the " or ' characters in order to allow it).
3. Is unique within the object type

Syntax

```
scli --add_user --username <NAME> --user_role {Monitor | Configure | BackEndConfigure |
FrontEndConfigure | Security | Administrator [--password <PASSWORD>]}
```

Parameters

--username *<NAME>*

Username to add to the system

--user_role {Monitor | Configure | BackEndConfigure | FrontEndConfigure | Security | Administrator}

Role of the user: Monitor, Configurator, Backend Configurator, Frontend Configurator, Security, or Administrator. For information on user roles, see the *Configure and Customize Dell PowerFlex*.

--password *<PASSWORD>*

User's password. If the password parameter is notincluded, you will be prompted to enter the password.

Example

```
scli --add_user --username siouser2 --user_role Configure --pasword 3snU8fw7M
```

delete_user

Delete the specified user from the system.

This command is available only to administrator users.

Syntax

```
scli --delete_user (--user_id <ID> | --username <NAME>)
```

Parameters

--user_id *<ID>*

ID of the user to be deleted

--username *<NAME>*

Username of the user to be deleted

Example

```
scli --delete_user --username siouser2
```

disable_admin

Disables the default Superuser.

The Superuser is the default user for setting up the system, and has all the privileges of all user roles. In some cases you may need to disable the Superuser in order to ensure that all users are associated with specific user roles.

NOTE: To re-enable the Superuser, use the `reset_admin` command.

Syntax

```
scli --disable_admin
  [--i_am_sure]
```

Parameters

--i_am_sure

Skip the safety questions for command execution.

Example

```
scli --disable_admin --i_am_sure
```

modify_user

Modify the user role of the specified user in the system.

This command is available only to administrator users.

Syntax

```
scli --modify_user (--user_id <ID> | --username <NAME>) --user_role {Monitor | Configure
| BackEndConfigure | FrontEndConfigure | Security | Administrator}
```

Parameters

--user_id *<ID>*

User ID of the user to modify

NOTE: The user ID is displayed when you create the user. To find this ID at a later time, use the `query_user` command.

--username *<NAME>*

User name of the user to modify

--user_role {Monitor | Configure | BackEndConfigure | FrontEndConfigure | Security | Administrator}

Role of the user: Monitor, Configurator, Backend Configurator, Frontend Configurator, Security, or Administrator. For information on user roles, see the Dell PowerFlex User Roles and LDAP Usage Technical Notes.

Example

```
scli --modify_user --username siouser3 --user_role Monitor
```

query_user

Display information about the specified user.

This command is available only to administrator users.

Syntax

```
scli --query_user (--user_id <ID> | --username <NAME>)
```

Parameters

--user_id *<ID>*

User's ID number

NOTE: The user ID is displayed when you create the user. To find this ID at a later time, use the `query_user` command.

--username *<NAME>*

Name of the user

Example

```
scli --query_user --username ps20210131124329
```

Sample output

```
ID: ddeaa1cf00000001 Role: Monitor, Need-New-Pass: No, User Name: ps20210131124329
```

query_user_authentication_properties

Retrieve information about LDAP services configured in the system.

Syntax

```
scli --query_user_authentication_properties
```

Parameters

None.

Example

```
scli --query_user_authentication_properties
```

query_users

Display all the users defined in the system, with their roles and user ID.

Syntax

```
scli --query_users
```

Parameters

None.

Example

```
scli --query_users
```

Sample output

```
System has 3 users
ID               Role     Need-New-Pass   User Name
-------------------------------------------------------------
ddea7abf00000000 SuperUser   No          admin
ddeaa1cf00000001 Monitor     No          ps20210131124329
ddeac8df00000002 Monitor     No          ps20210201100335
```

Reset the admin user password

You can reset the password of the default admin user (Superuser) using the combination of a file written to the MDM and the `reset_admin` CLI command.

Prerequisites

Ensure that you are using the admin user with Superuser permissions.

About this task

NOTE: The procedure refers only to the default admin user with Superuser permissions, which was created during the system setup.

Steps

1. Create a text file named `MDM_SERVICE_MODE` on the MDM in the location corresponding to your operating system:
   - Windows: `C:\Program Files\emc\scaleio\MDM\logs\MDM_SERVICE_MODE`
   - Linux: `/opt/emc/scaleio/mdm/logs/MDM_SERVICE_MODE`
2. In the body of the file, type the text **`Reset Admin`**, and save the file.
3. From the CLI, run the `reset_admin` command:

```
scli --reset_admin
```

Results

The admin user password is reset to `admin`.

reset_admin

Reset the default Superuser.

Reset the password of the default admin user with Superuser permissions.

Syntax

```
scli --reset_admin
  [--i_am_sure]
```

Parameters

`--i_am_sure`

Skip the safety questions for command execution.

Example

```
scli --reset_admin --i_am_sure
```

reset_password

Generate a new password for the specified user. The user must change the password again after logging in with the generated password.

This command is available only to administrator users.

Syntax

```
scli --reset_password (--user_id <ID> | --username <NAME>) [--password <PASSWORD>]
```

Parameters

--user_id *<ID>*

User ID of the user whose password will be reset

NOTE: The user ID is displayed when you create the user. To find this ID at a later time, use the `query_user` command.

--username *<NAME>*

Username of the user whose password will be reset

--new_password *<PASSWORD>*

User's new password. If password is missing, the admin will be prompted to enter the password.

Example

```
scli --reset_password --username siouser3
```

set_password

Change the password of the user currently logged in to the system.

This command is available only to administrator users.

Syntax

```
scli --set_password [--old_password <OLD_PASSWORD>] [--new_password <NEW_PASSWORD>]
```

Parameters

None.

--old_password *<OLD_PASSWORD>*

User's current password

--new_password *<NEW_PASSWORD>*

User's new password

NOTE: In Linux, to prevent the password from being recorded in the history log, omit the `old_password` or `new_password` flag and enter the password interactively.

Example

```
scli --set_password --old_password 1!2@3A --new_password P9*7&6
```

Password rules

The password must conform to the following rules:

1. Contains between six and 31 characters.
2. Contains characters from at least three of the following groups: [a-z], [A-Z], [0-9], special characters (!@#$ …)
3. The current password is not allowed.

set_user_authentication_method

Set the user authentication method for the system.

CAUTION: Use this command with caution. The operation is complex to roll back.

Syntax

```
scli --set_user_authentication_method (--ldap_authentication | --native_authentication |
--native_and_ldap_authentication | allow_ldap_without_admin)
  [--i_am_sure]
```

Parameters

--ldap_authentication

LDAP-based authentication method where users are managed on an LDAP-compliant server. Configure LDAP service and LDAP user before switching to this authentication method.

--native_authentication

Native authentication method where users are managed locally in the system

--native_and_ldap_authentication

A hybrid authentication method. Both LDAP and local users may log in to the system after it is set.

--allow_ldap_without_admin

Allow setting LDAP athentication method even if there is no LDAP service with administrator role.

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

Example

```
scli --set_user_authentication_method --native_and_ldap_authentication --i_am_sure
```

16

Volume Commands

This section contains commands for managing volumes.

add_volume

Add a volume to the PowerFlex system.

You can create a volume when the requested capacity is available and system capacity is balanced. To start allocating volumes, at least three SDS nodes must be configured in the system.

The created volume cannot be used until it is mapped to at least one SDC.

Syntax

```
scli --add_volume (((--protection_domain_id <ID> | --protection_domain_name <NAME>) --
storage_pool_name <NAME>) | --storage_pool_id <ID>) --size_gb <SIZE>
  [--volume_name <NAME>]
  [--compression_method none | normal]
  [--thin_provisioned | --thick_provisioned | <BLANK>]
  [--use_rmcache | --dont_use_rmcache | <BLANK>]
```

Parameters

--protection_domain_id *<ID>*
: Protection Domain ID

--protection_domain_name *<NAME>*
: Protection Domain name

--storage_pool_name *<NAME>*
: Storage Pool name

--storage_pool_id *<ID>*
: Storage Pool ID

--size_gb *<SIZE>*
: Volume size, in GB. Basic allocation granularity is 8 GB.

--volume_name *<NAME>*
: Name to be associated with the added volume

--compression_method {none | normal}
: Indicates the compression method to be used, relevant only for fine granularity data layouts.

--thin_provisioned | --thick_provisioned | <BLANK>
: Specifies whether the specified volume will be thin provisioned or thick provisioned (the default). If left blank, the default of the Storage Pool will be used (thick provisioned for medium granularity, thin provisioned for fine granularity).

--use_rmcache | --dont_use_rmcache | <BLANK>
: Specifies whether the specified volume will use Read RAM Cache for caching. Default: `--use_rmcache`. If left blank, use the value of the Storage Pool.

NOTE: If you use the `storage_pool_id` parameter in the command, you do not need to also specify the Protection Domain ID or name, and the Storage Pool name.

Example

```
scli --mdm_ip 192.168.1.200 --add_volume --size_gb 8 --volume_name vol_1 --
protection_domain_name pd1 --storage_pool_name sp1
```

Volume names

Assign each volume a meaningful name associated with its operational role. When a name has not been defined, the system may display default system-defined names that use the volume's ID.

Each volume name should conform to the following rules:

1. Contains fewer than 32 characters
2. Contains only alphanumeric and punctuation characters
3. Is unique within the object type

NOTE: PowerFlex objects are assigned a unique ID that can be used to identify the object in CLI commands. You can retrieve the ID via a query or through the object's property sheet in the PowerFlex GUI.

clear_scsi2_reservation

Clear SCSI2 reservation from an SDC.

Syntax

```
scli --clear_scsi2_reservation (--volume_ID <ID> | --volume_name <NAME>) [--wait] [--
i_am_sure]
```

Parameters

--volume_id *<ID>*

ID of the volume where SCSI2 reservation from the SDC will be reset

--volume_name *<NAME>*

Name of the volume where SCSI2 reservation from the SDC will be reset

--wait

Wait for the command to be executed. If specified, the return code is the actual return code of the command.

--i_am_sure

Skip the safety questions for command execution.

Example

```
scli --clear_scsi2_reservation --volume_name public --sdc_ip 10.76.60.233 --wait
```

clear_scsi3_reservation

Clear SCSI3 reservations from all SDCs.

Syntax

```
scli --clear_scsi3_reservation (--volume_id <ID> | --volume_name <NAME>) [--wait] [--
i_am_sure]
```

Parameters

--volume_id *<ID>*

ID of the volume for which SCSI3 reservation will be reset

--volume_name *<NAME>*

Name of the volume for which SCSI3 reservation will be reset

--wait

Wait for the command to be executed. If specified, the return code is the actual return code of the command.

--i_am_sure

Skip the safety questions for command execution.

Example

```
scli --clear_scsi3_reservation --volume_name temp1 --i_am_sure
```

map_volume_to_sdc

Map a volume to an SDC.

This command exposes the volume to the specified SDC, effectively creating a block device on the SDC.

For Linux devices, the `scini` device name may change on reboot. It is therefore recommended that you mount a mapped volume to the PowerFlex unique ID, which is a persistent device name, rather than to the `scini` device name.

To identify the unique ID, run the `ls -l /dev/disk/by-id/` command. For more information, see "Associate PowerFlex volumes with physical disks" in the *Configure and Customize Dell PowerFlex*.

NOTE: You can't map a volume if the volume is an auto snapshot that is not locked.

NOTE: The volume on the target of a peer system, can't be mapped if connected to an RCG.

Syntax

```
scli --map_volume_to_sdc (--volume_id <ID> | --volume_name <NAME>) (--sdc_id <ID> | --
sdc_name <NAME> | --sdc_guid <GUID> | --sdc_ip <IP>)
  [--access_mode <ACCESS_MODE>] [--allow_multi_map]
```

Parameters

--volume_id *<ID>*

Volume ID

--volume_name *<NAME>*

Volume name

--sdc_id *<ID>*

SDC ID

--sdc_name *<NAME>*

SDC name

--sdc_guid *<GUID>*

SDC GUID

--sdc_ip *<IP>*

SDC IP address

--access_mode {read_write | read_only | no_access}

Mapping access mode. Default is `read_write`.

--allow_multi_map

Allow mapping of additional SDCs to a volume that is already mapped. This flag is not relevant to the first volume mapping command, but should be specified in any subsequent mapping command.

Example

```
scli --mdm_ip 192.168.1.200 --map_volume_to_sdc --volume_name vol_1 --sdc_ip 192.168.1.3
```

modify_volume_capacity

Increase the capacity of a volume. You can increase (but not decrease) a volume capacity at any time, as long as there is enough capacity for the volume size to grow.

Syntax

```
scli --modify_volume_capacity (--volume_id <ID> | --volume_name <NAME>) --size_gb <SIZE>
```

Parameters

--volume_id *<ID>*

Volume ID

--volume_name *<NAME>*

Volume name

--size_gb *<SIZE>*

New volume size, in GB. Basic allocation granularity is 8 GB.

Example

```
scli --mdm_ip 192.168.1.200 --modify_volume_capacity
--volume_name vol_1 --size_gb 150000
```

overwrite_volume_content

Overwrite the destination volume content with the content of the source volume. Both volumes must belong to the same V-Tree.

Syntax

```
scli --overwrite_volume_content (--source_vol_id <ID> | --source_vol_name <NAME>) (--
destination_vol_id <ID> | --destination_vol_name <NAME>)
[--overwrite_locked_auto_snapshot]
```

Parameters

--source_vol_id *<ID>*
Source volume ID

--source_vol_name *<NAME>*
Source volume name

--destination_vol_id *<ID>*
Destination volume ID

--destination_vol_name *<NAME>*
Destination volume name

--overwrite_locked_auto_snapshot
Use this flag to overwrite volume content when the destination volume is a locked auto snapshot

Example

```
scli --overwrite_volume_content --source_vol_name vol1 --destination_vol_name vol2
```

query_all_volumes

Retrieve information about all volumes in the system. You can optionally filter the query results by Protection Domain, Storage Pool, or snapshot consistency group ID.

Syntax

```
scli --query_all_volumes
   [(((--protection_domain_id <ID> | --protection_domain_name <NAME>) --
storage_pool_name <NAME>) | --storage_pool_id <ID>) | --scg_id <ID>]
```

Parameters

--protection_domain_id *<ID>*
Protection Domain ID

--protection_domain_name *<NAME>*
Protection Domain name

--storage_pool_name *<NAME>*
Storage Pool name

--storage_pool_id *<ID>*
Storage Pool ID

--scg_id *<ID>*
Snapshot consistency group ID

Example

```
scli --mdm_ip 192.168.1.200 --query_all_volumes --protection_domain_id dc65bd9900000000
```

Sample output

```
Protection Domain fcc9af5500000000 Name: pd1
Storage Pool 3df1cd9600000000 Name: sp1
 Volume ID: f6351e3700000003 Name: vol1_114 Size: 16.0 GB (16384 MB) Mapped to 1 SDC
Thin-provisioned  intercepted for replication
 Volume ID: f6351e3800000004 Name: vol2_114 Size: 8.0 GB (8192 MB) Not mapped Thin-
provisioned
 Volume ID: f6351e3900000005 Name: vol3_pol1 Size: 8.0 GB (8192 MB) Not mapped Thin-
provisioned
 Volume ID: f6683c2700000002 Name: N/A Size: 8.0 GB (8192 MB) Not mapped Auto Snapshot
of vol3_pol1

Protection Domain fcc9af5600000001 Name: pd2
Storage Pool 3df1cd9700000001 Name: sp2
 <No volumes defined>

Protection Domain fcc9d66500000002 Name: rrr
 <No volumes defined>
Query-all-volumes returned 4 volumes
```

query_volume

Retrieve information about the specified volume.

Syntax

```
scli --query_volume (--volume_id <ID> | --volume_name <NAME>)
```

Parameters

--volume_id *<ID>*
Volume ID

--volume_name *<NAME>*
Volume name

Example

```
scli --query_volume --volume_id f6351e3900000005
```

Sample output

```
>> Volume ID: f6351e3900000005 Name: vol3_pol1
   Provisioning: Thin
   Size: 8.0 GB (8192 MB)
   Total base user data: 0 Bytes
   Storage Pool 3df1cd9600000000 Name: sp1
   Protection Domain fcc9af5500000000 Name: pd1
   Creation time: 2021-01-31 13:00:40
   Number of Direct Snapshots: 2
   Reads:  0 IOPS 0 Bytes per second
   Writes: 0 IOPS 0 Bytes per second
   Data layout: Medium granularity
   Mapping:
      SDC ID: c27ab52600000001 IP: 10.55.137.119 Name: N/A Access mode: READ_WRITE
   No SCSI reservations
   The volume is a source volume of Snapshot Policy: aaa4278700000000
   VTree migration info:
      Migration status: Paused
      Pause Reason: No space in destination Storage Pool
      Source:      Storage Pool: sp1 ID: 3df1cd9600000000 Protection Domain: pd1 ID:
fcc9af5500000000
      Destination: Storage Pool: sp2 ID: 3df1cd9700000001 Protection Domain: pd2 ID:
fcc9af5600000001
      Conversion type: No conversion
      Queue position: 1
      Progress percentage : 0%
   Doesn't use RAM Read Cache
   Replication info:
      Replication state: Not being Replicated
```

query_volume_sdc_meters

Query the IOPS, bandwidth, and latency counters for volume mappings. You can either specify a volume, an SDC, or both. For volume only, display metrics for all mappings of the specified volume. For SDC only, display metrics for all mappings of volumes to the specifed SDC. For both volume and SDC, display metrics for a specific mapping.

Syntax

```
scli --query_volume_sdc_meters [--volume_id <ID> | --volume_name <NAME>] [--sdc_id <ID>
| --sdc_name <NAME> | --sdc_guid <GUID> | --sdc_ip <IP>]
```

Parameters

--volume_id *<ID>*
: Volume ID

--volume_name *<NAME>*
: Volume name

--sdc_id *<ID>*
: SDC ID

--sdc_name *<NAME>*
: SDC name

--guid *<GUID>*
: SDC Global Unique Identifier

--sdc_ip *<IP>*
: SDC IP address

Example

```
scli --query_volume_sdc_meters --volume_id f6351e3700000003
```

Sample output

```
Information on 1 mapping:

Volume: f6351e3700000003 SDC: c27ab52600000001
Reads:  426 IOPS 212.8 MB (217907 KB) per-second
        213 IOPS 22.033 ms average latency
Writes: 450 IOPS 224.6 MB (230000 KB) per-second
        225 IOPS 261.137 ms average latency
```

remove_volume

Remove a PowerFlex volume.

NOTE: Removing a volume erases all the data on the corresponding volume.

Before removing a volume, you must ensure that it is not mapped to any SDCs. If the volume is mapped to SDCs, unmap the volume using the `unmap_volume_from_sdc` command or the PowerFlex GUI, or the vSphere plug-in (where applicable) before removing it.

Syntax

```
scli --remove_volume (--volume_id <ID> | --volume_name <NAME>)
{--remove_entire_snapshot_tree | --remove_with_descendant_snapshots | --
remove_descendant_snapshots_only}
  [--i_am_sure]
```

Parameters

--volume_id *<ID>*

ID of the volume to remove

--volume_name *<NAME>*

Name of the volume to remove

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

[Options]

Additional optional parameters:

--remove_entire_snapshot_tree

Remove the entire volume tree (V-Tree) attached to the specified volume, including parents and siblings.

--remove_with_descendant_snapshots

Remove the specified volume, and all volumes that were created as snapshots of the specified volume or one of its descendants.

--remove_descendant_snapshots_only

Remove snapshots created from the specified volume, but do not remove the specified volume.

NOTE: Before removing a V-Tree, you must unmap every volume in the V-Tree.

Example

```
scli --mdm_ip 192.168.1.200 --remove_volume --volume_name vol_1
```

rename_volume

Changes the name of a volume. You can use this command at any time.

Syntax

```
scli --rename_volume --volume_id <ID>| --volume_name <NAME> --new_name <NAME>
```

Parameters

--volume_id *<ID>*
Volume ID

--volume_name *<NAME>*
Volume name

--new_name *<NAME>*
New name to assign to the volume

Example

```
scli --mdm_ip 192.168.1.200 --rename_volume
--volume_name vol1 --new_name vol_new_1
```

set_volume_rmcache_usage

Control the use of RMcache in the specified volume.

You can only configure one volume at a time. If you want to ensure that all I/Os for this volume are cached, the relevant Storage Pool should be configured to use RMcache, and the relevant SDSs should all have RMcache enabled.

NOTE: You can also enable RMcache at the volume level during initial creation of the volume using the `--add_volume` command.

Syntax

```
scli --set_volume_rmcache_usage (--volume_id <ID>| --volume_name <NAME>)(--use_rmcache |
--dont_use_rmcache)
  [--i_am_sure]
```

Parameters

--volume_id *<ID>*
Volume ID

--volume_name *<NAME>*

Volume name

--use_rmcache

Use Read RAM Cache on the volume.

--dont_use_rmcache

Do not use RMcache on the volume. This is the default setting.

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

Example

```
scli --set_volume_rmcache_usage --volume_name vol_10 --use_rmcache
```

unmap_volume_from_sdc

Unmap a volume from one or all SDCs.

Syntax

```
scli --unmap_volume_from_sdc (--volume_id <ID>| --volume_name <NAME>) (--sdc_id <ID> |
--sdc_name <NAME> | --sdc_guid <GUID> | --sdc_ip <IP> | --all_sdc)
  [--i_am_sure]
```

Parameters

--volume_id *<ID>*

Volume ID

--volume_name *<NAME>*

Volume name

--sdc_id *<ID>*

SDC ID

--sdc_name *<NAME>*

SDC name

--sdc_guid *<GUID>*

SDC Global Unique Identifier

--sdc_ip *<IP>*

SDC IP address

--all_sdc

Unmap the volume from all SDCs.

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

Example (unmap volume from a single SDC)

```
scli --mdm_ip 192.168.1.200 --unmap_volume_from_sdc
--volume_name vol_1 --sdc_ip 192.168.1.3
```

Example (unmap volume from all SDCs)

```
scli --mdm_ip 192.168.1.200 --unmap_volume_from_sdc
--volume_name vol_1 --all_sdcs
```

set_volume_mapping_access_mode

Set the mapping access mode for the specified volume.

Syntax

```
scli --set_volume_mapping_access_mode (--volume_id <ID> | --volume_name <NAME>) (--
host_id <ID> | --host_name <NAME> | --nvme_host_nqn <NQN> | --sdc_guid <GUID> | --sdc_ip
<IP> | --all_host) --access_mode <ACCESS_MODE>
```

Parameters

--volume_id *<ID>*
Volume ID

--volume_name *<NAME>*
Volume name

--host_id *<ID>*
Host ID

--host_name *<NAME>*
host name

--nvme_host_nqn *<NQN>*
NVMe Host NQN

--sdc_guid *<GUID>*
SDC GUID

--sdc_ip *<IP>*
SDC IP address

--all_host
Use this flag to apply the volume access mode to all hosts

--access_mode {read_write | read_only | no_access}
The access mode

set_volume_access_mode_limit

Set the highest mapping access mode allowed for a volume.

Syntax

```
scli --set_volume_access_mode_limit (--volume_id <ID> | --volume_name <NAME>) --
access_mode_limit <ACCESS_MODE>
```

Parameters

--volume_id *<ID>*

Volume ID

--volume_name *<NAME>*

Volume name

--access_mode_limit {read_write | read_only}

Access mode limit.

Example

```
scli --mdm_ip 192.168.1.200 set_volume_access_mode_limit --volume_id a4b5862000000000 --
access_mode_limit read_only
```

17

V-Tree Commands

This section contains commands for managing V-Trees.

migrate_vtree

Migrate an entire V-Tree (a volume and all of its snapshots) from one Storage Pool to another Storage Pool, within the same Protection Domain or between two different Protection Domains.

During V-Tree migration, you can run other tasks such as creating snapshots, deleting snapshots, and entering maintenance mode.

When a V-Tree migration is requested, the MDM first estimates whether the destination Storage Pool has enough capacity for the migration to complete successfully. The MDM bases the estimation on its information of the current capacity of the V-Tree. If there is insufficient capacity at the destination based on that estimate, migration does not start. (You can use the `--ignore_destination_capacity` option to force the migration even if there is insufficient capacity at the destination, with the intention to increase the capacity as required during the migration.) The MDM does not reserve the estimated capacity at the destination (since the capacity of the source volume can grow during migration and the reserved capacity does not guarantee success). The MDM does not hold onto source capacity once it has been migrated, but releases it immediately.

V-Tree migration can take a long time, depending on the size of the V-Tree and the system workload. During migration, the V-Tree is fully available for user I/O. V-Tree migration is done volume block by volume block. When a single block has completed its migration, the capacity of the block at the source becomes available, and it becomes active in the destination Storage Pool. During migration, the V-Tree has some of its blocks active in the source Storage Pool and the remaining blocks active in the destination Storage Pool.

When migrating from a Medium Granularity Storage Pool to a Fine Granularity Storage Pool, volumes must be zero padded. For more information on zero padding, see "Storage Pools" in the *Getting to Know Dell PowerFlex Guide*.

NOTE: You cannot create snapshots when migrating from a Medium Granularity Storage Pool to a Fine Granularity Storage Pool.

Syntax

```
scli --migrate_vtree (--volume_id <ID> | --volume_name <NAME>) (((--
destination_protection_domain_id <ID> | --destination_protection_domain_name <NAME>) --
destination_storage_pool_name <NAME>) | --destination_storage_pool_id <ID>)
  [--migration_queue_position {head_of_queue | tail_of_queue}]
  [ --convert_to_thin_provisioned]
  [ --convert_to_thick_provisioned]
  [compression_method {none | normal}
  [--ignore_destination_capacity]
```

Parameters

--volume_id *<ID>*

Volume ID of the V-Tree to be migrated

--volume_name *<NAME>*

Volume name of the V-Tree to be migrated

--destination_protection_domain_id *<ID>*

ID of the destination Protection Domain

--destination_protection_domain_name *<NAME>*

Name of the destination Protection Domain

--destination_storage_pool_name *<NAME>*

Name of the destination Storage Pool

--destination_storage_pool_id *<ID>*

ID of the destination Storage Pool

--migration_queue_position {head_of_queue | tail_of_queue}

Define whether this migration should be placed at the head or the tail of the migration queue. By default, it is placed at the tail or it retains its position if the V-Tree is already in migration.

--convert_to_thin_provisioned

Convert a thick-provisioned V-Tree to be thin-provisioned.

NOTE: SDCs with a version earlier than v3.0 do not fully support converting a thick-provisioned V-Tree to a thin-provisioned V-Tree during migration; after migration, the V-Tree will be thin-provisioned, but the SDC will not be able to trim it. These volumes can be trimmed by unmapping and then remapping them, or by rebooting the SDC. The SDC version will not affect capacity allocation and a V-Tree converted from thick to thin provisioning will be reduced in size accordingly in the system.

--convert_to_thick_provisioned

Convert a thin-provisioned V-Tree to be thick-provisioned.

compression_method {none | normal}

Indicate the compression method to be used, relevant only for fine granularity data layouts. By default, the compression is according to the settings of the destination Storage Pool.

--ignore_destination_capacity

Force migration, even if currently there is not enough capacity at the destination.

Example

```
scli --set_vtree_migration_queue_position --volume_name vol1_test --
migration_queue_position head_of_queue
```

pause_vtree_migration

Pause migration of a V-Tree (a volume and all of its snapshots), either gracefully or forcefully. Once paused, you can choose to resume the V-Tree migration, or to roll back the migration and have all volume blocks returned to the original Storage Pool.

Syntax

```
scli --pause_vtree_migration (--volume_id <ID> | --volume_name <NAME>)
  [--pause_graceful]
  [--pause_forceful]
```

Parameters

--volume_id *<ID>*

Volume ID

--volume_name *<NAME>*

Volume name

--pause_graceful

The pause will be handled gracefully, allowing all data currently being moved to complete the migration.

--pause_forceful

The pause will be handled forcefully, aborting unfinished data and leaving it at the source, if possible.

Example

```
scli --pause_vtree_migration --volume_name vol1_test --pause_ graceful
```

query_all_vtree_migrations

Retrieve the current status of all V-Tree migrations.

Syntax

```
scli --query_all_vtree_migrations
```

Parameters

None.

Example

```
scli --query_all_vtree_migrations
```

Sample output

```
Protection Domain fcc9af5500000000 Name: pd1
Storage Pool 3df1cd9600000000 Name: sp1
VTree ID: 40e7f23600000005 (Name: vol3_pol1 Volume ID: f6351e3900000005)
VTree migration info:
        Migration status: Pending retry
        Pause Reason: No space in destination Storage Pool
        Source:      Storage Pool: sp1 ID: 3df1cd9600000000 Protection Domain: pd1 ID:
fcc9af5500000000
        Destination: Storage Pool: sp2 ID: 3df1cd9700000001 Protection Domain: pd2 ID:
fcc9af5600000001
        Conversion type: No conversion
        Queue position: 1
        Progress percentage : 0%

Protection Domain fcc9af5600000001 Name: pd2
Storage Pool 3df1cd9700000001 Name: sp2
 <No vtrees defined>

Protection Domain fcc9d66500000002 Name: rrr
 <No vtrees defined>
```

query_vtree

Retrieve information about the entire V-Tree (a volume and all of its snapshots) of the specified volume.

Syntax

```
scli --query_vtree (--vtree_id <ID> | --volume_id <ID> | --volume_name
```

Parameters

--vtree_id *<ID>*
V-Tree ID

--volume_id *<ID>*
Volume ID

--volume_name *<NAME>*
Volume name

Example

```
scli --query_vtree --volume_id f6351e3800000004
```

Sample output

```
VTree ID: 40e7f23500000004 Name: vol2_114
        Storage Pool 3df1cd9600000000 Name: sp1
        Protection Domain fcc9af5500000000 Name: pd1
        Data layout: Medium granularity
        Provisioning: Thin
        Total capacity in use: 0 Bytes
        Total user data: 0 Bytes
                Total base user data: 0 Bytes
                Total snapshots user data: 0 Bytes
        Number of Volumes: 2
        Migration status: Not in migration

>> Volume ID: f6351e3800000004 Name: vol2_114
   Provisioning: Thin
   Size: 8.0 GB (8192 MB)
   Total base user data: 0 Bytes
   Creation time: 2021-01-31 12:59:42
   Number of Direct Snapshots: 1
   Volume is not mapped
   No SCSI reservations
   The volume is a source volume of Snapshot Policy: aaa4278700000000
   Doesn't use RAM Read Cache
   Replication info:
      Replication state: Not being Replicated

   >> Volume ID: f66afb540000000a Name: N/A
      Auto Snapshot of vol2_114
      Size: 8.0 GB (8192 MB)
      Creation time: 2021-02-10 17:56:45
      SCG ID: 49645e1c00000003
      Volume is not mapped
      No SCSI reservations
      The volume is an auto snapshot of Snapshot Policy: aaa4278700000000
         Auto snapshot is unlocked
         Retention levels: 1
```

```
         Expected deletion time: 2021-02-10 18:02:46
      Doesn't use RAM Read Cache
      Replication info:
         Replication state: Not being Replicated
```

query_vtree_migration

Retrieve the current status of a V-Tree migration.

Syntax

```
scli --query_vtree_migration (--volume_id <ID> | --volume_name <NAME>)
```

Parameters

None.

--volume_id *<ID>*

Volume ID

--volume_name *<NAME>*

Volume name

Example

```
scli --query_vtree_migration --volume_name vol3_pol1
```

Sample output

```
VTree ID: 40e7f23600000005 Name: vol3_pol1
        Storage Pool 3df1cd9600000000 Name: sp1
        Protection Domain fcc9af5500000000 Name: pd1
        Data layout: Medium granularity
        Provisioning: Thin
        Total capacity in use: 0 Bytes
        Total user data: 0 Bytes
                Total base user data: 0 Bytes
                Total snapshots user data: 0 Bytes
        Number of Volumes: 3
        VTree migration info:
                Migration status: Paused
                Pause Reason: No space in destination Storage Pool
                Source:      Storage Pool: sp1 ID: 3df1cd9600000000 Protection Domain:
pd1 ID: fcc9af5500000000
                Destination: Storage Pool: sp2 ID: 3df1cd9700000001 Protection Domain:
pd2 ID: fcc9af5600000001
                Conversion type: No conversion
                Queue position: 1
                Progress percentage : 0%
```

resume_vtree_migration

Resume a paused V-Tree migration.

Syntax

```
scli --resume_vtree_migration (--volume_id <ID> | --volume_name <NAME>)
```

Parameters

None.

--volume_id *<ID>*

Volume ID of the V-Tree

--volume_name *<NAME>*

Volume name

Example

```
scli --resume_vtree_migration --volume_name vol1_test
```

rollback_vtree_migration

PowerFlex does not provide a dedicated SCLI command to roll back a V-Tree migration.

To roll back a paused V-Tree migration and return the volume and all associated snapshots to the original storage pool, reverse the migration direction and resume the migration. This workflow provides functionality equivalent to the rollback operation available in the PowerFlex GUI.

Prerequisites

Before you roll back a V-Tree migration:

- Ensure that a V-Tree migration is currently in progress.
- Pause the V-Tree migration before initiating the rollback procedure.
- Identify the original source Storage Pool to which the V-Tree will be migrated back.

Procedure

1. Query the current migration.

   Display the migration details and identify the source and destination storage pools.

   Syntax

   ```
   scli --query_vtree_migration --volume_name <VOL_NAME>
   ```

   Example

   ```
   scli --query_vtree_migration --volume_name vol1
   ```

2. Pause the migration.

   Pause the migration before initiating rollback. To force an immediate pause:

   ```
   scli --pause_vtree_migration --volume_name <VOL_NAME> --pause_forceful
   ```

To allow in-flight operations to complete before pausing:

```
scli --pause_vtree_migration --volume_name <VOL_NAME> --pause_graceful
```

3. Reverse the migration direction.

   Reverse the migration direction by migrating the V-Tree back to the original source storage pool.

   Syntax

   ```
   scli --migrate_vtree --volume_name <VOL_NAME> \
   --destination_storage_pool_name <ORIGINAL_SOURCE_SP>
   ```

   Example

   ```
   scli --migrate_vtree --volume_name vol1 \
   --destination_storage_pool_name sp_source
   ```

4. Optionaly, Move the rollback migration to the head of the migration queue.

   To prioritize the rollback migration:

   ```
   scli --set_vtree_migration_queue_position \
   --volume_name <VOL_NAME> \
   --migration_queue_position head_of_queue
   ```

5. Resume the migration.

   Resume the paused migration to begin moving the V-Tree back to the original storage pool.

   ```
   scli --resume_vtree_migration --volume_name <VOL_NAME>
   ```

6. Monitor rollback progress.

   Verify that the rollback migration is progressing and continue monitoring until the V-Tree is no longer reported as migrating.

   ```
   scli --query_vtree_migration --volume_name <VOL_NAME>
   ```

   NOTE: The rollback operation does not remove the migration object from the migration queue model. It reverses the migration direction and completes the migration back to the source storage pool.

set_vtree_migration_policy

Set the I/O priority policy for V-Tree migration for a specific Storage Pool. This feature affects system performance and should only be modified by advanced users.

Syntax

```
scli --set_vtree_migration_policy (((--protection_domain_id <ID> | --
protection_domain_name <NAME>) --storage_pool_name <NAME>) |--storage_pool_id <ID>)
--policy {optimize_migration | limit_concurrent_io | favor_application_io |
dynamic_bandwidth_throttling} (Options)
  [--i_am_sure]
```

Note about V-Tree migration throttling

During V-Tree migration, the source V-Tree is copied to two destination copies. This means that for each device in the source Storage Pool there are *X* MB/sec of reads, while in the destination Storage Pool there are 2*X* MB/sec of writes. If you set the `bandwidth_limit` to *X* MB/sec, it will remain that at the source, while it will increase to 2*X* MB/sec at the destination.

Parameters

--protection_domain_id *<ID>*

Protection Domain ID

--protection_domain_name *<NAME>*

Protection Domain name

--storage_pool_name *<NAME>*

Storage Pool name

--storage_pool_id *<ID>*

Storage Pool ID

--policy *<POLICY>*

Policy to use for I/O priority of V-Tree migration. Can be one of the following:

limit_concurrent_io

Limit the number of allowed concurrent V-Tree migration I/Os to the value defined for `concurrent_io_limit`.

favor_application_io

(Default) Limit the number of allowed concurrent V-Tree migration I/Os to the values defined for `concurrent_io_limit` and `bandwidth_limit`, regardless of user I/Os.

Options

Choose one or several:

--concurrent_io_limit *<LIMIT>*

The maximum number of concurrent V-Tree migration I/Os per device. Valid range: 1-20.

--bandwidth_limit *<BANDWIDTH>*

The maximum bandwidth of V-Tree migration I/Os, in KB/s, per device. The valid range is 1024-25600 (25 MB). This property will take effect only if the policy is set to `favor_application_io`.

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

Example

```
scli --set_vtree_migration_policy --protection_domain_name pd10 --storage_pool_name sp20
--policy favor_application_io --bandwidth_limit 5120
```

set_vtree_migration_queue_position

Set the position of a V-Tree migration to be at the head or the tail of the migration queue.

Syntax

```
scli --set_vtree_migration_queue_position (--volume_id <ID>| --volume_name <NAME>)  --
migration_queue_position {head_of_queue | tail_of_queue}
```

Parameters

--volume_id *<ID>*

Volume ID

--volume_name *<NAME>*

Volume name

--migration_queue_position

Queue position of the V-Tree migration:

- `head_of_queue`
- `tail_of_queue`

Example

```
scli --set_vtree_migration_queue_position --volume_name vol1_test --
migration_queue_position head_of_queue
```

Index

A

Acceleration Pool
- add_acceleration_pool 16
- query_acceleration_pool 17
- remove_acceleration_pool 18
- rename_acceleration_pool 18

access
- set_remote_read_only_limit_state 51

acknowledge scanner error counters
- acknowledge_scanner_error_counters 42

add_snapshot_policy 146
add_source_volume_to_snapshot_policy 147
admin
- disable_admin 219
- reset_admin 222

alerts
- set_capacity_alerts_threshold 170

authentication
- check_sdc_authentication_status 112
- generate_sdc_password 113
- query_user_authentication_properties 221
- set_component_authentication_properties 212
- set_sdc_authentication 119

C

certificates
- add_trusted_ca 43
- approve_all_mdm_certificates 184
- extract_root_ca 43
- generate_certificate 186
- generate_mdm_certificate 186
- list_trusted_ca 44
- remove_trusted_ca 46

clear_scsi3_reservation 226
counters
- query_oscillating_failure_counter_parameters 45
- reset_oscillating_failure_counters 47
- set_oscillating_failure_counter_parameters 50

D

devices
- query_all_device_latency_meters 25
- query_device_latency_meters 26

E

exit_maintenance_mode 128

F

Fault Set
- add_fault_set 37
- clear_fault_set 38
- query_all_fault_sets 38
- query_fault_set 39
- remove_fault_set 40

Fault Set *(continued)*
- rename_fault_set 40

fine granularity
- disable_fgl_metadata_cache 61
- enable_flg_metadata_cache 61
- set_default_fgl_metadata_cache_size 68
- set_fgl_metadata_cache_size 140

H

help 14

L

LDAP
- add_ldap_service 54
- assign_ldap_groups_to_roles 56
- reload_ldap_certificate 57
- remove_ldap_group_from_role_assignment 58
- remove_ldap_service 58
- rename_ldap_service 59

LED 33
- *See also* SDS, set_sds_device_led

license
- query_license 44
- set_license 49

lock_auto_snapshot 148
login
- set_cli_login_banner_acceptance_mode 48
- set_cli_login_banner_preemptive_acceptance 48
- set_login_banner 49

logout 14

M

maintenance mode
- enter_maintenance_mode 128

MDM
- add_standby_mdm 183
- approve_all_mdm_certificates 184
- create_mdm_cluster 184
- generate_mdm_certificate 186
- generate_mdm_csr_file 187
- modify_cluster_virtual_ips 188
- modify_management_ip 189
- modify_virtual_ip_interfaces 191
- query_cluster 199
- query_remote_read_only_limit_state 206
- refresh_mdm_cluster_capabilities 208
- remove_standby_mdm 209
- rename_mdm 209
- replace_cluster_mdm 210
- replace_mdm_security_files 212
- set_management_client_communication 213
- switch_cluster_mode 215
- switch_mdm_ownership 217

metadata acceleration
- modify_metadata_acceleration 16

modify_snapshot_policy 149

N

non-secure mode 12

O

object names 12

P

password
    reset_password 223
    set_password 223
pause_policy_snapshot 150
performance
    query_performance_parameters 200
    set_performance_parameters 213
persistent checksum
    disable_persistent_checksum 160
    enable_persistent_checksum 162
    modify_persistent_checksum 164
policy
    modify_spare_policy 190
protected maintenance mode
    set_protected_maintenance_mode_policy 179
Protection Domain
    activate 60
    add_protection_domain 60
    inactivate_protection_domain 62
    query_all 191
    query_port_flapping_status 63
    query_port_flapping_unsupported_components 64
    query_protection_domain 64
    remove_protection_domain 67
    rename_protection_domain 67

Q

query_all
    output description 195
query_all_snapshot_policies 150
query_disconnections 44
query_properties
    object presents 205
query_remote_syslog 45
query_sdc_to_sds_disconnections 137
query_snapshot_policy 151
query_volume_sdc_meters 231

R

Read RAM Cache
    set_rmcache_usage 175
    set_rmcache_write_handling_mode 176
rebuild/rebalance
    set_rebalance_mode 171
    set_rebalance_policy 172
    set_rebuild_mode 173
    set_rebuild_policy 174
    set_rebuild_rebalance_parallelism 175
remove_snapshot_policy 152
remove_source_volume_from_snapshot_policy 153
rename_snapshot_policy 153
replication
    abort_replication_consistency_group_snapshots_create 69
    activate_replication_consistency_group 69
    activate_replication_consistency_group_mappings 70
    add_replication_consistency_group 70
    add_replication_pair 72
    add_replication_peer_system 73
    add_sdr 73
    add_sdr_ip 74
    clear_broken_replication_consistency_group_alerts 75
    create_replication_consistency_group_snapshots 76
    enter_sdr_maintenance_mode 77
    exit_sdr_maintenance_mode 77
    failover_replication_consistency_group 78
    failover_test_start_replication_consistency_group. 78
    failover_test_stop_replication_consistency_group 79
    freeze_apply_replication_consistency_group 79
    inactivate_replication_consistency_group_mappings 80
    modify_replication_consistency_group_rpo 80
    modify_replication_consistency_group_target_volume_access_mo
    81
    modify_replication_peer_system_ip 82
    modify_replication_peer_system_port 82
    modify_sdr_ip_role 83
    modify_sdr_port 84
    pause_pair_initial_copy 85
    pause_replication_consistency_group 85
    print_recreate_replication_consistency_group_script 86
    query_all_replicated_volumes 86
    query_all_replication_consistency_groups 87
    query_all_replication_journal_capacity 88
    query_all_replication_pairs 89
    query_all_replication_peer_system 89
    query_all_sdr 90
    query_replication_consistency_group 90
    query_replication_pair 92
    query_replication_peer_system 92
    query_replication_peer_system_destination_volume_candidates 93
    query_replication_peer_system_protection_domains 94
    query_sdr 95
    query_sync_now_replication_consistency_group 96
    remove_replication_consistency_group 97
    remove_replication_pair 97
    remove_replication_peer_system 98
    remove_sdr 98
    remove_sdr_ip 99
    rename_replication_consistency_group 99
    rename_replication_peer_system 100
    rename_sdr 101
    restore_replication_consistency_group 102
    resume_pair_initial_copy 101
    resume_replication_consistency_group 102
    reverse_replication_consistency_group 103
    set_pair_initial_copy_queue_position 103
    set_replication_consistency_group_consistent 104
    set_replication_consistency_group_inconsistent 104
    set_replication_journal_capacity 105
    switchover_replication_consistency_group 106
    sync_now_replication_consistency_group 107
    terminate_replication_consistency_group 107
    unfreeze_apply_replication_consistency_group 108
reset_scsi2_reservation 226
resume_snapshot_policy 154
RFcache
    disable_rfcache 109

RFcache *(continued)*
enable_rfcache 110
set_rfcache_parameters 110
roles
assign_ldap_groups_to_roles 56

S

SCLI
format 10
syntax 11
SDC
add_sdc 112
query_all_approved_sdc 113
query_all_sdc 114
query_restricted_sdc_mode 115
query_sdc 115
query_sdc_volume_limits 116
remove_sdc 117
rename_sdc 117
reset_sdc_sds_connections 118
restricted mode 112
set_restricted_sdc_mode 118
set_sdc_volume_limits 120
SDR
generate_certificate 186
show_certificate 143
SDS
abort_remove_sds 122
abort_remove_sds_device 20
activate_sds_device 20
add_sds 122
add_sds_device 21
add_sds_ip 125
clear_sds_device_error 23
disable_sds_rmcache 126
enable_sds_rmcache 127
enter_protected_maintenance_mode 129
exit_protected_maintenance_mode 130
generate_certificate 186
modify_sds_device_capacity 24
modify_sds_ip_role 131
modify_sds_port 132
query_all_sds 132
query_device_test 27
query_sds 134
query_sds_connectivity_status 133, 136
query_sds_device_info 28
query_sds_network_test_results 137
remove_sds 138
remove_sds_device 30
remove_sds_ip 138
rename_device 31
rename_sds 139
set_drl_properties 140
set_sds_device_external_acceleration_type 32
set_sds_device_led 33
set_sds_device_media_type 34
set_sds_network_limits 141
set_sds_rmcache_size 142
show_certificate 143
start_device_test 34
start_sds_network_test 144
set_approved_sdc_ips 121
set_checksum_mode 179
snapshot
clear_snapshot_security 148
set_snapshot_security 154
snapshot policy
align_snapshot_policy_to_current_time 147
snapshots
remove_snapshots_consistency_group 152
Storage Pool
add_storage_pool 158
disable_background_device_scanner 160
enable_background_device_scanner 161
modify_compression_method 163
modify_zero_padding_policy 164
query_inaccessible_user_address_space 165
query_storage_pool 166
remove_storage_pool 168
rename_storage_pool 168
reset_scanner_error_counters 169
set_storage_pool_external_acceleration_type 177
set_storage_pool_media_type 178
test_inaccessible_data 180
Superuser
reset 222
syslog
set_syslog_facility 52
start_remote_syslog 52
stop_remote_syslog 53
system
query_system_limits 207
rename_system 210

T

trusted certificate
add_certificate 182

U

unlock_auto_snapshot 156
upgrade
abort_upgrade 182
allow_commands_during_upgrade 184
finalize_upgrade 185
query_upgrade 208
start_upgrade 215
users
add_user 218
delete_user 218
modify_user 219
query_user 220
query_users 221
set_user_authentication_method 56, 224

V

V-Tree
migrate_vtree 237, 242
pause_vtree_migration 238
query_all_vtree_migrations 239
query_vtree 240
query_vtree_migration 241
resume_vtree_migration 241
set_vtree_migration_policy 243

V-Tree *(continued)*
set_vtree_migration_queue_position 244
volume
set_volume_access_mode_limit 235
snapshot_volume 155
unmap_volume_from_sdc 234
volumes
add_volume 225
map_volume_to_sdc 227
modify_volume_capacity 228
overwrite_volume_content 229
query_all_volumes 229
query_volume 230
remove_volume 232
rename_volume 233
set_volume_rmcache_usage 233

Z

zero padding, *See* Storage Pool, modify_zero_padding_policy